



```
In [1]: from pyspark import SparkConf
from pyspark.sql import SparkSession
from pyspark.ml.feature import VectorAssembler
from pyspark.ml.stat import Correlation
from pyspark.sql.types import NumericType, DoubleType
from concurrent.futures import ThreadPoolExecutor, as_completed
from pyspark.sql.functions import countDistinct, abs, stddev, col as spark_col
from pyspark.sql import functions as F
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from pyspark.sql.functions import input_file_name, regexp_extract, lit, create
from itertools import chain
import os
import tarfile
import glob
from datetime import datetime, timedelta
```

```
In [2]: BASE_KPI_PATH = "/persistent/RAN_fault_v1/KPI_data"

# -----
# Extract all tar.gz files
# -----

for file in os.listdir(BASE_KPI_PATH):
    if file.endswith(".tar.gz"):
        tar_path = os.path.join(BASE_KPI_PATH, file)
        extract_path = os.path.join(BASE_KPI_PATH, file.replace(".tar.gz", ""))

        if not os.path.exists(extract_path):
            os.makedirs(extract_path)
            print(f"Extracting {file}")

            with tarfile.open(tar_path, "r:gz") as tar:
                tar.extractall(extract_path)

print("Extraction completed")

# -----
# Find all parquet files
# -----

parquet_files = []

for root, dirs, files in os.walk(BASE_KPI_PATH):
    for f in files:
        if f.endswith(".parquet"):
            parquet_files.append(os.path.join(root, f))

print("Total parquet files:", len(parquet_files))

# -----
# Read parquet files using pandas (Pure Python)
```

```
# -----  
  
dataframes = []  
  
for parquet_file in parquet_files:  
    try:  
        df = pd.read_parquet(parquet_file)  
        dataframes.append(df)  
        print(f"Loaded: {parquet_file}")  
    except Exception as e:  
        print(f"Error loading {parquet_file}: {e}")  
  
# Combine all dataframes  
if dataframes:  
    combined_df = pd.concat(dataframes, ignore_index=True)  
    print(f"\nCombined DataFrame shape: {combined_df.shape}")  
    print(combined_df.head())  
else:  
    print("No parquet files loaded successfully")
```

Extraction completed

Total parquet files: 607

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=2/MIN_ID=0/part-00000-fd2b9ccd-c006-4ac8-b5f1-82f2a5a877dd.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=2/MIN_ID=45/part-00000-11524111-e005-4373-859d-4f4eca262645.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=2/MIN_ID=15/part-00000-1911b7da-b950-4a7f-843e-11e359d1decf.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=2/MIN_ID=30/part-00000-990e4828-1a6f-48be-a665-c27f2e1c9bcd.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=9/MIN_ID=0/part-00000-3884118b-caf5-4f69-9e0b-0b3177e28f57.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=9/MIN_ID=45/part-00000-5bf753ae-e91f-4023-9d42-a56ba2e55763.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=9/MIN_ID=15/part-00000-2089c673-f67d-4662-8f6f-20dc4231216b.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=9/MIN_ID=30/part-00000-b4352490-c4d9-4b51-8392-94e0d82d328a.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=16/MIN_ID=0/part-00000-839ba787-1e04-4e3e-9907-14dbbea74ee1.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=16/MIN_ID=45/part-00000-34a8d32f-a354-4288-b4fa-f1b8d63ea6f5.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=16/MIN_ID=15/part-00000-e8fdeec6-837e-4d2f-a883-718ca60d2b04.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=16/MIN_ID=30/part-00000-539a5aa0-b705-48f3-a99c-f5a678ae636e.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=19/MIN_ID=45/part-00000-3eb0127e-03e8-4367-b606-206e38a6d52a.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=19/MIN_ID=15/part-00000-d76ce0ce-a028-4f6a-9c2e-80a0c171c937.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=19/MIN_ID=30/part-00000-21fc745b-c464-4c81-a41d-a3fd5468c3ef.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=17/MIN_ID=45/part-00000-34672cb4-6e0f-4ecd-abc8-127c6c66ff03.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=17/MIN_ID=15/part-00000-8fc93967-3fc6-4fd6-be0c-41a9254f02f9.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=17/MIN_ID=30/part-00000-8fc93967-3fc6-4fd6-be0c-41a9254f02f9.c000.snappy.parquet

6-07-02/HOUR_ID=17/MIN_ID=30/part-00000-1b6cbdb2-3765-475a-a337-8aae429a5887.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=11/MIN_ID=45/part-00000-1b4be95d-e7a2-4607-b39e-805fee6aaa3a.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=11/MIN_ID=15/part-00000-5acdfeb20-6320-4b45-947e-936eb1228d80.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=11/MIN_ID=30/part-00000-96eaf413-d4e7-4c0c-b364-0ff30903d1bf.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=12/MIN_ID=0/part-00000-7edb9f45-3e85-487b-8a3e-f22f4edb56aa.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=12/MIN_ID=45/part-00000-d652a736-29b4-4752-955c-a5aebad24e14.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=12/MIN_ID=15/part-00000-9e103549-5dac-4111-bdff-caa26924bb1a.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=12/MIN_ID=30/part-00000-3b309154-61d9-4237-8043-7b440f92112a.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=7/MIN_ID=0/part-00000-5209ca7e-87f5-41dc-8d3d-14af7cf2e761.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=7/MIN_ID=45/part-00000-002f97bd-40d3-4434-aef4-493c20dbd49b.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=7/MIN_ID=15/part-00000-36939434-6f9b-44c6-bddd-42d2187056b3.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=7/MIN_ID=30/part-00000-288cbaf1-5f8b-4842-b11e-1a093d41ab78.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=21/MIN_ID=0/part-00000-belle724-3d5e-4e7b-bcf9-5ef357727f8c.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=21/MIN_ID=45/part-00000-3ebfce39-cd9f-4e19-86e6-4f1de3faefd8.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=21/MIN_ID=15/part-00000-2b012391-39fc-456f-a89b-fbdf768d8595.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=21/MIN_ID=30/part-00000-5259f1c9-aca9-4b1b-b8e5-5d62c4dd13fa.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=20/MIN_ID=45/part-00000-ceedf5f9-29b5-4679-8ffc-d66d80d4b0ac.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=20/MIN_ID=15/part-00000-d7456828-2342-4cef-98bb-21bf6f09f9b2.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=20/MIN_ID=30/part-00000-5259f1c9-aca9-4b1b-b8e5-5d62c4dd13fa.c000.snappy.parquet

6-07-02/HOUR_ID=20/MIN_ID=30/part-00000-ff841a18-9427-4528-a353-dbeebb1f032e.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=13/MIN_ID=45/part-00000-a210ffee-acd4-4e30-bf25-5112d6c2185b.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=13/MIN_ID=15/part-00000-53b09721-df74-4e9d-946e-5e7413eeaa2c.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=13/MIN_ID=30/part-00000-d2c71e65-b970-4e03-90ff-1cad6b364289.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=0/MIN_ID=45/part-00000-a246e031-a49a-4c22-95a9-1f1c795fcd73.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=0/MIN_ID=30/part-00000-8b10b3e7-2d0e-4263-9b74-3a40fd2ca392.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=8/MIN_ID=45/part-00000-4ff0750a-e5a2-4d08-b43b-e0b8b20a3c47.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=8/MIN_ID=15/part-00000-claf9228-31ca-4f5c-a66c-ceb6d4406d59.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=8/MIN_ID=30/part-00000-4f316883-e4bb-4818-8ad8-4abbbfb03c32c.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=6/MIN_ID=45/part-00000-bad47d01-5465-4bce-abf5-a1db6a6b8192.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=6/MIN_ID=15/part-00000-0e00bf92-a2c8-4ae8-b36f-97ce2ef8e57d.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=6/MIN_ID=30/part-00000-333572f8-213b-4c93-b8ff-442c6c86c1eb.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=3/MIN_ID=45/part-00000-1247e74b-dc5e-4742-a478-d5c8c7beee6b.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=3/MIN_ID=15/part-00000-dbe4e8c6-fd43-4fea-bdae-5870ccfa836d.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=3/MIN_ID=30/part-00000-be4545b1-8acf-40be-b813-da9b3bb2fc74.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=18/MIN_ID=0/part-00000-bb57bf12-f66f-4104-bb93-0ce1564863ab.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=18/MIN_ID=45/part-00000-15e61343-2c88-41aa-ba6b-18090ac8fa08.c000.snappy.parquet
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Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=18/MIN_ID=15/part-00000-200a5dbf-3423-4047-9bf4-f30e62a3cf54.c000.snappy.parquet

6-07-02/HOUR_ID=18/MIN_ID=30/part-00000-56752c59-15bd-4608-877b-ddc68ec354f3.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=23/MIN_ID=0/part-00000-a14520a3-287e-457e-a0f2-80a7bd67c2d6.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=23/MIN_ID=45/part-00000-1c4ff279-5acf-4bca-b673-951e2316af8c.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=23/MIN_ID=15/part-00000-8d26c8d9-cf25-4773-aa44-0e0347ee2b72.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=23/MIN_ID=30/part-00000-8f44493d-8b00-41ee-b6da-b3134f67bc17.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=22/MIN_ID=0/part-00000-e7391af0-2e58-4ddb-a6e2-12576d7054fc.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=22/MIN_ID=45/part-00000-d78c0e3a-2080-4803-bf29-14fc962fa81d.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=22/MIN_ID=15/part-00000-26693de6-1d12-428a-9961-396b80ad242d.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=22/MIN_ID=30/part-00000-8add267d-4dc0-4ae6-bb97-526f395389a2.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=15/MIN_ID=0/part-00000-9eb25992-c6a4-494a-a019-f41d9416c108.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=15/MIN_ID=45/part-00000-74bf185e-9a5f-4224-8463-dd609b36f636.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=15/MIN_ID=15/part-00000-3c512f5a-4570-4683-9735-cce7efdb2566.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=15/MIN_ID=30/part-00000-c33fbe15-27b9-45cc-b339-09101e911091.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=14/MIN_ID=45/part-00000-b4356b33-2007-42e9-b4be-fd6415e824b6.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=14/MIN_ID=15/part-00000-29a934ae-c5c6-4792-b42c-c1b986ff3f3f.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=14/MIN_ID=30/part-00000-1c60d107-305f-4415-9645-1c8d12278ecc.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=10/MIN_ID=0/part-00000-0b070d7c-644e-455a-b26f-904e3acac784.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=10/MIN_ID=45/part-00000-d032300a-e948-41d1-a363-a0a99672404d.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=10/MIN_ID=15/part-00000-7684e32d-8483-4fb9-8840-b13e85e03c5e.c000.snappy.parquet
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Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-02/NRCell/DATE_ID=2026-07-02/HOUR_ID=1/MIN_ID=0/part-00000-d162bd94-ad52-4484-b59e-935718803987.c000.snappy.parquet
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Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-04/NRCell/DATE_ID=2026-07-04/HOUR_ID=0/MIN_ID=30/part-00000-c008b4f5-3ba3-4dc6-b22d-ee0feb37f49.c000.snappy.parquet

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Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-04/NRCell/DATE_ID=2026-07-04/HOUR_ID=8/MIN_ID=15/part-00000-d9d0462f-c273-4532-81c0-4741bb9f6eec.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-04/NRCell/DATE_ID=2026-07-04/HOUR_ID=8/MIN_ID=30/part-00000-3b8feed8-28a4-4306-aaf6-4c1e0e3751e0.c000.snappy.parquet

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Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-04/NRCell/DATE_ID=2026-07-04/HOUR_ID=6/MIN_ID=45/part-00000-a24f31c2-000a-49e2-a4f6-6c1bb2dbffa9.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-04/NRCell/DATE_ID=2026-07-04/HOUR_ID=6/MIN_ID=15/part-00000-8d9dc99e-8634-42d9-baab-990b5ef30e54.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-04/NRCell/DATE_ID=2026-07-04/HOUR_ID=6/MIN_ID=30/part-00000-964601b2-19ba-46c5-a242-6965848b55b5.c000.snappy.parquet

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Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-04/NRCell/DATE_ID=2026-07-04/HOUR_ID=3/MIN_ID=15/part-00000-05e5be18-aa0f-42e9-b49b-9216a8d88dcd.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-04/NRCell/DATE_ID=2026-07-04/HOUR_ID=3/MIN_ID=30/part-00000-6f3cede2-3319-4e81-a762-70c35aa519f5.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-04/NRCell/DATE_ID=2026-07-04/HOUR_ID=18/MIN_ID=45/part-00000-97305f81-c97f-4c95-ac90-a7d8955ec6e0.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-04/NRCell/DATE_ID=2026-07-04/HOUR_ID=18/MIN_ID=15/part-00000-76f1b8c4-58b0-45ae-945e-4bc8b5163d7e.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-04/NRCell/DATE_ID=2026-07-04/HOUR_ID=18/MIN_ID=30/part-00000-4bd64a53-bfd6-401e-b370-e2cde9bb0473.c000.snappy.parquet

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Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-04/NRCell/DATE_ID=2026-07-04/HOUR_ID=14/MIN_ID=30/part-00000-2434d9a8-4a7c-439d-8b68-011d35e49aa8.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-04/NRCell/DATE_ID=2026-07-04/HOUR_ID=10/MIN_ID=0/part-00000-2976e910-345c-4d51-9768-bcef2b8a564c.c000.snappy.parquet
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Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-05/NRCell/DATE_ID=2026-07-05/HOUR_ID=2/MIN_ID=0/part-00000-11144744-e037-4a11-b766-b73d983ff1c0.c000.snappy.parquet
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Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-05/NRCell/DATE_ID=2026-07-05/HOUR_ID=2/MIN_ID=30/part-00000-129653de-d68c-4b1e-8373-05f37a86586a.c000.snappy.parquet

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Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-05/NRCell/DATE_ID=2026-07-05/HOUR_ID=9/MIN_ID=45/part-00000-ec654dfc-414f-49e5-baa4-f4a1a1ca6436.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-05/NRCell/DATE_ID=2026-07-05/HOUR_ID=9/MIN_ID=15/part-00000-c413ee3f-5db4-4cdc-b02d-273ee7727157.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-05/NRCell/DATE_ID=2026-07-05/HOUR_ID=9/MIN_ID=30/part-00000-0724f525-8b11-40e4-8543-3691e955aebf.c000.snappy.parquet
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Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-05/NRCell/DATE_ID=2026-07-05/HOUR_ID=0/MIN_ID=45/part-00000-4a7637cf-dc06-4c1d-a03a-d0d70ffccd2b.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-05/NRCell/DATE_ID=2026-07-05/HOUR_ID=0/MIN_ID=15/part-00000-ee1d7140-d649-4726-8b17-b83fb4cd4186.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-05/NRCell/DATE_ID=2026-07-05/HOUR_ID=0/MIN_ID=30/part-00000-84eb7e76-12cf-4cc9-bfd4-824f7dcfb938.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-05/NRCell/DATE_ID=2026-07-05/HOUR_ID=8/MIN_ID=0/part-00000-00b58e8d-2eae-4627-b405-5e9d6219e669.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-05/NRCell/DATE_ID=2026-07-05/HOUR_ID=8/MIN_ID=45/part-00000-4e268501-4039-4321-860a-8f8bb1bf0855.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-05/NRCell/DATE_ID=2026-07-05/HOUR_ID=8/MIN_ID=15/part-00000-00097459-a581-4d3e-a893-e7c23922eb50.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-05/NRCell/DATE_ID=2026-07-05/HOUR_ID=8/MIN_ID=30/part-00000-900fbf60-9a30-4766-8aa0-d7c0cf2230ba.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-05/NRCell/DATE_ID=2026-07-05/HOUR_ID=6/MIN_ID=0/part-00000-7e323b7c-fe26-4306-a3eb-12829f9beeea.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-05/NRCell/DATE_ID=2026-07-05/HOUR_ID=6/MIN_ID=45/part-00000-9398c274-fb4e-4236-90dc-182e859d3884.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-05/NRCell/DATE_ID=2026-07-05/HOUR_ID=6/MIN_ID=15/part-00000-40f0ea11-d0a8-4f5f-9840-861900450f65.c000.snappy.parquet

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Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-05/NRCell/DATE_ID=2026-07-05/HOUR_ID=3/MIN_ID=0/part-00000-6e7bc3df-57a6-4f45-88e8-c77f434c2c57.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-05/NRCell/DATE_ID=2026-07-05/HOUR_ID=3/MIN_ID=45/part-00000-129c5ca7-4c02-411f-bfef-f9b9fad0e005.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-05/NRCell/DATE_ID=2026-07-05/HOUR_ID=3/MIN_ID=15/part-00000-ccf9ef2c-326f-43f8-9300-929c47f99660.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-05/NRCell/DATE_ID=2026-07-05/HOUR_ID=3/MIN_ID=30/part-00000-c2bd5b51-cff9-41e7-bfae-58ee94ed80e5.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-05/NRCell/DATE_ID=2026-07-05/HOUR_ID=18/MIN_ID=0/part-00000-ee3ab2c9-2703-463e-8f6d-860cafebf39b.c000.snappy.parquet

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Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-05/NRCell/DATE_ID=2026-07-05/HOUR_ID=22/MIN_ID=15/part-00000-b6adaf97-c33f-4073-8f84-4d3da557e002.c000.snappy.parquet

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Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-06/NRCell/DATE_ID=2026-07-06/HOUR_ID=2/MIN_ID=45/part-00000-1dde76a4-92cf-49bd-bbdf-5b783cd234dc.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-06/NRCell/DATE_ID=2026-07-06/HOUR_ID=2/MIN_ID=15/part-00000-15c74a5f-5d70-40a8-a9ae-454f98f83920.c000.snappy.parquet
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Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-06/NRCell/DATE_ID=2026-07-06/HOUR_ID=9/MIN_ID=45/part-00000-8a65afa9-3752-4748-a5ea-b1d5e4c7ea55.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-06/NRCell/DATE_ID=2026-07-06/HOUR_ID=9/MIN_ID=15/part-00000-470147c6-9053-4cf0-be6b-f19ef67a99c6.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-06/NRCell/DATE_ID=2026-07-06/HOUR_ID=9/MIN_ID=30/part-00000-5ab4c3d9-67e8-40dd-acfe-320a89b72840.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-06/NRCell/DATE_ID=2026-07-06/HOUR_ID=16/MIN_ID=0/part-00000-9c2cfc55-7363-4b84-a6ba-6cb4d8eb904d.c000.snappy.parquet

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Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-06/NRCell/DATE_ID=2026-07-06/HOUR_ID=7/MIN_ID=30/part-00000-bba16906-fea0-4e63-91ef-efa6eff220ac.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-06/NRCell/DATE_ID=2026-07-06/HOUR_ID=21/MIN_ID=0/part-00000-3137c33c-b0cd-4c08-bd57-eb66c26c039c.c000.snappy.parquet

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Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-06/NRCell/DATE_ID=2026-07-06/HOUR_ID=0/MIN_ID=45/part-00000-3154522e-4584-4abb-9ce9-37c2da64da6a.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-06/NRCell/DATE_ID=2026-07-06/HOUR_ID=0/MIN_ID=15/part-00000-e1446ed7-dbc7-419e-9153-55b4f1bde6f6.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-06/NRCell/DATE_ID=2026-07-06/HOUR_ID=0/MIN_ID=30/part-00000-0c09da1d-235c-4a0c-9651-d1a5411a6c3d.c000.snappy.parquet

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Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-06/NRCell/DATE_ID=2026-07-06/HOUR_ID=8/MIN_ID=15/part-00000-64b447e7-e333-46be-8c67-3c18b242a0be.c000.snappy.parquet
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Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-01/NRCell/DATE_ID=2026-07-01/HOUR_ID=9/MIN_ID=15/part-00000-726470ce-47c4-4f2a-ad3c-f49868270ed7.c000.snappy.parquet
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Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-01/NRCell/DATE_ID=2026-07-01/HOUR_ID=18/MIN_ID=0/part-00000-dce569d9-2784-4de5-83e2-840b450a2395.c000.snappy.parquet

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Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-01/NRCell/DATE_ID=2026-07-01/HOUR_ID=18/MIN_ID=15/part-00000-7708e55f-e16d-4fe1-9452-38710d9fddb.b.c000.snappy.parquet

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Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-01/NRCell/DATE_ID=2026-07-01/HOUR_ID=22/MIN_ID=15/part-00000-46e7f71e-6749-4ff7-9eb6-45ad376b7f03.c000.snappy.parquet

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Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-01/NRCell/DATE_ID=2026-07-01/HOUR_ID=4/MIN_ID=0/part-00000-c7401cd1-6564-4f6b-ad3e-e965251f6def.c000.snappy.parquet
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Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-01/NRCell/DATE_ID=2026-07-01/HOUR_ID=5/MIN_ID=30/part-00000-12cbf634-4859-47a1-a6e5-d64b2124b018.c000.snappy.parquet
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Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-07/NRCell/DATE_ID=2026-07-07/HOUR_ID=2/MIN_ID=15/part-00000-481e1555-0c04-4258-a153-5e05029afbe7.c000.snappy.parquet
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Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-07/NRCell/DATE_ID=2026-07-07/HOUR_ID=11/MIN_ID=0/part-00000-54e16811-7007-4d56-bbf3-81470c2ed123.c000.snappy.parquet
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Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=8/MIN_ID=15/part-00000-241f4adc-b026-4f16-a419-bc3dfb5303d0.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=8/MIN_ID=30/part-00000-536d0e79-37b0-4c7c-ae9-57923be5efc3.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=6/MIN_ID=0/part-00000-60db7f26-7b78-4bb7-bbf6-6434eb054ab0.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=6/MIN_ID=45/part-00000-c83b0e97-bf6f-4e9c-816c-62145209df04.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=6/MIN_ID=15/part-00000-1cc40402-9074-4ae9-92b6-63a4b272afc0.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=6/MIN_ID=30/part-00000-6f5f86d7-5fa3-4445-89f4-50f2b820072c.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=3/MIN_ID=45/part-00000-34ca8c6e-4bc5-4262-88e2-1c2ea6b8a50a.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=3/MIN_ID=15/part-00000-22e618e2-447f-4263-a298-57b8eecd202.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=3/MIN_ID=30/part-00000-8c7ce910-5832-4b7e-8887-bd0594b51380.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=18/MIN_ID=0/part-00000-ab95f500-ac19-4918-aa26-4dd8a8b9922b.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=18/MIN_ID=45/part-00000-ff494af9-1044-4dbc-817c-747a449ad462.c000.snappy.parquet
Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=18/MIN_ID=15/part-00000-3e85745a-569d-4dca-84ee-37d241b9c825.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=18/MIN_ID=30/part-00000-ab2edf84-c461-4d3e-b4fb-c30a1f583022.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=23/MIN_ID=45/part-00000-139826ee-de5c-4e47-906f-2e9379f7ead7.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=23/MIN_ID=15/part-00000-36ed1fe2-fa01-4587-aa2e-5f327ff60d14.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=23/MIN_ID=30/part-00000-95abd6eb-0287-43c5-99b3-44d72254f51f.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=22/MIN_ID=45/part-00000-c5fa97ce-9070-43c8-95c1-2edc4661d016.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=22/MIN_ID=15/part-00000-63bbe074-6a5c-4216-a503-3f5131c80998.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=22/MIN_ID=30/part-00000-be6d705f-dcd0-4f6a-b55d-0e7a8c51d771.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=15/MIN_ID=45/part-00000-81baa33d-f9cd-4821-95cf-788d2fa71269.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=15/MIN_ID=15/part-00000-abe1098c-5b83-45d2-a733-23726ec419d6.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=15/MIN_ID=30/part-00000-d14e499c-717e-4b96-8b37-f71737544891.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=14/MIN_ID=0/part-00000-4e7a245e-10d7-4661-8dae-4383412e617b.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=14/MIN_ID=45/part-00000-bea90df7-b223-4ec7-9550-c6f2e14bc6f3.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=14/MIN_ID=15/part-00000-bfb9cd20-34ac-4d25-9e65-93ce687d9870.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=14/MIN_ID=30/part-00000-c2ef7be6-ffbb-4dfa-b58f-395ac35667bf.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=10/MIN_ID=45/part-00000-ca8df467-cb73-40ae-b4f8-600e6d7c149c.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=10/MIN_ID=15/part-00000-1c01d27b-5d0e-498e-ab7a-1c1683bd48ee.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=10/MIN_ID=30/part-00000-97a46aff-61eb-43fa-b7ee-23ab84eba8a9.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=1/MIN_ID=45/part-00000-0cb680e3-a881-4f25-a80b-c4a78b42c070.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=1/MIN_ID=15/part-00000-9c08e395-21fb-4d5f-bebb-deed54dd93b7.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=1/MIN_ID=30/part-00000-e38dac7a-7f5d-4640-8f1c-baae6548b63a.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=4/MIN_ID=0/part-00000-05e118f4-5ada-4bcf-87a3-2f13e1e0df1d.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=4/MIN_ID=45/part-00000-ac864e66-7942-48ad-acfe-7555b416cc82.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=4/MIN_ID=15/part-00000-35d464db-10a4-401c-880d-8ac64dc4bc6a.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=4/MIN_ID=30/part-00000-2398d415-2e82-4170-976b-79c145afda88.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=5/MIN_ID=0/part-00000-c9dc7382-fb4b-4967-837b-c05e2f90cd32.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=5/MIN_ID=45/part-00000-448f7b25-024f-4d87-a8c6-c88caac4d790.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=5/MIN_ID=15/part-00000-9a5e1e6d-3862-4676-9988-5e776589a1e7.c000.snappy.parquet

Loaded: /persistent/RAN_fault_v1/KPI_data/DATE_ID=2026-07-03/NRCell/DATE_ID=2026-07-03/HOUR_ID=5/MIN_ID=30/part-00000-3d3adaa4-e6c6-4ec3-a72d-a797f4be85b6.c000.snappy.parquet

Combined DataFrame shape: (18030814, 82)

	COLLECTTIME	CELLID	SITE \
0	20260702020000	ATG0001H_7NB02_S01	ATG0001P_9NB02
1	20260702020000	ATG0001H_7NB02_S04	ATG0001P_9NB02
2	20260702020000	ATG0003H_7NB01_S01	ATG0003T_2NB01
3	20260702020000	ATG0003H_7NB01_S03	ATG0003T_2NB01
4	20260702020000	ATG0006H_7NB01_S02	ATG0006T_2NB01

	NR_SA_Paging_Discard_Rate	NR_DL_Active_UEs	NR_UL_Active_UEs \
0	NaN	1.132	1.727
1	NaN	1.551	1.944
2	NaN	1.811	1.390
3	NaN	1.770	1.479
4	NaN	1.046	1.073

	NR_DL_MAC_Volume_MB	NR_UL_MAC_Volume_MB	NR_DL_Active_UEs_True \
0	99.548	50.048	0.252
1	800.701	83.931	0.914
2	516.914	70.622	1.363
3	736.355	41.639	1.242
4	68.105	7.764	0.112

	NR_UL_Active_UEs_True ...	NR_ENDC_RRC_CU_Max	NR_SA_RRC_CU_Max \
0	1.619 ...	19.0	0.0
1	1.569 ...	32.0	0.0
2	0.945 ...	21.0	0.0
3	0.918 ...	28.0	0.0
4	0.280 ...	11.0	0.0

	NR_ENDC_RRC_CU_Avg	NR_SA_DRB_EST_SR	NR_SA_Drop_Rate	NR_SgNB_Abnormal_RR
0	13.833	NaN	NaN	0.0
1	23.922	NaN	NaN	0.0
2	15.317	NaN	NaN	0.0
3	21.206	NaN	NaN	0.0
4	5.922	NaN	NaN	0.0

	NR_RRC_SSR	ESS_TOTAL_PRB_AVG_UTIL_DL	ESS_NR_PRB_AVG_UTIL_DL	essEnabled
0	100.0	7.486420	4.033320	1
1	100.0	24.622996	18.494840	1
2	100.0	35.661805	19.811150	1
3	NaN	29.576583	24.543550	1
4	NaN	17.533979	2.369099	1

[5 rows x 82 columns]

In [3]: BASE_TICKET_PATH = "/persistent/RAN_fault_v1/ITSM_Ticket_Data"

```
# -----
# Extract all tar.gz files
# -----

for file in os.listdir(BASE_TICKET_PATH):

    if file.endswith(".tar.gz"):

        tar_path = os.path.join(BASE_TICKET_PATH, file)

        extract_path = os.path.join(
            BASE_TICKET_PATH,
            file.replace(".tar.gz", "")
        )

        if not os.path.exists(extract_path):

            os.makedirs(extract_path, exist_ok=True)

            print(f"Extracting {file}")

            with tarfile.open(tar_path, "r:gz") as tar:
                tar.extractall(extract_path)

print("Extraction completed.")

# -----
# Read all CSVs into one Pandas DataFrame
# -----

dfs = []

for root, dirs, files in os.walk(BASE_TICKET_PATH):

    for file in files:
```

```

        if file.lower().endswith(".csv"):

            csv_path = os.path.join(root, file)

            print(f"Reading {csv_path}")

            temp_df = pd.read_csv(
                csv_path,
                low_memory=False
            )

            dfs.append(temp_df)
    if dfs:
        # Concatenate all DataFrames
        ticket_df = pd.concat(
            dfs,
            ignore_index=True
        )

        print(f"\nTotal Rows    : {ticket_df.shape[0]:,}")
        print(f"Total Columns : {ticket_df.shape[1]}")
        ticket_df.head()
    else:
        print("No CSV files found.")
    print(len(dfs))

```

[illegible]

[illegible]

a_hourly/2026-07-02/23/itsm_hpd_tms_data.csv
Reading /persistent/RAN_fault_v1/ITSM_Ticket_Data/2026-07-02/itsm_hpd_tms_data_hourly/2026-07-02/04/itsm_hpd_tms_data.csv
Reading /persistent/RAN_fault_v1/ITSM_Ticket_Data/2026-07-02/itsm_hpd_tms_data_hourly/2026-07-02/07/itsm_hpd_tms_data.csv
Reading /persistent/RAN_fault_v1/ITSM_Ticket_Data/2026-07-02/itsm_hpd_tms_data_hourly/2026-07-02/17/itsm_hpd_tms_data.csv
Reading /persistent/RAN_fault_v1/ITSM_Ticket_Data/2026-07-02/itsm_hpd_tms_data_hourly/2026-07-02/00/itsm_hpd_tms_data.csv
Reading /persistent/RAN_fault_v1/ITSM_Ticket_Data/2026-07-02/itsm_hpd_tms_data_hourly/2026-07-02/22/itsm_hpd_tms_data.csv
Reading /persistent/RAN_fault_v1/ITSM_Ticket_Data/2026-07-02/itsm_hpd_tms_data_hourly/2026-07-02/06/itsm_hpd_tms_data.csv
Reading /persistent/RAN_fault_v1/ITSM_Ticket_Data/2026-07-02/itsm_hpd_tms_data_hourly/2026-07-02/16/itsm_hpd_tms_data.csv
Reading /persistent/RAN_fault_v1/ITSM_Ticket_Data/2026-07-02/itsm_hpd_tms_data_hourly/2026-07-02/21/itsm_hpd_tms_data.csv
Reading /persistent/RAN_fault_v1/ITSM_Ticket_Data/2026-07-02/itsm_hpd_tms_data_hourly/2026-07-02/18/itsm_hpd_tms_data.csv
Reading /persistent/RAN_fault_v1/ITSM_Ticket_Data/2026-07-02/itsm_hpd_tms_data_hourly/2026-07-02/12/itsm_hpd_tms_data.csv
Reading /persistent/RAN_fault_v1/ITSM_Ticket_Data/2026-07-02/itsm_hpd_tms_data_hourly/2026-07-02/05/itsm_hpd_tms_data.csv
Reading /persistent/RAN_fault_v1/ITSM_Ticket_Data/2026-07-02/itsm_hpd_tms_data_hourly/2026-07-02/08/itsm_hpd_tms_data.csv
Reading /persistent/RAN_fault_v1/ITSM_Ticket_Data/2026-07-02/itsm_hpd_tms_data_hourly/2026-07-02/11/itsm_hpd_tms_data.csv

[illegible]

Total Rows : 2,835,299

Total Columns : 31
120

In [121]: ticket_df

	INCIDENT_NUMBER	VENDOR_TICKET_NUMBER	ASSIGNED_GROUP
16410	INC000101958091	SCCD_TT202607100807_A23447892	TRUE-TH-BBT-NOR1-CMI1-NOP
16566	INC000101959839	SCCD_TT202607101447_A23448518	TRUE-TH-WW-CEN-CNT-NOP
16853	INC000101962626	SCCD_TT202607102893_A23449985	TRUE-TH-WW-EAS-CCS-NOP
16961	INC000101963557	SCCD_TT202607103555_A23450645	TRUE-TH-BBT-NOR1-CMI2-NOP
17146	INC000101965073	SCCD_TT202607104396_A23451491	TRUE-TH-WW-CEN-LBR-NOP
...	...	...	...
55871	INC000102118254	SCCD_TT202607161267_A23508142	TRUE-TH-WW-BMAW1-NOP
55954	INC000102119012	SCCD_TT202607161367_A23508238	TRUE-TH-BBT-NOR1-CMI2-NOP
56006	INC000102118961	SCCD_TT202607161427_A23508298	TRUE-TH-BBT-NOR1-LPN-NOP
56007	INC000102119260	SCCD_TT202607161395_A23508267	TRUE-TH-WW-CEN-ATG-NOP
56061	INC000102119370	SCCD_TT202607161443_A23508316	TRUE-TH-BBT-NOR1-CMI2-NOP

329 rows × 39 columns

```
In [5]: BASE_CLOSED_TICKET_PATH = "/persistent/RAN_fault_v1/ITSM_Closed_Ticket_data"

# -----
# Extract all tar.gz files
# -----

for file in os.listdir(BASE_CLOSED_TICKET_PATH):
```

```

if file.endswith(".tar.gz"):

    tar_path = os.path.join(BASE_CLOSED_TICKET_PATH, file)

    extract_path = os.path.join(
        BASE_CLOSED_TICKET_PATH,
        file.replace(".tar.gz", "")
    )

    if not os.path.exists(extract_path):

        os.makedirs(extract_path, exist_ok=True)

        print(f"Extracting {file}")

        with tarfile.open(tar_path, "r:gz") as tar:
            tar.extractall(extract_path)

print("Extraction completed.")

# -----
# Read all CSVs into one Pandas DataFrame
# -----

closed_ticket_dfs = []

for root, dirs, files in os.walk(BASE_CLOSED_TICKET_PATH):

    for file in files:

        if file.lower().endswith(".csv"):

            csv_path = os.path.join(root, file)

            print(f"Reading {csv_path}")

            temp_df = pd.read_csv(
                csv_path,
                low_memory=False
            )

            closed_ticket_dfs.append(temp_df)

# -----
# Concatenate all DataFrames
# -----

if closed_ticket_dfs:

    tickets_closed_df = pd.concat(
        closed_ticket_dfs,
        ignore_index=True
    )

```

```

)

print(f"\nTotal Rows      : {tickets_closed_df.shape[0],}")
print(f"Total Columns : {tickets_closed_df.shape[1]}")

display(tickets_closed_df.head())

else:

    print("No CSV files found.")

```

Extraction completed.

Reading /persistent/RAN_fault_v1/ITSM_Closed_Ticket_data/2026-07-04/itsm_closed_data/2026-07-04/itsm_closed_data.csv
 Reading /persistent/RAN_fault_v1/ITSM_Closed_Ticket_data/2026-07-03/itsm_closed_data/2026-07-03/itsm_closed_data.csv
 Reading /persistent/RAN_fault_v1/ITSM_Closed_Ticket_data/2026-07-05/itsm_closed_data/2026-07-05/itsm_closed_data.csv
 Reading /persistent/RAN_fault_v1/ITSM_Closed_Ticket_data/2026-07-06/itsm_closed_data/2026-07-06/itsm_closed_data.csv
 Reading /persistent/RAN_fault_v1/ITSM_Closed_Ticket_data/2026-07-02/itsm_closed_data/2026-07-02/itsm_closed_data.csv
 Reading /persistent/RAN_fault_v1/ITSM_Closed_Ticket_data/2026-07-01/itsm_closed_data/2026-07-01/itsm_closed_data.csv
 Reading /persistent/RAN_fault_v1/ITSM_Closed_Ticket_data/2026-07-09/itsm_closed_data/2026-07-09/itsm_closed_data.csv
 Reading /persistent/RAN_fault_v1/ITSM_Closed_Ticket_data/2026-07-07/itsm_closed_data/2026-07-07/itsm_closed_data.csv
 Reading /persistent/RAN_fault_v1/ITSM_Closed_Ticket_data/2026-07-08/itsm_closed_data/2026-07-08/itsm_closed_data.csv

Total Rows : 9,669

Total Columns : 31

	INCIDENT_NUMBER	VENDOR_TICKET_NUMBER	ASSIGNED_GROUP	IMPACT
0	INC000102012688	NaN	TRUE-TH-MS-BO-TRP_IIG	400
1	INC000102016351	SCCD_TT202607125934_A23472935	TRUE-TH-BBT-NOE2-ANC-NOP	400
2	INC000101994345	SCCD_TT202607116536_A23463496	TRUE-TH-WW-CEN-ATG-NOP	300
3	INC000101916687	SCCD_TT202606321312_A23431166	TRUE-TH-WW-CEN-ATG-NOP	300
4	INC000102018339	SCCD_TT202607126793_A23473776	TRUE-TH-WW-CEN-AYT-NOP	300

5 rows × 31 columns

```
In [6]: tickets_closed_df.columns
```

```
Out[6]: Index(['INCIDENT_NUMBER', 'VENDOR_TICKET_NUMBER', 'ASSIGNED_GROUP', 'IMPACT',  
             'PRIORITY', 'URGENCY', 'COMPANY', 'DESCRIPTION', 'DETAILED_DESCRIPTION',  
             'HPD_CI', 'NODETYPE', 'OP_TIME', 'OP_TYPE', 'SUBMIT_DATE',  
             'submit_date.UTC', 'submit_date.BKK', 'Categorization_Tier_1',  
             'Categorization_Tier_2', 'Categorization_Tier_3', 'RESOLUTION_CATEGORY_TIER_1',  
             'RESOLUTION_CATEGORY_TIER_2', 'RESOLUTION_CATEGORY_TIER_3',  
             'INC_STATUS', 'STATUS', 'LAST_RESOLVED_DATE.UTC',  
             'LAST_RESOLVED_DATE.BKK', 'CLOSED_DATE.UTC', 'CLOSED_DATE.BKK',  
             'REPORTED_SOURCE', 'rn', 'SEVERITY'],  
             dtype='object')
```

```
In [7]: conf = SparkConf().setAll([  
        ('spark.executor.memory', '22g'),  
        ('spark.executor.cores', '3'),  
        ('spark.driver.memory', '8G'),  
        ('spark.cores.max', '6'),  
        ('spark.app.name', 'Data Quality'),  
        ('spark.sql.files.ignoreCorruptFiles', 'true'),  
        ('spark.master', 'spark://spark-master-svc:7077')  
    ])  
  
    spark = SparkSession.builder.config(conf=conf).getOrCreate()
```

Setting default log level to "WARN".

To adjust logging level use `sc.setLogLevel(newLevel)`. For SparkR, use `setLogLevel(newLevel)`.

26/07/16 05:10:14 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

```
In [8]: spark
```

```
Out[8]: SparkSession - in-memory
```

SparkContext

Spark UI

Version v3.3.1

Master spark://spark-master-svc:7077

AppName Data Quality

```

In [9]: # Base directory path
base_path = "/persistent/RAN_fault_v1/Alarm_Data"

# Date range
start_date = datetime(2026, 7, 1)
end_date = datetime(2026, 7, 7)

# Generate dates
date_list = []
current_date = start_date

while current_date <= end_date:
    date_list.append(current_date.strftime("%Y-%m-%d"))
    current_date += timedelta(days=1)

print(f"Dates to process: {date_list}")

alarm_df = None
total_files = 0

for date in date_list:

    print("="*60)
    print(f"Processing Date : {date}")
    print("="*60)

    for hour in range(24):

        hour_str = f"{hour:02d}"

        directory_path = os.path.join(
            base_path,
            date,
            "persistent/ingestion_files/onefm",
            date,
            hour_str
        )

        if not os.path.exists(directory_path):
            print(f"Directory not found : {hour_str}")
            continue

        try:

            temp_df = (
                spark.read
                .option("header", "true")
                .option("delimiter", "~")
                .option("mode", "PERMISSIVE")
                .csv(directory_path)
                .withColumn("DATE_ID", lit(date))
                .withColumn("HOUR_ID", lit(hour_str))
            )

```

```

        rows = temp_df.count()

        if rows > 0:

            if alarm_df is None:
                alarm_df = temp_df
            else:
                alarm_df = alarm_df.unionByName(temp_df, allowMissingColumns=True)

            total_files += 1
            print(f"Loaded Hour {hour_str} : {rows} rows")

        except Exception as e:
            print(f"Error reading {directory_path}")
            print(e)

print("="*60)
print("FINAL SUMMARY")
print("="*60)

if alarm_df is not None:

    print(f"Total directories loaded : {total_files}")
    print(f"Total rows : {alarm_df.count()}")
    print(f"Total columns : {len(alarm_df.columns)}")

    alarm_df.printSchema()

    alarm_df.show(5, truncate=False)

else:
    print("No data found.")

```

Dates to process: ['2026-07-01', '2026-07-02', '2026-07-03', '2026-07-04', '2026-07-05', '2026-07-06', '2026-07-07']

=====
Processing Date : 2026-07-01
=====

Loaded Hour 00 : 344172 rows

Loaded Hour 01 : 349448 rows

Loaded Hour 02 : 402857 rows

Loaded Hour 03 : 387328 rows

Loaded Hour 04 : 397161 rows

Loaded Hour 05 : 412717 rows

Loaded Hour 06 : 429124 rows

Loaded Hour 07 : 409669 rows

Loaded Hour 08 : 434493 rows

Loaded Hour 09 : 437216 rows

Loaded Hour 10 : 391172 rows

Loaded Hour 11 : 376040 rows

Loaded Hour 12 : 364355 rows

Loaded Hour 13 : 332934 rows

Loaded Hour 14 : 332261 rows

Loaded Hour 15 : 309287 rows

Loaded Hour 16 : 297671 rows

Loaded Hour 17 : 399452 rows

Loaded Hour 18 : 400092 rows

Loaded Hour 19 : 403792 rows

Loaded Hour 20 : 337356 rows

Loaded Hour 21 : 262510 rows

[Stage 89:=====>
7]

(2 + 5) /

Loaded Hour 22 : 264121 rows

[Stage 93:=====>
6]

(2 + 4) /

Loaded Hour 23 : 322580 rows

=====
Processing Date : 2026-07-02
=====

Loaded Hour 00 : 384713 rows

Loaded Hour 01 : 326613 rows

Loaded Hour 02 : 331586 rows

Loaded Hour 03 : 400228 rows

Loaded Hour 04 : 390624 rows

Loaded Hour 05 : 419948 rows

Loaded Hour 06 : 383979 rows

Loaded Hour 07 : 381930 rows

Loaded Hour 08 : 393356 rows

Loaded Hour 09 : 695368 rows

Loaded Hour 10 : 388732 rows

Loaded Hour 11 : 399612 rows

Loaded Hour 12 : 378795 rows

Loaded Hour 13 : 357397 rows

Loaded Hour 14 : 353074 rows

Loaded Hour 15 : 307395 rows

Loaded Hour 16 : 305785 rows

Loaded Hour 17 : 328581 rows

Loaded Hour 18 : 293542 rows

Loaded Hour 19 : 276100 rows

[Stage 177:=====> (6 + 1) /
7]

Loaded Hour 20 : 238956 rows

Loaded Hour 21 : 229336 rows

Loaded Hour 22 : 250672 rows

Loaded Hour 23 : 323501 rows

Processing Date : 2026-07-03

Loaded Hour 00 : 303620 rows

Loaded Hour 01 : 285382 rows

Loaded Hour 02 : 331100 rows

Loaded Hour 03 : 395422 rows

Loaded Hour 04 : 387138 rows

Loaded Hour 05 : 389081 rows

Loaded Hour 06 : 436652 rows

Loaded Hour 07 : 442981 rows

Loaded Hour 08 : 428569 rows

Loaded Hour 09 : 361903 rows

Loaded Hour 10 : 366144 rows

Loaded Hour 11 : 352088 rows

Loaded Hour 12 : 346477 rows

[Stage 245:=====>
7]

(4 + 3) /

Loaded Hour 13 : 316643 rows

[Stage 249:=====>
7]

(3 + 4) /

Loaded Hour 14 : 301687 rows

[Stage 253:=====>
7]

(3 + 4) /

Loaded Hour 15 : 299809 rows

[Stage 257:=====>
6]

(2 + 4) /

Loaded Hour 16 : 239039 rows
Loaded Hour 17 : 239865 rows
Loaded Hour 18 : 241693 rows
Loaded Hour 19 : 232457 rows
Loaded Hour 20 : 224582 rows
Loaded Hour 21 : 236614 rows
Loaded Hour 22 : 234083 rows

[Stage 285:=====>
6]

(4 + 2) /

Loaded Hour 23 : 293935 rows

=====

Processing Date : 2026-07-04

=====

Loaded Hour 00 : 316356 rows

Loaded Hour 01 : 336822 rows

Loaded Hour 02 : 333323 rows

Loaded Hour 03 : 356804 rows

Loaded Hour 04 : 406572 rows

Loaded Hour 05 : 383348 rows

Loaded Hour 06 : 400828 rows

Loaded Hour 07 : 381026 rows

Loaded Hour 08 : 390345 rows

Loaded Hour 09 : 350893 rows

Loaded Hour 10 : 320462 rows

Loaded Hour 11 : 314452 rows

Loaded Hour 12 : 358134 rows

[Stage 341:=====>
6]

(4 + 2) /

Loaded Hour 13 : 279253 rows

Loaded Hour 14 : 272973 rows

Loaded Hour 15 : 257734 rows

[Stage 353:=====> (4 + 3) /
7]

Loaded Hour 16 : 243087 rows
Loaded Hour 17 : 243936 rows
Loaded Hour 18 : 224499 rows
Loaded Hour 19 : 220629 rows
Loaded Hour 20 : 222074 rows
Loaded Hour 21 : 229413 rows

[Stage 377:=====> (6 + 1) /
7]

Loaded Hour 22 : 234432 rows

[Stage 381:=====> (3 + 4) /
7]

Loaded Hour 23 : 306631 rows

=====
Processing Date : 2026-07-05
=====

Loaded Hour 00 : 324358 rows

Loaded Hour 01 : 299688 rows

Loaded Hour 02 : 299959 rows

Loaded Hour 03 : 337791 rows

Loaded Hour 04 : 360341 rows

Loaded Hour 05 : 365186 rows

Loaded Hour 06 : 355544 rows

Loaded Hour 07 : 352684 rows

Loaded Hour 08 : 346023 rows

Loaded Hour 09 : 336849 rows

[Stage 425:=====> (2 + 5) /
7]

Loaded Hour 10 : 314056 rows

Loaded Hour 11 : 315161 rows

Loaded Hour 12 : 314714 rows

[Stage 437:=====> (2 + 5) /
7]

Loaded Hour 13 : 300618 rows

[Stage 441:=====> (3 + 3) /
6]

Loaded Hour 14 : 278117 rows

[Stage 445:=====> (2 + 4) /
6]

Loaded Hour 15 : 253539 rows

Loaded Hour 16 : 250252 rows

Loaded Hour 17 : 246800 rows

Loaded Hour 18 : 230144 rows

Loaded Hour 19 : 228297 rows

[Stage 465:=====> (5 + 2) /
7]

Loaded Hour 20 : 232766 rows

[Stage 469:=====> (5 + 2) /
7]

Loaded Hour 21 : 220785 rows

Loaded Hour 22 : 238199 rows

[Stage 477:=====> (2 + 5) /
7]

Loaded Hour 23 : 277007 rows

=====
Processing Date : 2026-07-06
=====

Loaded Hour 00 : 316134 rows

Loaded Hour 01 : 273950 rows

[Stage 489:=====> (1 + 6) /
7]

Loaded Hour 02 : 304094 rows

Loaded Hour 03 : 366205 rows

Loaded Hour 04 : 378980 rows

Loaded Hour 05 : 360934 rows

Loaded Hour 06 : 385519 rows

Loaded Hour 07 : 396456 rows

Loaded Hour 08 : 425871 rows

Loaded Hour 09 : 390784 rows

Loaded Hour 10 : 397896 rows

Loaded Hour 11 : 360218 rows

Loaded Hour 12 : 384256 rows

Loaded Hour 13 : 349068 rows

Loaded Hour 14 : 328201 rows

Loaded Hour 15 : 318427 rows

[Stage 545:=====> (5 + 2) /
7]

Loaded Hour 16 : 277554 rows

Loaded Hour 17 : 328461 rows

[Stage 553:=====> (4 + 2) /
6]

Loaded Hour 18 : 273011 rows

Loaded Hour 19 : 276355 rows

[Stage 561:=====> (4 + 2) /
6]

Loaded Hour 20 : 235356 rows

Loaded Hour 21 : 233424 rows

Loaded Hour 22 : 249102 rows

Loaded Hour 23 : 279993 rows

Processing Date : 2026-07-07

Loaded Hour 00 : 306162 rows

Loaded Hour 01 : 286540 rows

Loaded Hour 02 : 331144 rows

Loaded Hour 03 : 421311 rows

Loaded Hour 04 : 418538 rows

Loaded Hour 05 : 427037 rows

Loaded Hour 06 : 408131 rows

Loaded Hour 07 : 396764 rows

Loaded Hour 08 : 418221 rows

Loaded Hour 09 : 418047 rows

Loaded Hour 10 : 406041 rows

Loaded Hour 11 : 403943 rows

Loaded Hour 12 : 355027 rows

Loaded Hour 13 : 338106 rows

Loaded Hour 14 : 310372 rows

[Stage 637:=====> (6 + 1) / 7]

Loaded Hour 15 : 277005 rows

Loaded Hour 16 : 297928 rows

Loaded Hour 17 : 302380 rows

[Stage 649:=====> (6 + 1) /
7]

Loaded Hour 18 : 288857 rows
Loaded Hour 19 : 254783 rows
Loaded Hour 20 : 247783 rows

[Stage 661:=====> (3 + 3) /
6]

Loaded Hour 21 : 243607 rows
Loaded Hour 22 : 267676 rows

[Stage 669:=====> (5 + 1) /
6]

Loaded Hour 23 : 287278 rows

=====

FINAL SUMMARY

=====

Total directories loaded : 168

Total rows : 55490429

Total columns : 46

root

```
|-- RECTIMESTAMP: string (nullable = true)
|-- FIRSTOCCURRENCE: string (nullable = true)
|-- IDENTIFIER: string (nullable = true)
|-- LASTOCCURRENCE: string (nullable = true)
|-- SERVICE: string (nullable = true)
|-- EMS_NAME: string (nullable = true)
|-- NODESTATUS: string (nullable = true)
|-- SERVICEAFFECTING: string (nullable = true)
|-- CLEARTIME: string (nullable = true)
|-- OBJECT: string (nullable = true)
|-- NODETYPE: string (nullable = true)
|-- SEVERITY: string (nullable = true)
|-- EXTENDEDATTR: string (nullable = true)
|-- ELEMENTMANAGERIP: string (nullable = true)
|-- EVENTID: string (nullable = true)
|-- ADDITIONALSTRING: string (nullable = true)
|-- ALERTKEY: string (nullable = true)
|-- SITENAME: string (nullable = true)
|-- ORIGINALSEVERITY: string (nullable = true)
|-- SITESTATUS: string (nullable = true)
|-- SITEID: string (nullable = true)
|-- CLASS: string (nullable = true)
|-- OBJECTSTATUS: string (nullable = true)
|-- CLEAREDDBY: string (nullable = true)
|-- X733SPECIFICPROB: string (nullable = true)
|-- MANAGER: string (nullable = true)
|-- NODEALIAS: string (nullable = true)
|-- SUMMARY: string (nullable = true)
|-- NODE: string (nullable = true)
|-- TT_FLAG: string (nullable = true)
|-- SERVERSERIAL: string (nullable = true)
|-- TMTRANSSTATE: string (nullable = true)
|-- TTCREATIONTIME: string (nullable = true)
|-- TMTRANSBEGINTIME: string (nullable = true)
|-- TT_ID: string (nullable = true)
|-- RECEIVEDATPROBE: string (nullable = true)
|-- physicalCard: string (nullable = true)
|-- OP_TYPE: string (nullable = true)
|-- OP_TIME_DWH: string (nullable = true)
|-- CIRCLE: string (nullable = true)
|-- REGION: string (nullable = true)
|-- LOCATION: string (nullable = true)
|-- BACKHAULTYPE: string (nullable = true)
|-- CONTROLNE: string (nullable = true)
|-- DATE_ID: string (nullable = false)
|-- HOUR_ID: string (nullable = false)
```



```
26/07/16 05:14:03 WARN package: Truncated the string representation of a plan s
ince it was too large. This behavior can be adjusted by setting 'spark.sql.debu
g.maxToStringFields'.
26/07/16 05:14:07 WARN DAGScheduler: Broadcasting large task binary with size
3.6 MiB
```

[illegible]

```

+|vertiv-siteweb-socket@150.236.236.43:6791|07/01/2026 00:07:56|0|SCADA_S
iteweb_New|ACTIVE|0|2026-07-01 00:31:53 UTC|null|PAC (Wind
Cooling)|0|null
|10.125.4.11|0074465275|null|null|RANGSIT|3
|Live|RST|1010802|null|Deduplication Automation|PAC (Wind Cool
ing)|vertiv-siteweb-socket@150.236.236.43:6791|RST-PAC_301_Liebert_PEX2060|Retu
rn Humidity Low Hum 44|RST-PAC_301_Liebert_PEX2060|0
|500821599|0|1970-01-01 00:00:00 UTC|1970-01-01 00:00:00 UTC|null
|2026-07-01 00:07:57 UTC|null|U|2026-07-01 00:33:38.321005 UTC|BA
NGKOK WEST1|BANGKOK|THANYABURI|SCADA|RST-PAC_301_Liebert_PEX2060|2026-0
7-01|00|
|07/01/2026 00:40:03|07/01/2026 00:07:58|0074465278|RST-PAC_308_Liebert_PEX206
0|vertiv-siteweb-socket@150.236.236.43:6791|07/01/2026 00:07:58|0|SCADA_S
iteweb_New|ACTIVE|0|2026-07-01 00:28:39 UTC|null|PAC (Wind
Cooling)|0|null
|10.125.4.11|0074465278|null|null|RANGSIT|3
|Live|RST|1010802|null|Deduplication Automation|PAC (Wind Cool
ing)|vertiv-siteweb-socket@150.236.236.43:6791|RST-PAC_308_Liebert_PEX2060|Retu
rn Humidity Low Hum 44|RST-PAC_308_Liebert_PEX2060|0
|500821604|0|1970-01-01 00:00:00 UTC|1970-01-01 00:00:00 UTC|null
|2026-07-01 00:08:01 UTC|null|U|2026-07-01 00:30:39.604007 UTC|BA
NGKOK WEST1|BANGKOK|THANYABURI|SCADA|RST-PAC_308_Liebert_PEX2060|2026-0
7-01|00|
|07/01/2026 00:40:03|07/01/2026 00:11:13|0074465337|RST-PAC_319_Liebert_PEX206
0|vertiv-siteweb-socket@150.236.236.43:6791|07/01/2026 00:11:13|0|SCADA_S
iteweb_New|ACTIVE|0|2026-07-01 00:28:39 UTC|null|PAC (Wind
Cooling)|0|null
|10.125.4.11|0074465337|null|null|RANGSIT|3
|Live|RST|1010802|null|Deduplication Automation|PAC (Wind Cool
ing)|vertiv-siteweb-socket@150.236.236.43:6791|RST-PAC_319_Liebert_PEX2060|Retu
rn Temperature High Temp 22.100000381469727|RST-PAC_319_Liebert_PEX2060|0
|500822009|0|1970-01-01 00:00:00 UTC|1970-01-01 00:00:00 UTC|null
|2026-07-01 00:11:11 UTC|null|U|2026-07-01 00:30:39.604008 UTC|BA
NGKOK WEST1|BANGKOK|THANYABURI|SCADA|RST-PAC_319_Liebert_PEX2060|2026-0
7-01|00|
|07/01/2026 00:40:03|07/01/2026 00:17:35|0074465389|RST-RAC_404_Liebert_CRV4|ve
rtiv-siteweb-socket@150.236.236.43:6791|07/01/2026 00:17:35|0|SCADA_Si
teweb_New|ACTIVE|0|2026-07-01 00:31:33 UTC|null|RAC (Wind
Cooling)|0|"MATE info=""Updated by rule: main_Rule, module: SiteWeb_RanI
nfra_Correlation_Rule, instance id: 729, instance name: DTAC-OF-TH_ALL_ALL_K2
8.""|10.125.4.11|0074465389|null|null|RANGSIT|4
|Live|RST|1010802|null|Deduplication Automation|RAC (Wind Cool
ing)|vertiv-siteweb-socket@150.236.236.43:6791|RST-RAC_404_Liebert_CRV4|Supp
ly Temperature Low Temp 16.799999237060547|RST-RAC_404_Liebert_CRV4|2
|500822680|125|2026-07-01 00:25:40 UTC|1970-01-01 00:00:00 UTC|INC0
00101960432|2026-07-01 00:17:34 UTC|null|U|2026-07-01 00:31:38.69
4008 UTC|BANGKOK WEST1|BANGKOK|THANYABURI|SCADA|RST-RAC_404_Liebert_CRV4
|2026-07-01|00|
+-----+
+-----+
+-----+-----+-----+
+-----+-----+-----+
+-----+-----+-----+
+-----+-----+-----+
+-----+

```

```

+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
only showing top 5 rows

```

```

In [10]: print("Total Rows      :", len(ticket_df))
print("Unique Incident IDs   :", ticket_df["INCIDENT_NUMBER"].nunique())
print("Open Tickets :", ticket_df.shape[0])
print("Closed Tickets :", tickets_closed_df.shape[0])

common = ticket_df["INCIDENT_NUMBER"].isin(
    tickets_closed_df["INCIDENT_NUMBER"]
).sum()

print("Common Incident IDs :", common)

```

```

Total Rows      : 2835299
Unique Incident IDs   : 56148
Open Tickets : 2835299
Closed Tickets : 9669
Common Incident IDs : 61740

```

```

In [11]: # Keep latest record for each incident in ticket_df
ticket_df = (
    ticket_df
    .sort_values("submit_date_utc")
    .drop_duplicates(subset="INCIDENT_NUMBER", keep="last")
)

# Keep latest record for each incident in closed tickets
tickets_closed_df = (
    tickets_closed_df
    .sort_values("LAST_RESOLVED_DATE_utc")
    .drop_duplicates(subset="INCIDENT_NUMBER", keep="last")
)

```

```

In [12]: closed_cols = [
    "INCIDENT_NUMBER",
    "LAST_RESOLVED_DATE_utc",
    "LAST_RESOLVED_DATE_BKK",
    "CLOSED_DATE_utc",
    "CLOSED_DATE_BKK"
]

merged_ticket_df = ticket_df.merge(
    tickets_closed_df[closed_cols],
    on="INCIDENT_NUMBER",

```

```
    how="left"  
)
```

In [13]: merged_ticket_df.columns

```
Out[13]: Index(['INCIDENT_NUMBER', 'VENDOR_TICKET_NUMBER', 'ASSIGNED_GROUP', 'IMPACT',  
              'PRIORITY', 'URGENCY', 'COMPANY', 'DESCRIPTION', 'DETAILED_DESCRIPTION',  
              'HPD_CI', 'NODETYPE', 'OP_TIME', 'OP_TYPE', 'SUBMIT_DATE',  
              'submit_date.UTC', 'submit_date.BKK', 'Categorization_Tier_1',  
              'Categorization_Tier_2', 'Categorization_Tier_3', 'RESOLUTION_CATEGORY',  
              'RESOLUTION_CATEGORY_TIER_2', 'RESOLUTION_CATEGORY_TIER_3',  
              'INC_STATUS', 'STATUS', 'REPORTED_SOURCE', 'EVENT_END_TIME',  
              'EVENT_START_TIME', 'PENDING_TIME', 'PENDING_DUE_TIME', 'rn',  
              'SEVERITY', 'LAST_RESOLVED_DATE.UTC', 'LAST_RESOLVED_DATE.BKK',  
              'CLOSED_DATE.UTC', 'CLOSED_DATE.BKK'],  
              dtype='object')
```

```
In [14]: print("Open Tickets :", ticket_df.shape)  
         print("Closed Tickets :", tickets_closed_df.shape)  
         print("Merged :", merged_ticket_df.shape)  
  
         print(  
             merged_ticket_df[  
                 "LAST_RESOLVED_DATE.UTC"  
             ].notna().sum(),  
             "tickets have resolution information."  
         )
```

```
Open Tickets : (56148, 31)  
Closed Tickets : (9669, 31)  
Merged : (56148, 35)  
5689 tickets have resolution information.
```

```
In [15]: degraded_cells = (  
    combined_df  
    .groupby("CELLID")["NR_Cell_Availability"]  
    .min()  
    .reset_index(name="min_availability")  
)  
  
degraded_cells = degraded_cells[  
    degraded_cells["min_availability"] < 100  
][["CELLID"]]  
  
print(degraded_cells.head())  
print("Number of degraded cells:", len(degraded_cells))  
  
filtered_df = combined_df.merge(  
    degraded_cells,  
    on="CELLID",  
    how="inner"  
)
```

```

CELLID
5    ATG0003H_7NB01_S02
14   ATG0011H_7NB04_S01
15   ATG0011H_7NB04_S02
16   ATG0011H_7NB04_S03
17   ATG0011L_6MM02_S11
Number of degraded cells: 10490

```

```
In [16]: print("Number of unique sites:", filtered_df["SITE"].nunique())
```

Number of unique sites: 3689

```
In [17]: cells_per_site = (
    filtered_df
    .groupby("SITE")["CELLID"]
    .nunique()
    .sort_values(ascending=False)
)

print(cells_per_site)
```

```

SITE
BKFTS02L_6NB04      9
CRI6315L_6NB02      6
CMI6931X_2NB10      6
SPB0019L_6NB02      6
SPB6024L_6NB02      6
..
RCB9001L_6NB02      1
CRI0204P_9NB02      1
CRI0199P_9NB01      1
CRI0163P_9NB01      1
ATG0003T_2NB01      1
Name: CELLID, Length: 3689, dtype: int64

```

```
In [18]: filtered_df["NR_Cell_Availability"].describe()
```

```

Out[18]: count      6.298297e+06
         mean       9.949459e+01
         std        6.669821e+00
         min        -1.110000e-01
         25%         1.000000e+02
         50%         1.000000e+02
         75%         1.000000e+02
         max         1.000000e+02
         Name: NR_Cell_Availability, dtype: float64

```

```
In [19]: merged_ticket_df["HPD_CI"]
```

```
Out[19]: 0      NPT0332
1      SKA0357
2      RYG0203
3      BKK0400
4      BKK0378
...
56143   TRT7148
56144   TAK6737
56145   PTTA033
56146   SSK0307
56147   CRI6459
Name: HPD_CI, Length: 56148, dtype: object
```

```
In [20]: filtered_df[["SITE", "CELLID"]].drop_duplicates().head(20)
```

```
Out[20]:
```

	SITE	CELLID
0	ATG0011H_7NB04	ATG0011H_7NB04_S02
473	ATG0011H_7NB04	ATG0011H_7NB04_S03
946	ATG0011L_6NB02	ATG0011L_6MM02_S13
1475	ATG0019T_2NB01	ATG0019H_7NB01_S01
2078	ATG0019T_2NB01	ATG0019H_7NB01_S03
2681	ATG0022P_9NB03	ATG0022H_7NB03_S02
3281	ATG0022P_9NB03	ATG0022H_7NB03_S04
3881	ATG0022L_6NB01	ATG0022L_6MM01_S13
4481	ATG0022L_6NB01	ATG0022L_6MM01_S14
5081	ATG0024T_2NB01	ATG0024H_7NB01_S01
5688	ATG0067P_9NB01	ATG0067H_7NB01_S01
6295	ATG0067P_9NB01	ATG0067H_7NB01_S03
6902	ATG0096T_2NB01	ATG0096H_7NB01_S02
7509	ATG0114T_2NB01	ATG0114H_7NB01_S03
8116	ATG1101P_9NB02	ATG1101H_7NB02_S01
8721	ATG1101P_9NB02	ATG1101H_7NB02_S02
9326	ATG1101P_9NB02	ATG1101H_7NB02_S03
9931	ATG1532P_9NB01	ATG1532H_7NB01_S01
10536	ATG6003P_9NB01	ATG6003H_7NB01_S01
11140	ATG6003P_9NB01	ATG6003H_7NB01_S03

```
In [21]: merged_ticket_df[["HPD_CI"]].drop_duplicates().head(20)
```

```
Out[21]: 0      NPT0332
        1      SKA0357
        2      RYG0203
        3      BKK0400
        4      BKK0378
        5      KPT0201
        6      SNI0139
        7      PNA0095
        8      BKK3835
        9      CBI0891
       10      BKK3009
       11      CC00478
       12      CBI1060
       13      NRT0090
       14      CBI0920
       15      BKK4829
       16      CRI0289
       17      BKK4265
       18      BKK0286
       19      NPM0043
      Name: HPD_CI, dtype: object
```

```
In [22]: print(merged_ticket_df["SEVERITY"].value_counts(dropna=False))
```

```
SA4.24      20072
NSA3.24      10823
SA1.4        6082
NSA4.168     5608
SA3.14       4299
NSA2.12      2694
SA3.12       1534
NSA2.8       1389
NSA5.720     1224
SA3.4        747
SA2.4        559
NSA1.4       489
PSA1.12      278
PSA3.168     189
SA2.6        99
PSA2.48      62
      Name: SEVERITY, dtype: int64
```

```
In [23]: # Assigned Group: Filter BBT or WW
        # Severity: Filter 'SA1', SA2', or 'SA3'
        # Categorization_Tier_1: Filter NOC-NW-RAN'
        # Categorization_Tier_2: Filter 'RAN-ERICSSON SITE DOWN', 'RAN-ERICSSON SITE U
        # NODETYPE: Filter for eNode B/LTE Cell (for 4G) and NR GNodeB – Ericsson (for
        # *****filtered based on above conditon for itsm_ticket_data*****

        # Map node type to technology
        nodetype_tech_mapping = {
            'eNode B': '4G',
            'LTE Cell': '4G',
            'NR GNodeB - Ericsson': '5G'
```



```

}

filtered_ticket_df = merged_ticket_df

filtered_ticket_df["Technology"] = (
    filtered_ticket_df["NODETYPE"]
    .map(nodetype_tech_mapping)
)

filtered_ticket_df = filtered_ticket_df[
    # Keep only 4G and 5G Ericsson records
    filtered_ticket_df["Technology"].isin(["4G", "5G"])

    # Assigned Group contains BBT or WW
    & filtered_ticket_df["ASSIGNED_GROUP"].str.contains(
        "BBT|WW",
        case=False,
        na=False
    )

    # Severity contains SA1 or SA2 or SA3
    & filtered_ticket_df["SEVERITY"].str.contains(
        "SA1|SA2|SA3",
        case=False,
        na=False,
        regex=True
    )

    # Categorization Tier 1
    & (filtered_ticket_df["Categorization_Tier_1"] == "NOC-NW-RAN")

    # Categorization Tier 2
    & filtered_ticket_df["Categorization_Tier_2"].isin([
        "RAN-ERICSSON SITE DOWN",
        "RAN-ERICSSON SITE UP/DOWN",
        "RAN-ERICSSON ROUTE SITE DOWN"
    ])
].copy()

print(filtered_ticket_df["Technology"].value_counts())

```

```

4G      344
5G       28
Name: Technology, dtype: int64

```

```

In [24]: duplicate_hpd_ci = (
    filtered_ticket_df["HPD_CI"]
    .value_counts()
    .loc[lambda x: x > 1]
)

print(duplicate_hpd_ci)

```

CMI7658	4
NAN6778	4
UTR6701	3
SKT7233	3
UTT0614	3
NAN6926	3
CMI7628	2
LPG6739	2
NKT8591	2
CMI6858	2
SMS8554	2
PCR0198	2
SMK6704	2
CMI6764	2
SRB0067	2
MHS6508	2
MHS7119	2
SPB1205	2
NAN0138	2
CMIA220	2
RYG0355	2
CMI7624	2
SMK0084	2
TAK6709	2
TAK6704	2
CMI7534	2
RCB1003	2
PSN6716	2
MHS7225	2
PCT7402	2
NKT8609	2
TAK7125	2
LPG0062	2
PCR0158	2
KCN0361	2

Name: HPD_CI, dtype: int64

```
In [25]: site_list = filtered_ticket_df["HPD_CI"].dropna().unique().tolist()
         print(site_list)
```

```
['CMI7658', 'CNT0039', 'CCS0331', 'CMIA002', 'LBR8549', 'PCR0036', 'CMI8935',
'TAK7125', 'TAK0241', 'SPB6004', 'RYG0775', 'PTT8617', 'TAK7207', 'PSN8804', 'B
KC7493', 'LPG0062', 'AYT1307', 'BKK1418', 'NAN6778', 'KCN0361', 'KCN6723', 'NAN
6926', 'PY00027', 'CMI7644', 'SPB0122', 'SRB6752', 'CRI6716', 'PCR0158', 'LPN67
17', 'KPP0224', 'CMI1936', 'SPB0067', 'TAK6708', 'CMI6865', 'PCBC009', 'BKE727
8', 'RCB0114', 'AYT7156', 'RCB0116', 'NAN6931', 'ATG8505', 'NKT0255', 'SPB120
5', 'NTB7147', 'PCT0174', 'SPB1020', 'LPG0115', 'LPN0160', 'PCT7402', 'TAK020
5', 'KCN6748', 'TAK7655', 'TAK8803', 'LPG6739', 'SPB0126', 'NAN6923', 'CRI673
7', 'KCN8516', 'NKT8591', 'NAN6731', 'CMIA279', 'BKF1212', 'PSN6712', 'PCB863
9', 'SMK0039', 'SKT0176', 'PSN0363', 'CMIA069', 'CMI6858', 'PY07244', 'NKT764
1', 'SMP5744', 'SMS8554', 'CCS8609', 'NKW8806', 'CCS0383', 'PCR0198', 'UTT061
4', 'BKD6742', 'SKT0045', 'NAN7138', 'SMK6704', 'LPG6757', 'CMI6487', 'PCB673
0', 'PSN8501', 'MHS0073', 'SRB1004', 'CRI6747', 'LPG0259', 'NAN6854', 'CCS741
2', 'SMP8629', 'KCN1301', 'SRB0067', 'MHS6508', 'NKT7916', 'SBR0060', 'MHS623
6', 'CMI8777', 'PSN8581', 'PTT7167', 'LBR0235', 'MHS0086', 'CMI7419', 'CRI769
3', 'RCB8614', 'LPN7604', 'CRI6317', 'AYT1411', 'CRI6743', 'RYG7159', 'CRI719
7', 'SRB0018', 'CMI7628', 'BKAC086', 'PY00105', 'NKT0112', 'PHE8544', 'NKT140
7', 'CMI8583', 'MHS7119', 'SMK6716', 'CMI7842', 'NAN7145', 'CMI6755', 'SRB856
4', 'SMK7919', 'NAN0138', 'CMIA220', 'CMI7609', 'NKT7309', 'UTR6751', 'SRB726
9', 'PSN7193', 'SRB0093', 'SRB7298', 'SRB7277', 'BKF8673', 'UTT0049', 'SMK018
7', 'CCS0429', 'SPB7655', 'NKT1104', 'TAK6514', 'CMI6268', 'LPG7405', 'CMI762
4', 'CBR1120', 'BKF7512', 'BKC2001', 'SMK0084', 'CCS1633', 'NKT0307', 'NAN004
4', 'SMK0438', 'SMK0465', 'AYT7674', 'PY07246', 'CMI0672', 'BKAC387', 'MHS651
1', 'NAN6935', 'SMK0059', 'CMI8738', 'LPN6525', 'BKE7287', 'AYT6769', 'SKT723
3', 'UTR6701', 'UTT6703', 'MHS0059', 'NAN0003', 'NAN0024', 'SRB7173', 'MHS722
5', 'PHE6801', 'CRI7189', 'NKT8560', 'NKT8609', 'PSN6716', 'TAK6744', 'CMI791
2', 'KCN7906', 'NKW1090', 'AYT8594', 'KCN6035', 'AYT7153', 'NAN6983', 'MHS684
2', 'LPG6759', 'RCB1003', 'CRI6880', 'LPG6754', 'CMIA237', 'CCS7651', 'CMI753
4', 'LPG8533', 'TAK6704', 'TAK6709', 'BKE1130', 'MHS7255', 'MHS0018', 'NAN002
7', 'PTT8701', 'TAK0164', 'CMI0900', 'RCBC019', 'PCT8504', 'PCB0055', 'AYT100
7', 'RYG0355', 'RYG7947', 'MHS7240', 'CMI7605', 'UTT0029', 'RCB8529', 'PCB761
2', 'PCB7679', 'RCB0005', 'CRI0469', 'KCN0011', 'BKA8631', 'CMI1819', 'AYT036
9', 'RCB8585', 'RCB7184', 'NKT0147', 'CMI7836', 'CMI7659', 'KCN7918', 'CMI889
5', 'MHS6257', 'LBR7309', 'LPG7172', 'CBR1125', 'LPN8561', 'SPB7104', 'MHS005
0', 'NTB8615', 'KCN6768', 'RCB8534', 'RCB0102', 'RCBC011', 'CMI7838', 'MHS673
8', 'PSN7400', 'CRI6244', 'CRI6432', 'PCB7676', 'CRI6756', 'PCB6729', 'CMI886
3', 'CRI1644', 'CMI6764', 'MHS0123', 'CMIC081', 'LBR1001', 'RCB0332', 'CMI681
4', 'CMI7128', 'CMI8509', 'NKT0138', 'RCB0341', 'CRI0103', 'AYT6773', 'RCB024
3', 'SPB8540', 'AYT8592', 'NKW6753', 'CRI3013', 'SMK7671', 'SMK7423', 'KCN676
6', 'CMI6779', 'CRI6708', 'SKT1923', 'UTR6738', 'CMI0942', 'CMI0801', 'SMP705
3', 'AYT0550', 'CMI6741', 'CRI7198', 'CRI7196', 'NKW8803', 'AYT0405', 'AYT034
7', 'UTR6744', 'KCN7624', 'SPB0198', 'RYG7246', 'SMK6701', 'SPB0019', 'LPG678
4', 'SMK1007', 'PCB6722', 'LBR0240', 'KPP8538', 'AYT8658', 'BKCZ035', 'TAK671
0', 'BKE8707', 'BKBC232', 'BKF1042', 'SMPA044', 'SMP2652', 'BKE8607', 'SMP618
9', 'BKFC912', 'BKB7535', 'SMP7496', 'SMP2047', 'BKBC323', 'BKB1022', 'BKC763
9', 'CCS8588', 'CMI6883', 'CMI0868', 'LBR0176', 'SMP2507', 'KCN5904', 'LBR740
2', 'CMIC059', 'NTB7211', 'CMI3189', 'LPN1141', 'ATG6722', 'CMI6772']
```

```
In [26]: # Convert dictionary to Spark map expression
mapping_expr = create_map(
    [lit(x) for x in chain(*nodetype_tech_mapping.items())]
)

# Create Technology column
```

```

alarm_df = alarm_df.withColumn(
    "Technology",
    mapping_expr[col("NODETYPE")]
)

# Filter only 4G and 5G
alarm_df = alarm_df.filter(
    col("Technology").isin("4G", "5G")
)

# Count records by Technology
alarm_df.groupBy("Technology").count().show()

```

26/07/16 05:14:40 WARN DAGScheduler: Broadcasting large task binary with size 1503.3 KiB

[Stage 676:=====>(1155 + 6) / 1161]

Technology	count
4G	6665399
5G	611576

In [27]: `alarm_df.show()`

26/07/16 05:15:37 WARN DAGScheduler: Broadcasting large task binary with size 4.1 MiB

```

+-----+-----+-----+-----+
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+-----+-----+-----+-----+
|      RECTIMESTAMP|      FIRSTOCCURRENCE|      IDENTIFIER|      LASTOCCURREN
CE|      SERVICE|      EMS_NAME|NODESTATUS|SERVICEAFFECTING|
CLEAR TIME|      OBJECT|      NODETYPE|SEVERITY|EXTENDEDATTR|ELEMENT
MANAGERIP|EVENTID|      ADDITIONALSTRING|      ALERTKEY|      SITENAM
E|ORIGINALSEVERITY|SITESTATUS| SITEID|      CLASS|OBJECTSTATUS|      CLEARED B
Y|      X733SPECIFICPROB|      MANAGER|      NODEALIAS|      SUM
MARY|      NODE|TT_FLAG|SERVERSERIAL|TMTRANSSTATE|      TTCREATIONTIME|
TMTRANSBEGIN TIME|TT_ID|      RECEIVEDATPROBE|physicalCard|OP_TYPE|      OP_TI
ME_DWH|      CIRCLE| REGION|      LOCATION|BACKHAULTYPE|      CONTROLNE|
DATE_ID|HOUR_ID|Technology|
+-----+-----+-----+-----+
+-----+-----+-----+-----+
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+-----+-----+-----+-----+
|07/01/2026 00:40:03|06/30/2026 23:36:28|AYT0170E_1NB04_S0...|06/30/2026 23:5
1:20|G09,L07,L09,L18,L...|True-PMS-QoS-Huaw...|      ACTIVE|      0|202
6-07-01 00:21:...|AYT0170E_1NB04_S01|      LTE Cell|      0|      nul
l|      10.50.238.226|      null|PSA3 - 20260701T0...|AYT0170E_1NB04:Q0...|
TBD|      3|      Live|AYT0170|1010895|      null|Deduplication Au
t...|QOS_RRCSuccNoLTET...|pms-monitoring-so...|AYT0170E_1NB04_S01|4G-ERI-QOS_RR
CSuc...|AYT0170E_1NB04|      0|      213202008|      0|1970-01-01 00:00:...|1
970-01-01 00:00:...|      null|2026-06-30 23:47:...|      null|      U|2026-07-01
00:33:...|PHRA NAKHON SI AY...|CENTRAL|      UTHAI|      |True-L18|AYT0170X_2NB0
4|2026-07-01|      00|      4G|
|07/01/2026 00:40:03|02/26/2026 10:55:30|BKA9116-L23 | L23...|02/26/2026 10:5
5:30|HAM04_0529 | AN-H...|      PMT00L_KPI_Data|      ACTIVE|      0|202
6-07-01 00:31:...|      L23-BKA9116-1D|      eNode B|      0|      nul
l|      null|      null|Value=Temp locked...|      BKA9116-L23|      Big
C Huamark|      3|      Live|BKA9116|      7955|      null|      Expire Au
tomation|      Locked Cell|pmtool-system-glf...|      BKA9116-L23|
Locked Cell|      BKA9116-L23|      0|      213213764|      0|1970-01-01 00:0
0:...|1970-01-01 00:00:...|      null|2026-07-01 00:31:...|      null|      U|202
6-07-01 00:33:...|      BANGKOK EAST1|BANGKOK|      BANG KAPI|      |RAN-4G||
4G-L23|2026-07-01|      00|      4G|
|07/01/2026 00:40:03|02/26/2026 10:55:30|BKA9116-L23 | L23...|02/26/2026 10:5
5:30|HAM04_0529 | AN-H...|      PMT00L_KPI_Data|      ACTIVE|      0|202

```

```

6-07-01 00:31:...| L23-BKA9116-1D| eNode B| 0| nul
l| null| null|Value=Temp locked...| BKA9116-L23| Big
C Huamark| 3| Live|BKA9116| 7955| null| Expire Au
tomation| Locked Cell|pmttool-system-glf...| BKA9116-L23|
Locked Cell| BKA9116-L23| 0| 213213764| 0|1970-01-01 00:0
0:...|1970-01-01 00:00:...| null|2026-07-01 00:31:...| null| I|202
6-07-01 00:31:...| BANGKOK EAST1|BANGKOK| BANG KAPI| |RAN-4G||
4G-L23|2026-07-01| 00| 4G|
|07/01/2026 00:40:03|05/12/2026 10:00:00|BKB7641-L23 | L23...|05/12/2026 10:0
0:00|TMM05_0215 | AN-T...| PMT00L_KPI_Data| ACTIVE| 0|202
6-07-01 00:31:...| L23-BKB7641-4B| eNode B| 0| nul
l| null| null| Value=Unknown| BKB7641-L23|HRB / Vara
korntra...| 3| Live|BKB7641| 7955| null| Expire A
utomation| Locked Cell|pmttool-system-glf...| BKB7641-L23|
Locked Cell| BKB7641-L23| 0| 213214093| 0|1970-01-01 00:0
0:...|1970-01-01 00:00:...| null|2026-07-01 00:31:...| null| I|202
6-07-01 00:31:...| BANGKOK EAST4|BANGKOK|KHLONG TOEI| RAN-BSC| BKB7
641-L23|2026-07-01| 00| 4G|
|07/01/2026 00:40:03|05/12/2026 10:00:00|BKB7641-L23 | L23...|05/12/2026 10:0
0:00|TMM05_0215 | AN-T...| PMT00L_KPI_Data| ACTIVE| 0|202
6-07-01 00:31:...| L23-BKB7641-4B| eNode B| 0| nul
l| null| null| Value=Unknown| BKB7641-L23|HRB / Vara
korntra...| 3| Live|BKB7641| 7955| null| Expire A
utomation| Locked Cell|pmttool-system-glf...| BKB7641-L23|
Locked Cell| BKB7641-L23| 0| 213214093| 0|1970-01-01 00:0
0:...|1970-01-01 00:00:...| null|2026-07-01 00:31:...| null| U|202
6-07-01 00:33:...| BANGKOK EAST4|BANGKOK|KHLONG TOEI| RAN-BSC| BKB7
641-L23|2026-07-01| 00| 4G|
|07/01/2026 00:40:03|02/12/2026 07:45:00|BKBC074-L23 | L23...|02/12/2026 07:4
5:00|L07,L21,L23,L26,N...| PMT00L_KPI_Data| ACTIVE| 0|202
6-07-01 00:31:...| L23-BKBC074-4A| eNode B| 0| nul
l| null| null| Value=Unknown| BKBC074-L23| PRAKHA
NONGMARKET| 3| Live|BKBC074| 7955| null| Expire A
utomation| Locked Cell|pmttool-system-glf...| BKBC074-L23|
Locked Cell| BKBC074-L23| 0| 213214065| 0|1970-01-01 00:0
0:...|1970-01-01 00:00:...| null|2026-07-01 00:31:...| null| I|202
6-07-01 00:31:...| BANGKOK EAST4|BANGKOK| VADHANA| |RAN-4G||
4G-L23|2026-07-01| 00| 4G|
|07/01/2026 00:40:03|02/12/2026 07:45:00|BKBC074-L23 | L23...|02/12/2026 07:4
5:00|L07,L21,L23,L26,N...| PMT00L_KPI_Data| ACTIVE| 0|202
6-07-01 00:31:...| L23-BKBC074-4A| eNode B| 0| nul
l| null| null| Value=Unknown| BKBC074-L23| PRAKHA
NONGMARKET| 3| Live|BKBC074| 7955| null| Expire A
utomation| Locked Cell|pmttool-system-glf...| BKBC074-L23|
Locked Cell| BKBC074-L23| 0| 213214065| 0|1970-01-01 00:0
0:...|1970-01-01 00:00:...| null|2026-07-01 00:31:...| null| U|202
6-07-01 00:33:...| BANGKOK EAST4|BANGKOK| VADHANA| |RAN-4G||
4G-L23|2026-07-01| 00| 4G|
|07/01/2026 00:40:03|10/09/2025 07:45:00|BKFA198-L23 | L23...|10/09/2025 07:4
5:00|DNM32_0447 | AN-D...| PMT00L_KPI_Data| ACTIVE| 0|202
6-07-01 00:31:...| L23-BKFA198-4B| eNode B| 0| nul
l| null| null| Value=Unknown| BKFA198-L23|GRF/Soi Fa
ctory, ...| 3| Live|BKFA198| 7955| null| Expire A
utomation| Locked Cell|pmttool-system-glf...| BKFA198-L23|

```

Locked Cell| BKFA198-L23| 0| 213214064| 0|1970-01-01 00:0
0:...|1970-01-01 00:00:...| null|2026-07-01 00:31:...| null| U|202
6-07-01 00:33:...| Nonthaburi|Bangkok| LAM LUK KA| |RAN-4G|| BKFA
198-L23|2026-07-01| 00| 4G|
|07/01/2026 00:40:03|10/09/2025 07:45:00|BKFA198-L23 | L23...|10/09/2025 07:4
5:00|DNM32_0447 | AN-D...| PMT00L_KPI_Data| ACTIVE| 0|202
6-07-01 00:31:...| L23-BKFA198-4B| eNode B| 0| nul
l| null| null| Value=Unknown| BKFA198-L23|GRF/Soi Fa
ctory, ...| 3| Live|BKFA198| 7955| null| Expire A
utomation| Locked Cell|pmt00l-system-glf...| BKFA198-L23|
Locked Cell| BKFA198-L23| 0| 213214064| 0|1970-01-01 00:0
0:...|1970-01-01 00:00:...| null|2026-07-01 00:31:...| null| I|202
6-07-01 00:31:...| Nonthaburi|Bangkok| LAM LUK KA| |RAN-4G|| BKFA
198-L23|2026-07-01| 00| 4G|
|07/01/2026 00:40:03|10/09/2025 07:45:00|BKFA198-L23 | L23...|10/09/2025 07:4
5:00|DNM32_0447 | AN-D...| PMT00L_KPI_Data| ACTIVE| 0|202
6-07-01 00:31:...| L23-BKFA198-4D| eNode B| 0| nul
l| null| null| Value=Unknown| BKFA198-L23|GRF/Soi Fa
ctory, ...| 3| Live|BKFA198| 7955| null| Expire A
utomation| Locked Cell|pmt00l-system-glf...| BKFA198-L23|
Locked Cell| BKFA198-L23| 0| 213214061| 0|1970-01-01 00:0
0:...|1970-01-01 00:00:...| null|2026-07-01 00:31:...| null| U|202
6-07-01 00:33:...| Nonthaburi|Bangkok| LAM LUK KA| |RAN-4G|| BKFA
198-L23|2026-07-01| 00| 4G|
|07/01/2026 00:40:03|10/09/2025 07:45:00|BKFA198-L23 | L23...|10/09/2025 07:4
5:00|DNM32_0447 | AN-D...| PMT00L_KPI_Data| ACTIVE| 0|202
6-07-01 00:31:...| L23-BKFA198-4D| eNode B| 0| nul
l| null| null| Value=Unknown| BKFA198-L23|GRF/Soi Fa
ctory, ...| 3| Live|BKFA198| 7955| null| Expire A
utomation| Locked Cell|pmt00l-system-glf...| BKFA198-L23|
Locked Cell| BKFA198-L23| 0| 213214061| 0|1970-01-01 00:0
0:...|1970-01-01 00:00:...| null|2026-07-01 00:31:...| null| I|202
6-07-01 00:31:...| Nonthaburi|Bangkok| LAM LUK KA| |RAN-4G|| BKFA
198-L23|2026-07-01| 00| 4G|
|07/01/2026 00:40:03|10/09/2025 07:45:00|BKFA198-L23 | L23...|10/09/2025 07:4
5:00|DNM32_0447 | AN-D...| PMT00L_KPI_Data| ACTIVE| 0|202
6-07-01 00:31:...| L23-BKFA198-4E| eNode B| 0| nul
l| null| null| Value=Unknown| BKFA198-L23|GRF/Soi Fa
ctory, ...| 3| Live|BKFA198| 7955| null| Expire A
utomation| Locked Cell|pmt00l-system-glf...| BKFA198-L23|
Locked Cell| BKFA198-L23| 0| 213214062| 0|1970-01-01 00:0
0:...|1970-01-01 00:00:...| null|2026-07-01 00:31:...| null| I|202
6-07-01 00:31:...| Nonthaburi|Bangkok| LAM LUK KA| |RAN-4G|| BKFA
198-L23|2026-07-01| 00| 4G|

|07/01/2026 00:40:03|10/09/2025 07:45:00|BKFA198-L23 | L23...|10/09/2025 07:4
5:00|DNM32_0447 | AN-D...| PMT00L_KPI_Data| ACTIVE| 0|202
6-07-01 00:31:...| L23-BKFA198-4F| eNode B| 0| nul
l| null| null| Value=Unknown| BKFA198-L23|GRF/Soi Fa
ctory, ...| 3| Live|BKFA198| 7955| null| Expire A
utomation| Locked Cell|pmtool-system-glf...| BKFA198-L23|
Locked Cell| BKFA198-L23| 0| 213214063| 0|1970-01-01 00:0
0:...|1970-01-01 00:00:...| null|2026-07-01 00:31:...| null| U|202
6-07-01 00:33:...| Nonthaburi|BANGKOK| LAM LUK KA| |RAN-4G|| BKFA
198-L23|2026-07-01| 00| 4G|

|07/01/2026 00:40:03|10/09/2025 07:45:00|BKFA198-L23 | L23...|10/09/2025 07:4
5:00|DNM32_0447 | AN-D...| PMT00L_KPI_Data| ACTIVE| 0|202
6-07-01 00:31:...| L23-BKFA198-4F| eNode B| 0| nul
l| null| null| Value=Unknown| BKFA198-L23|GRF/Soi Fa
ctory, ...| 3| Live|BKFA198| 7955| null| Expire A
utomation| Locked Cell|pmtool-system-glf...| BKFA198-L23|
Locked Cell| BKFA198-L23| 0| 213214063| 0|1970-01-01 00:0
0:...|1970-01-01 00:00:...| null|2026-07-01 00:31:...| null| I|202
6-07-01 00:31:...| Nonthaburi|BANGKOK| LAM LUK KA| |RAN-4G|| BKFA
198-L23|2026-07-01| 00| 4G|

|07/01/2026 00:40:03|11/02/2025 00:40:30|BKK1417-N07 | N07...|11/02/2025 00:4
0:30|G09,G18,L07,L21,L...| PMT00L_KPI_Data| ACTIVE| 0|202
6-07-01 00:31:...| N07-BKK1417-1A|NR GNodeB - Ericsson| 0| nul
l| null| null|Value=Temp locked...| BKK1417-N07| M
ALL_THAPRA| 3| Live|BKK1417| 7955| null| Expire A
utomation| Locked Cell|pmtool-system-glf...| BKK1417-N07|
Locked Cell| BKK1417-N07| 0| 213213686| 0|1970-01-01 00:0
0:...|1970-01-01 00:00:...| null|2026-07-01 00:31:...| null| I|202
6-07-01 00:31:...| BANGKOK WEST2|BANGKOK| THON BURI| |RAN-5G|| BKK1
417-N07|2026-07-01| 00| 5G|

|07/01/2026 00:40:03|11/02/2025 00:40:30|BKK1417-N07 | N07...|11/02/2025 00:4
0:30|G09,G18,L07,L21,L...| PMT00L_KPI_Data| ACTIVE| 0|202
6-07-01 00:31:...| N07-BKK1417-1A|NR GNodeB - Ericsson| 0| nul
l| null| null|Value=Temp locked...| BKK1417-N07| M
ALL_THAPRA| 3| Live|BKK1417| 7955| null| Expire A
utomation| Locked Cell|pmtool-system-glf...| BKK1417-N07|
Locked Cell| BKK1417-N07| 0| 213213686| 0|1970-01-01 00:0
0:...|1970-01-01 00:00:...| null|2026-07-01 00:31:...| null| U|202
6-07-01 00:33:...| BANGKOK WEST2|BANGKOK| THON BURI| |RAN-5G|| BKK1
417-N07|2026-07-01| 00| 5G|

|07/01/2026 00:40:03|11/02/2025 00:40:30|BKK1417-N07 | N07...|11/02/2025 00:4
0:30|G09,G18,L07,L21,L...| PMT00L_KPI_Data| ACTIVE| 0|202
6-07-01 00:31:...| N07-BKK1417-1B|NR GNodeB - Ericsson| 0| nul
l| null| null|Value=Temp locked...| BKK1417-N07| M
ALL_THAPRA| 3| Live|BKK1417| 7955| null| Expire A
utomation| Locked Cell|pmtool-system-glf...| BKK1417-N07|
Locked Cell| BKK1417-N07| 0| 213213688| 0|1970-01-01 00:0
0:...|1970-01-01 00:00:...| null|2026-07-01 00:31:...| null| I|202
6-07-01 00:31:...| BANGKOK WEST2|BANGKOK| THON BURI| |RAN-5G|| BKK1
417-N07|2026-07-01| 00| 5G|

|07/01/2026 00:40:03|11/02/2025 00:40:30|BKK1417-N07 | N07...|11/02/2025 00:4
0:30|G09,G18,L07,L21,L...| PMT00L_KPI_Data| ACTIVE| 0|202
6-07-01 00:31:...| N07-BKK1417-1B|NR GNodeB - Ericsson| 0| nul
l| null| null|Value=Temp locked...| BKK1417-N07| M

Out[29]:	INCIDENT_NUMBER	VENDOR_TICKET_NUMBER	ASSIGNED_GROUP
16410	INC000101958091	SCCD_TT202607100807_A23447892	TRUE-TH-BBT-NOR1-CMI1-NOP
16566	INC000101959839	SCCD_TT202607101447_A23448518	TRUE-TH-WW-CEN-CNT-NOP
16853	INC000101962626	SCCD_TT202607102893_A23449985	TRUE-TH-WW-EAS-CCS-NOP
16961	INC000101963557	SCCD_TT202607103555_A23450645	TRUE-TH-BBT-NOR1-CMI2-NOP
17146	INC000101965073	SCCD_TT202607104396_A23451491	TRUE-TH-WW-CEN-LBR-NOP

5 rows × 36 columns

```
In [30]: filtered_df.loc[
    filtered_df["SITE"].str.contains("CMIA002", case=False, na=False),
    ["SITE", "CELLID", "COLLECTTIME", "NR_Cell_Availability"]
]
```

Out[30]:	SITE	CELLID	COLLECTTIME	NR_Cell_Availab
6295134	CMIA002H_7NB02	CMIA002H_7NB02_S01	20260704020000	1
6295135	CMIA002H_7NB02	CMIA002H_7NB02_S01	20260704024500	1
6295136	CMIA002H_7NB02	CMIA002H_7NB02_S01	20260704021500	1
6295137	CMIA002H_7NB02	CMIA002H_7NB02_S01	20260704023000	1
6295138	CMIA002H_7NB02	CMIA002H_7NB02_S01	20260704090000	1
...	...	...	...	...
6297322	CMIA002H_7NB02	CMIA002H_7NB02_S03	20260703231500	1
6297323	CMIA002H_7NB02	CMIA002H_7NB02_S03	20260703233000	1
6297324	CMIA002H_7NB02	CMIA002H_7NB02_S03	20260703224500	1
6297325	CMIA002H_7NB02	CMIA002H_7NB02_S03	20260703221500	1
6297326	CMIA002H_7NB02	CMIA002H_7NB02_S03	20260703223000	1

1299 rows × 4 columns

```
In [31]: filtered_ticket_df.isna().sum()
```

```
Out[31]: INCIDENT_NUMBER      0
VENDOR_TICKET_NUMBER          0
ASSIGNED_GROUP                 0
IMPACT                         0
PRIORITY                      0
URGENCY                       0
COMPANY                       0
DESCRIPTION                    0
DETAILED_DESCRIPTION           0
HPD_CI                        0
NODETYPE                      0
OP_TIME                       0
OP_TYPE                       0
SUBMIT_DATE                   0
submit_date.UTC               0
submit_date.BKK               0
Categorization_Tier_1         0
Categorization_Tier_2         0
Categorization_Tier_3         0
RESOLUTION_CATEGORY           370
RESOLUTION_CATEGORY_TIER_2     370
RESOLUTION_CATEGORY_TIER_3     370
INC_STATUS                    0
STATUS                        0
REPORTED_SOURCE               0
EVENT_END_TIME                225
EVENT_START_TIME              0
PENDING_TIME                  370
PENDING_DUE_TIME              372
rn                            0
SEVERITY                      0
LAST_RESOLVED_DATE.UTC        358
LAST_RESOLVED_DATE.BKK        358
CLOSED_DATE.UTC               358
CLOSED_DATE.BKK               358
Technology                    0
dtype: int64
```

```
In [32]: filtered_df.columns
```

```

Out[32]: Index(['COLLECTTIME', 'CELLID', 'SITE', 'NR_SA_Paging_Discard_Rate',
               'NR_DL_Active_UEs', 'NR_UL_Active_UEs', 'NR_DL_MAC_Volume_MB',
               'NR_UL_MAC_Volume_MB', 'NR_DL_Active_UEs_True', 'NR_UL_Active_UEs_True',
               'NR_DL_RBSym_Util', 'NR_UL_RBSym_Util', 'NR_Msg2_Attempt_SR',
               'NR_Msg2_Attempts', 'NR_RACH_SR', 'NR_RACH_Att',
               'NR_UL_MAC_Cell_Thp_total_time_Mbps',
               'NR_DL_MAC_Cell_Thp_total_time_Mbps', 'NR_DL_HARQ_BLER_256QAM',
               'NR_DL_RLC_BLER', 'NR_Avg_DL_MAC_Thp_Mbps', 'NR_Avg_UL_MAC_UE_Thp_Mbps',
               'NR_DL_MAC_Cell_Thp_Mbps', 'NR_UL_MAC_Cell_Thp_Mbps',
               'NR_UL_HARQ_BLER_64_QAM', 'NR_UL_RLC_BLER', 'NR_UL_HARQ_BLER',
               'NR_DL_HARQ_BLER', 'NR_UL_DTX_Rate', 'NR_DL_DTX_Rate',
               'NR_Cell_Availability', 'NR_Avg_UL_PUCCH_SINR_dB',
               'NR_Avg_UL_PUSCH_SINR_dB', 'NR_RANK_Below3_Rate', 'NR_Avg_CQI',
               'NR_Avg_CQI_64QAM', 'NR_Avg_CQI_256QAM', 'NR_Avg_UL_Pathloss_dB',
               'NR_Avg_UL_Power_Headroom_dB', 'NR_Avg_UL_Pathloss_95percentile_dB',
               'NR_UL_RSSI', 'NR_Avg_MCS_Table1', 'NR_Avg_MCS_Table2', 'NR_TA',
               'DL_Thp_Bad_Sessions', 'DL_Thp_Fair_Sessions', 'DL_Thp_Good_Sessions',
               'DL_Thp_Excellent_Sessions', 'UL_Thp_Bad_Sessions',
               'UL_Thp_Fair_Sessions', 'UL_Thp_Good_Sessions',
               'UL_Thp_Excellent_Sessions', 'NR_Avg_UL_MAC_Thp_EBS_Mbps',
               'NR_Avg_DL_MAC_Thp_EBS_Mbps', 'DL_Total_Sessions', 'UL_Total_Sessions',
               'NR_Avg_DL_Latency_ms', 'NR_Cell_Availability_Num', 'NR_ENDC_Setup_Attempt',
               'NR_ENDC_Setup_SR', 'NR_SA_RRC_ATT', 'NR_SA_RRC_SSSR',
               'NR_SA_NG_Sig_SR', 'NR_ENDC_Inter_sgNodeB_PSCell_Change_SR',
               'NR_ENDC_Intra_sgNodeB_PSCell_Change_SR', 'NR_SA_NR_to_LTE_RWR_Ratio',
               'NR_SA_INTER_GNB_HO_SR', 'NR_SA_INTRA_GNB_HO_SR',
               'NR_SA_HO_Prep_Succ_Rate', 'NR_SA_HO_Exec_Succ_Rate', 'NR_SA_HO_SR',
               'NR_ENDC_SARR', 'NR_ENDC_RRC_CU_Max', 'NR_SA_RRC_CU_Max',
               'NR_ENDC_RRC_CU_Avg', 'NR_SA_DRB_EST_SR', 'NR_SA_Drop_Rate',
               'NR_SgNB_Abnormal_RR', 'NR_RRC_SSR', 'ESS_TOTAL_PRB_AVG_UTIL_DL',
               'ESS_NR_PRB_AVG_UTIL_DL', 'essEnabled'],
              dtype='object')

```

```

In [33]: filtered_df['COLLECTTIME'] = pd.to_datetime(filtered_df['COLLECTTIME'])
         filtered_df['COLLECTTIME'] = filtered_df['COLLECTTIME'].dt.tz_localize('Asia/E

```

```

In [34]: filtered_df['COLLECTTIME'].head()

```

```

Out[34]: 0    2026-07-01 19:00:00+00:00
         1    2026-07-01 19:45:00+00:00
         2    2026-07-01 19:30:00+00:00
         3    2026-07-02 12:45:00+00:00
         4    2026-07-02 12:15:00+00:00
         Name: COLLECTTIME, dtype: datetime64[ns, UTC]

```

```

In [35]: alarm_df

```

```
Out[35]: DataFrame[RECTIMESTAMP: string, FIRSTOCCURRENCE: string, IDENTIFIER: string,
LASTOCCURRENCE: string, SERVICE: string, EMS_NAME: string, NODESTATUS: string,
SERVICEAFFECTING: string, CLEARTIME: string, OBJECT: string, NODETYPE: string,
SEVERITY: string, EXTENDEDATTR: string, ELEMENTMANAGERIP: string, EVENTID: string,
ADDITIONALSTRING: string, ALERTKEY: string, SITENAME: string, ORIGINALSEVERITY: string,
SITESTATUS: string, SITEID: string, CLASS: string, OBJECTSTATUS: string, CLEARED BY: string,
X733SPECIFICPROB: string, MANAGER: string, NODEALIAS: string, SUMMARY: string, NODE: string,
TT_FLAG: string, SERIAL: string, TMTRANSSTATE: string, TTCREATIONTIME: string, TMTRANSBEGINTIME: string,
TT_ID: string, RECEIVEDATPROBE: string, physicalCard: string, OP_TYPE: string, OP_TIME_DWH: string,
CIRCLE: string, REGION: string, LOCATION: string, BACKHAULTYPE: string, CONTROLNE: string,
DATE_ID: string, HOUR_ID: string, Technology: string]
```

```
In [36]: print("Total rows:", len(filtered_ticket_df))
print("Unique HPD_CI:", filtered_ticket_df["HPD_CI"].nunique())
```

```
Total rows: 372
Unique HPD_CI: 329
```

```
In [37]: filtered_ticket_df = filtered_ticket_df.drop_duplicates(
subset=["HPD_CI"]
)
```

```
In [38]: kpi_sites = filtered_df["SITE"].dropna().astype(str).unique()

ticket_mapping = []

for hpd_ci in filtered_ticket_df["HPD_CI"].dropna().astype(str).unique():
    matches = [kpi_site for kpi_site in kpi_sites if hpd_ci in kpi_site]

    ticket_mapping.append({
        "HPD_CI": hpd_ci,
        "Matched_KPI_Sites": matches,
        "Match_Count": len(matches)
    })

ticket_mapping_df = pd.DataFrame(ticket_mapping)
```

```
In [39]: ticket_mapping_df
```

Out[39]:

	HPD_CI	Matched_KPI_Sites	Match_Count
0	CMI7658	[]	0
1	CNT0039	[CNT0039H_7NB01]	1
2	CCS0331	[]	0
3	CMIA002	[CMIA002H_7NB02]	1
4	LBR8549	[]	0
...	...	...	...
324	NTB7211	[]	0
325	CMI3189	[]	0
326	LPN1141	[]	0
327	ATG6722	[ATG6722P_9NB01]	1
328	CMI6772	[]	0

329 rows × 3 columns

In [40]: `filtered_df.shape`

Out[40]: (6298302, 82)

In [41]:

```

site_status = (
    filtered_df
    .groupby(["SITE", "COLLECTTIME"])
    .agg(
        total_cells=("CELLID", "nunique"),
        down_cells=("NR_Cell_Availability", lambda x: (x == 0).sum())
    )
    .reset_index()
)

site_status["COMPLETE_SITE_OUTAGE"] = (
    site_status["total_cells"] == site_status["down_cells"]
)

```

In [42]:

```

complete_site_outages = site_status[
    site_status["COMPLETE_SITE_OUTAGE"]
]

```

In [43]: `complete_site_outages.count()`

```
Out[43]: SITE                8244
COLLECTTIME                8244
total_cells                8244
down_cells                 8244
COMPLETE_SITE_OUTAGE       8244
dtype: int64
```

```
In [44]: kpi_complete_outage_df = filtered_df.merge(
    complete_site_outages[["SITE", "COLLECTTIME"]],
    on=["SITE", "COLLECTTIME"],
    how="inner"
)
```

```
In [45]: kpi_complete_outage_df.shape
```

```
Out[45]: (21230, 82)
```

```
In [46]: kpi_complete_outage_df.head()
```

```
Out[46]:
```

	COLLECTTIME	CELLID	SITE	NR_SA_Paging_Discard_
0	2026-07-02 09:00:00+00:00	ATG0019H_7NB01_S01	ATG0019T_2NB01	
1	2026-07-02 09:00:00+00:00	ATG0019H_7NB01_S03	ATG0019T_2NB01	
2	2026-07-02 09:00:00+00:00	ATG0019H_7NB01_S02	ATG0019T_2NB01	
3	2026-07-02 09:45:00+00:00	ATG0019H_7NB01_S01	ATG0019T_2NB01	
4	2026-07-02 09:45:00+00:00	ATG0019H_7NB01_S03	ATG0019T_2NB01	

5 rows × 82 columns

```
In [47]: #Alarm filtering
alarm_summary = (
    alarm_df
    .groupBy("SITEID", "NODE")
    .agg(
        F.count("*").alias("Alarm_Count")
    )
    .orderBy(F.desc("Alarm_Count"))
)

# Display the result
alarm_summary.show(truncate=False)
```

```
26/07/16 05:24:14 WARN DAGScheduler: Broadcasting large task binary with size 1470.9 KiB
[Stage 685:> (0 + 6) / 7]
```

```
+-----+-----+-----+
|SITEID |NODE           |Alarm_Count|
+-----+-----+-----+
|null    |null           |332803     |
|BKK3596 |BKK3596-L23    |56617      |
|KLS7155 |KLS7155P_9NB01|56080      |
|SKT3127 |SKT3127-L23    |49769      |
|UTT0620 |UTT0620-L23    |49557      |
|BKK3816 |BKK3816-L23    |37701      |
|NKR0237 |NKR0237-L23    |36347      |
|SK00098 |SK00098-L23    |28290      |
|SK00221 |SK00221-L23    |28265      |
|BKK7715 |BKK7715-N07    |26814      |
|BKK7715 |BKK7715-L21    |26674      |
|BKK7478 |BKK7478-L21    |25104      |
|KKNC044 |KKNC044-L23    |24838      |
|SK00148 |SK00148-L23    |23825      |
|SK00090 |SK00090-L23    |23584      |
|CMI0114 |CMI0114-L23    |23093      |
|SK00162 |SK00162-L23    |21955      |
|AYT0101 |AYT0101T_2NB02|20929      |
|SK00223 |SK00223B-L23   |20771      |
|SK00215 |SK00215-L23    |18933      |
+-----+-----+-----+
only showing top 20 rows
```

```
In [48]: # Count NULL values in each column
null_counts = alarm_df.select([
    F.count(F.when(F.col(c).isNull(), c)).alias(c)
    for c in alarm_df.columns
])

null_counts.show(truncate=False)
```

```
26/07/16 05:25:19 WARN DAGScheduler: Broadcasting large task binary with size 4.1 MiB
[Stage 686:=====>(1155 + 6) / 1161]
```



```

+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
|RECTIMESTAMP|FIRSTOCCURRENCE|IDENTIFIER|LASTOCCURRENCE|SERVICE|EMS_NAME|NODEST
ATUS|SERVICEAFFECTING|CLEARTIME|OBJECT |NODETYPE|SEVERITY|EXTENDEDATTR|ELEMENTM
ANAGERIP|EVENTID|ADDITIONALSTRING|ALERTKEY|SITENAME|ORIGINALSEVERITY|SITESTATU
S|SITEID|CLASS |OBJECTSTATUS|CLEAREDDBY|X733SPECIFICPROB|MANAGER|NODEALIAS|SUMMA
RY|NODE |TT_FLAG|SERVERSERIAL|TMTRANSSTATE|TTCREATIONTIME|TMTRANSBEGINTIME|T
T_ID |RECEIVEDATPROBE|physicalCard|OP_TYPE|OP_TIME_DWH|CIRCLE|REGION|LOCATIO
N|BACKHAULTYPE|CONTROLNE|DATE_ID|HOUR_ID|Technology|
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
|0|0|0|0|38948|2512|36404
|0|0|1585860|0|0|6828809|621128
|662262|362145|332808|1079598|332803|335082|33293
1|332803|5468264|2350126|332950|332803|332803|332924|3328
15|332815|332815|332815|332815|332815|6982279|33
2815|7040777|332815|332815|333798|333714|338837|633741
|334969|0|0|0|
+-----+-----+-----+-----+-----+-----+
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+-----+-----+-----+-----+-----+-----+
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+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+

```

In [49]: *# Keep only rows where both SITEID and NODE are not NULL*

```

alarm_df = alarm_df.filter(
    F.col("SITEID").isNotNull() &
    F.col("NODE").isNotNull()
)

print(f"Number of rows after dropping NULL SITEID/NODE: {alarm_df.count()}")

```

26/07/16 05:27:10 WARN DAGScheduler: Broadcasting large task binary with size 167.0 KiB

[Stage 689:=====>(1155 + 6) / 1161]

Number of rows after dropping NULL SITEID/NODE: 6944032

In [50]: *##finding the duplicates rows by grouping on column and removing the column "C*

```
cols_to_check = [c for c in alarm_df.columns if c != "OP_TIME_DWH"]

duplicates = (
    alarm_df
    .groupBy(cols_to_check)
    .count()
    .filter(F.col("count") > 1)
    .orderBy(F.desc("count"))
)

duplicates.show(100, truncate=False)
```

```
26/07/16 05:28:03 WARN DAGScheduler: Broadcasting large task binary with size
5.1 MiB
26/07/16 05:30:11 WARN DAGScheduler: Broadcasting large task binary with size
2.9 MiB
[Stage 694:=====> (65 + 2) / 6
7]
```

[illegible]


```
-----
-----
-----+-----+-----
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+-----
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-----+-----+-----+-----+-----
+-----+-----+-----+-----
+-----+-----+-----+-----+-----
+-----+-----+-----+-----+-----
+-----+-----+-----+
|07/05/2026 22:40:02|07/05/2026 22:28:44|1758084173|ericsson-enm-corba@150.23
6.236.3:6795|PSN6744P_9NB01|07/05/2026 22:29:20|PLK-B_0218 | PS
N0013|True_Ericsson_ENM_UTRAN_SOUTH_3A|ACTIVE
|0|2026-07-05 22:29:41 UTC|null
|NR GNodeB - Ericsson|0|null
|10.11.6.8|1758084173|Contact next level of support and provide log f
iles required by Data Collection Guideline.
|SubNetwork=RadioNode,MeContext=PSN6744P_9NB01,ManagedElement=PSN6744P_9NB01,EN
odeBFunction=1,SectorCarrier=L07G-S1C1|Nakhonchum
|3|Live|PSN6744|1409|RETAINT|Deduplication Automat
ion|SW Error|ericsson-enm-corba@150.236.236.3:6795
|PSN6744P_9NB01|SW Error
|PSN6744P_9NB01|0|691266826|0|1970-01-01 00:00:00 UTC|1
970-01-01 00:00:00 UTC|null|2026-07-05 22:28:51 UTC|null|U
|PHITSANULOK|NORTH|NAKHON THAI|True-NR FDD|NR FDD
|2026-07-05|22|5G|4|
|07/06/2026 11:30:03|07/06/2026 11:18:46|1760017177|ericsson-enm-corba@150.23
6.236.3:6795|PSN6744P_9NB01|07/06/2026 11:19:58|PLK-B_0218 | PS
N0013|True_Ericsson_ENM_UTRAN_SOUTH_3A|ACTIVE
|0|2026-07-06 11:20:06 UTC|null
|NR GNodeB - Ericsson|0|null
|10.11.6.8|1760017177|Contact next level of support and provide log f
iles required by Data Collection Guideline.
|SubNetwork=RadioNode,MeContext=PSN6744P_9NB01,ManagedElement=PSN6744P_9NB01,EN
odeBFunction=1,SectorCarrier=L07G-S1C1|Nakhonchum
|3|Live|PSN6744|1409|RETAINT|Deduplication Automat
ion|SW Error|ericsson-enm-corba@150.236.236.3:6795
|PSN6744P_9NB01|SW Error
|PSN6744P_9NB01|0|691587170|0|1970-01-01 00:00:00 UTC|1
970-01-01 00:00:00 UTC|null|2026-07-06 11:18:51 UTC|null|U
```

|PHITSANULOK |NORTH |NAKHON THAI |True-NR FDD |NR FDD
|2026-07-06|11 |5G |4 |
|07/05/2026 19:05:02|07/05/2026 18:53:53|1758084173|ericsson-enm-corba@150.23
6.236.3:6795|PSN6744P_9NB01 |07/05/2026 18:54:37|PLK-B_0218 | PS
N0013 |True_Ericsson_ENM_UTRAN_SOUTH_3A |ACTIVE
|0 |2026-07-05 18:54:42 UTC|null
|NR GNodeB - Ericsson|0 |null
|10.11.6.8 |1758084173 |Contact next level of support and provide log f
iles required by Data Collection Guideline.
|SubNetwork=RadioNode,MeContext=PSN6744P_9NB01,ManagedElement=PSN6744P_9NB01,EN
odeBFunction=1,SectorCarrier=L07G-S1C1 |Nakhonchum
|3 |Live |PSN6744|1409 |RETAINT |Deduplication Automat
ion|SW Error |ericsson-enm-corba@150.236.236.3:6795
|PSN6744P_9NB01 |SW Error
|PSN6744P_9NB01 |0 |691219034 |0 |1970-01-01 00:00:00 UTC|1
970-01-01 00:00:00 UTC|null |2026-07-05 18:54:00 UTC|null |U
|PHITSANULOK |NORTH |NAKHON THAI |True-NR FDD |NR FDD
|2026-07-05|19 |5G |4 |
|07/05/2026 17:30:02|07/05/2026 17:19:06|1758084173|ericsson-enm-corba@150.23
6.236.3:6795|PSN6744P_9NB01 |07/05/2026 17:20:32|PLK-B_0218 | PS
N0013 |True_Ericsson_ENM_UTRAN_SOUTH_3A |ACTIVE
|0 |2026-07-05 17:20:53 UTC|null
|NR GNodeB - Ericsson|0 |null
|10.11.6.8 |1758084173 |Contact next level of support and provide log f
iles required by Data Collection Guideline.
|SubNetwork=RadioNode,MeContext=PSN6744P_9NB01,ManagedElement=PSN6744P_9NB01,EN
odeBFunction=1,SectorCarrier=L07G-S1C1 |Nakhonchum
|3 |Live |PSN6744|1409 |RETAINT |Deduplication Automat
ion|SW Error |ericsson-enm-corba@150.236.236.3:6795
|PSN6744P_9NB01 |SW Error
|PSN6744P_9NB01 |0 |691199788 |0 |1970-01-01 00:00:00 UTC|1
970-01-01 00:00:00 UTC|null |2026-07-05 17:19:12 UTC|null |U
|PHITSANULOK |NORTH |NAKHON THAI |True-NR FDD |NR FDD
|2026-07-05|17 |5G |4 |
|07/07/2026 19:50:03|07/07/2026 19:39:18|1760717219|ericsson-enm-corba@150.23
6.236.3:6795|PSN6744P_9NB01 |07/07/2026 19:39:49|PLK-B_0218 | PS
N0013 |True_Ericsson_ENM_UTRAN_SOUTH_3A |ACTIVE
|0 |2026-07-07 19:39:55 UTC|null
|NR GNodeB - Ericsson|0 |null
|10.11.6.8 |1760717219 |Contact next level of support and provide log f
iles required by Data Collection Guideline.
|SubNetwork=RadioNode,MeContext=PSN6744P_9NB01,ManagedElement=PSN6744P_9NB01,EN
odeBFunction=1,SectorCarrier=L07G-S1C1 |Nakhonchum
|3 |Live |PSN6744|1409 |RETAINT |Deduplication Automat
ion|SW Error |ericsson-enm-corba@150.236.236.3:6795
|PSN6744P_9NB01 |SW Error
|PSN6744P_9NB01 |0 |692321513 |0 |1970-01-01 00:00:00 UTC|1
970-01-01 00:00:00 UTC|null |2026-07-07 19:39:24 UTC|null |U
|PHITSANULOK |NORTH |NAKHON THAI |True-NR FDD |NR FDD
|2026-07-07|19 |5G |4 |
|07/07/2026 00:30:03|07/07/2026 00:18:40|1760717219|ericsson-enm-corba@150.23
6.236.3:6795|PSN6744P_9NB01 |07/07/2026 00:19:20|PLK-B_0218 | PS
N0013 |True_Ericsson_ENM_UTRAN_SOUTH_3A |ACTIVE
|0 |2026-07-07 00:19:27 UTC|null

|NR GNodeB - Ericsson|0 |null
|10.11.6.8 |1760717219 |Contact next level of support and provide log files required by Data Collection Guideline.
|SubNetwork=RadioNode,MeContext=PSN6744P_9NB01,ManagedElement=PSN6744P_9NB01,ENodeBFunction=1,SectorCarrier=L07G-S1C1 |Nakhonchum
|3 |Live |PSN6744|1409 |RETAINT |Deduplication Automation|SW Error |ericsson-enm-corba@150.236.236.3:6795
|PSN6744P_9NB01 |SW Error
|PSN6744P_9NB01 |0 |691864733 |0 |1970-01-01 00:00:00 UTC|1970-01-01 00:00:00 UTC|null |2026-07-07 00:18:48 UTC|null |U
|PHITSANULOK |NORTH |NAKHON THAI |True-NR FDD |NR FDD
|2026-07-07|00 |5G |4 |
|07/06/2026 00:05:02|07/05/2026 23:42:06|1758084173|ericsson-enm-corba@150.236.236.3:6795|PSN6744P_9NB01 |07/05/2026 23:54:41|PLK-B_0218 |PSN0013 |True_Ericsson_ENM_UTRAN_SOUTH_3A |ACTIVE
|0 |2026-07-05 23:54:47 UTC|null
|NR GNodeB - Ericsson|0 |null
|10.11.6.8 |1758084173 |Contact next level of support and provide log files required by Data Collection Guideline.
|SubNetwork=RadioNode,MeContext=PSN6744P_9NB01,ManagedElement=PSN6744P_9NB01,ENodeBFunction=1,SectorCarrier=L07G-S1C1 |Nakhonchum
|3 |Live |PSN6744|1409 |RETAINT |Deduplication Automation|SW Error |ericsson-enm-corba@150.236.236.3:6795
|PSN6744P_9NB01 |SW Error
|PSN6744P_9NB01 |0 |691289771 |0 |1970-01-01 00:00:00 UTC|1970-01-01 00:00:00 UTC|null |2026-07-05 23:42:12 UTC|null |U
|PHITSANULOK |NORTH |NAKHON THAI |True-NR FDD |NR FDD
|2026-07-06|00 |5G |4 |
|07/05/2026 16:40:02|07/05/2026 16:29:04|1758084173|ericsson-enm-corba@150.236.236.3:6795|PSN6744P_9NB01 |07/05/2026 16:30:23|PLK-B_0218 |PSN0013 |True_Ericsson_ENM_UTRAN_SOUTH_3A |ACTIVE
|0 |2026-07-05 16:30:30 UTC|null
|NR GNodeB - Ericsson|0 |null
|10.11.6.8 |1758084173 |Contact next level of support and provide log files required by Data Collection Guideline.
|SubNetwork=RadioNode,MeContext=PSN6744P_9NB01,ManagedElement=PSN6744P_9NB01,ENodeBFunction=1,SectorCarrier=L07G-S1C1 |Nakhonchum
|3 |Live |PSN6744|1409 |RETAINT |Deduplication Automation|SW Error |ericsson-enm-corba@150.236.236.3:6795
|PSN6744P_9NB01 |SW Error
|PSN6744P_9NB01 |0 |691187679 |0 |1970-01-01 00:00:00 UTC|1970-01-01 00:00:00 UTC|null |2026-07-05 16:29:09 UTC|null |U
|PHITSANULOK |NORTH |NAKHON THAI |True-NR FDD |NR FDD
|2026-07-05|16 |5G |4 |
|07/01/2026 05:20:02|07/01/2026 05:08:14|1101746962|ericsson-enm-corba@150.236.236.78:6796|LBR0092T_2NB01 |07/01/2026 05:09:26|G09,L07,L21,N07,U09,U21 |True_Ericsson_ENM_UTRAN_CENTRAL_4A |ACTIVE
|0 |2026-07-01 05:09:39 UTC|null
|NR GNodeB - Ericsson|0 |null
|10.23.17.8 |1101746962 |Contact next level of support and provide log files required by Data Collection Guideline.
|SubNetwork=RadioNode,MeContext=LBR0092T_2NB01,ManagedElement=LBR0092T_2NB01,ENodeBFunction=1,SectorCarrier=L07G-S2C1 |null
|3 |Live |LBR0092|1409 |null |Deduplication Automation

ion|SW Error |ericsson-enm-corba@150.236.236.78:6796
|LBR0092T_2NB01 |SW Error
|LBR0092T_2NB01 |0 |688896073 |0 |1970-01-01 00:00:00 UTC|1
970-01-01 00:00:00 UTC|null |2026-07-01 05:08:21 UTC|null |U
|LOP BURI |CENTRAL |PHATTHANA NIKHOM |True-NR FDD |NR FDD
|2026-07-01|05 |5G |4 |
|07/03/2026 21:55:02|07/03/2026 14:23:01|1497829213|ericsson-enm-corba@150.23
6.236.58:6794|CMI6717E_1NB04_S01 |07/03/2026 15:35:46|CMI-A_0377 |CM
I1000 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-03 20:41:42 UTC|CMI6717E_1NB04_S01
|LTE Cell |0 |"MATE info="Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1497829213 |Preambles, no activity, recov succ
|SubNetwork=RadioNode,MeContext=CMI6717T_2NB04,ManagedElement=CMI6717T_2NB04,EN
odeBFunction=1,EUtranCellFDD=CMI6717E_1NB04_S01 |Tasatoi
|4 |Live |CMI6717|1010400|null |Deduplication Automat
ion|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794
|CMI6717E_1NB04_S01 |Suspected Sleeping Cell
|CMI6717E_1NB04_S01|10 |43133487 |125 |2026-07-03 20:23:49 UTC|1
970-01-01 00:00:00 UTC|INC000102014349|2026-07-03 14:23:07 UTC|null |U
|CHIANG MAI1 |NORTH |T_Retain_FAC ||True-L18| |CMI671
7T_2NB04 |2026-07-03|21 |4G |3 |
|07/05/2026 21:05:03|07/05/2026 13:23:05|1502722068|ericsson-enm-corba@150.23
6.236.58:6794|CMI6272E_1NB05_S03 |07/05/2026 13:23:05|G09,L07,L09,L1
8,L21,L23,N07,U08,U09,U21 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-05 19:54:26 UTC|CMI6272E_1NB05_S03
|LTE Cell |0 |"MATE info="Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1502722068 |No activity, recov fail
|SubNetwork=RadioNode,MeContext=CMI6272T_2NB05,ManagedElement=CMI6272T_2NB05,EN
odeBFunction=1,EUtranCellFDD=CMI6272E_1NB05_S03 |null
|4 |Live |CMI6272|1010400|null |Deduplication Automat
ion|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794
|CMI6272E_1NB05_S03 |Suspected Sleeping Cell
|CMI6272E_1NB05_S03|10 |44035160 |125 |2026-07-05 19:23:42 UTC|1
970-01-01 00:00:00 UTC|INC000102042347|2026-07-05 13:23:10 UTC|null |U
|CHIANG MAI1 |NORTH |MUEANG CHIANG MAI ||True-L18| |CMI627
2T_2NB05 |2026-07-05|21 |4G |3 |
|07/04/2026 20:35:02|07/04/2026 14:14:08|1499953875|ericsson-enm-corba@150.23
6.236.58:6794|CMI1616E_2NB01_S03 |07/04/2026 14:14:08|LPN-A_0528 |LP
N3052 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-04 19:24:12 UTC|CMI1616E_2NB01_S03
|LTE Cell |0 |"MATE info="Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1499953875 |No activity, recov fail
|SubNetwork=RadioNode,MeContext=CMI1616T_2NB01,ManagedElement=CMI1616T_2NB01,EN
odeBFunction=1,EUtranCellFDD=CMI1616E_2NB01_S03 |GRF-TOT Doi In
thanon |4 |Live |CMI1616|1010400|null
|Deduplication Automation|Suspected Sleeping Cell |ericsson-enm-cor
ba@150.236.236.58:6794 |CMI1616E_2NB01_S03 |Suspected Sleeping Cell
|CMI1616E_2NB01_S03|10 |43626331 |125 |2026-07-04 18:57:45 UTC|1

970-01-01 00:00:00 UTC|INC000102028753|2026-07-04 14:14:13 UTC|null |U
|CHIANG MAI1 |NORTH |MAE CHAEM ||True-L21| |CMI161
6T_2NB01 |2026-07-04|20 |4G |3 |
|07/02/2026 09:05:02|07/02/2026 01:23:02|1492827950|ericsson-enm-corba@150.23
6.236.58:6794|CMI6741E_1NB03_S01 |07/02/2026 01:23:02|MHS-B_0170 |MH
S(CMI1000) |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-02 07:53:23 UTC|CMI6741E_1NB03_S01
|LTE Cell |0 |"MATE info=""Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1492827950 |No activity, recov fail
|SubNetwork=RadioNode,MeContext=CMI6741T_2NB03,ManagedElement=CMI6741T_2NB03,EN
odeBFunction=1,EUtranCellFDD=CMI6741E_1NB03_S01 |GRF-Muangket
|4 |Live |CMI6741|1010400|null |Deduplication Automat
ion|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794
|CMI6741E_1NB03_S01 |Suspected Sleeping Cell
|CMI6741E_1NB03_S01|10 |42256727 |125 |2026-07-02 07:24:14 UTC|1
970-01-01 00:00:00 UTC|INC000101984158|2026-07-02 01:23:07 UTC|null |U
|CHIANG MAI |NORTH |MAE TAENG ||True-L18| |CMI674
1T_2NB03 |2026-07-02|09 |4G |3 |
|07/06/2026 16:55:03|07/06/2026 16:41:02|1760717219|ericsson-enm-corba@150.23
6.236.3:6795|PSN6744P_9NB01 |07/06/2026 16:43:31|PLK-B_0218 |PS
N0013 |True_Ericsson_ENM_UTRAN_SOUTH_3A |ACTIVE
|0 |2026-07-06 16:43:37 UTC|null
|NR GNodeB - Ericsson|0 |null
|10.11.6.8 |1760717219 |Contact next level of support and provide log f
iles required by Data Collection Guideline.
|SubNetwork=RadioNode,MeContext=PSN6744P_9NB01,ManagedElement=PSN6744P_9NB01,EN
odeBFunction=1,SectorCarrier=L07G-S1C1 |Nakhonchum
|3 |Live |PSN6744|1409 |RETAINT |Deduplication Automat
ion|SW Error |ericsson-enm-corba@150.236.236.3:6795
|PSN6744P_9NB01 |SW Error
|PSN6744P_9NB01 |0 |691729317 |0 |1970-01-01 00:00:00 UTC|1
970-01-01 00:00:00 UTC|null |2026-07-06 16:41:09 UTC|null |U
|PHITSANULOK |NORTH |NAKHON THAI |True-NR FDD |NR FDD
|2026-07-06|16 |5G |3 |
|07/07/2026 19:05:02|07/07/2026 11:28:09|1507115744|ericsson-enm-corba@150.23
6.236.58:6794|CMI0943E_1NB03_S02 |07/07/2026 11:28:09|L07,L18,L21,N0
7,U09,U21 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-07 17:53:19 UTC|CMI0943E_1NB03_S02
|LTE Cell |0 |"MATE info=""Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1507115744 |No activity, recov fail
|SubNetwork=RadioNode,MeContext=CMI0943T_2NB03,ManagedElement=CMI0943T_2NB03,EN
odeBFunction=1,EUtranCellFDD=CMI0943E_1NB03_S02 |BAN PA NGEW
|4 |Live |CMI0943|1010400|null |Deduplication Automat
ion|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794
|CMI0943E_1NB03_S02 |Suspected Sleeping Cell
|CMI0943E_1NB03_S02|10 |44988490 |125 |2026-07-07 17:28:58 UTC|1
970-01-01 00:00:00 UTC|INC000102085524|2026-07-07 11:28:16 UTC|null |U
|CHIANG MAI2 |NORTH |PHRAO ||True-L18| |CMI094
3T_2NB03 |2026-07-07|19 |4G |3 |
|07/07/2026 02:05:02|07/06/2026 18:23:22|1505628092|ericsson-enm-corba@150.23

6.236.58:6794|CMI0593E_1NB03_S02 |07/06/2026 18:23:22|CMI-A_0219 | CM
I1000 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-07 00:52:46 UTC|CMI0593E_1NB03_S02
|LTE Cell |0 |"MATE info="Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1505628092 |No activity, recov succ
|SubNetwork=RadioNode,MeContext=CMI0593T_2NB03,ManagedElement=CMI0593T_2NB03,EN
odeBFunction=1,EUtranCellFDD=CMI0593E_1NB03_S02 |VIRABDA RESORT
|4 |Live |CMI0593|1010400|null |Deduplication Automat
ion|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794
|CMI0593E_1NB03_S02 |Suspected Sleeping Cell
|CMI0593E_1NB03_S02|10 |44567409 |125 |2026-07-07 00:24:23 UTC|1
970-01-01 00:00:00 UTC|INC000102066916|2026-07-06 18:23:28 UTC|null |U
|CHIANG MAI1 |NORTH |HANG DONG ||True-L18| |CMI059
3T_2NB03 |2026-07-07|02 |4G |3 |
|07/02/2026 02:05:02|07/01/2026 18:20:31|1492882533|ericsson-enm-corba@150.23
6.236.58:6794|CMI7430E_1NB03_S02 |07/01/2026 18:20:31|G09,L07,L09,L1
8,L21,N07,U09,U21 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-02 00:54:35 UTC|CMI7430E_1NB03_S02
|LTE Cell |0 |"MATE info="Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1492882533 |No activity, recov fail
|SubNetwork=RadioNode,MeContext=CMI7430T_2NB03,ManagedElement=CMI7430T_2NB03,EN
odeBFunction=1,EUtranCellFDD=CMI7430E_1NB03_S02 |T_CMI9304
|4 |Live |CMI7430|1010400|null |Deduplication Automat
ion|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794
|CMI7430E_1NB03_S02 |Suspected Sleeping Cell
|CMI7430E_1NB03_S02|10 |42118555 |125 |2026-07-02 00:20:59 UTC|1
970-01-01 00:00:00 UTC|INC000101978219|2026-07-01 18:20:40 UTC|null |U
|CHIANG MAI1 |NORTH |MUEANG CHIANG MAI ||True-L18| |CMI743
0T_2NB03 |2026-07-02|02 |4G |3 |
|07/01/2026 03:35:01|06/30/2026 19:54:33|1490515276|ericsson-enm-corba@150.23
6.236.58:6794|CMI7423E_2NB03_S07 |06/30/2026 19:54:33|G09,L09,L18,L2
1,L23,U08,U21 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-01 02:21:52 UTC|CMI7423E_2NB03_S07
|LTE Cell |0 |"MATE info="Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1490515276 |No activity, recov fail
|SubNetwork=RadioNode,MeContext=CMI7423T_2NB03,ManagedElement=CMI7423T_2NB03,EN
odeBFunction=1,EUtranCellFDD=CMI7423E_2NB03_S07 |KAD DHARA DHEV
I |4 |Live |CMI7423|1010400|null
|Deduplication Automation|Suspected Sleeping Cell |ericsson-enm-cor
ba@150.236.236.58:6794 |CMI7423E_2NB03_S07 |Suspected Sleeping Cell
|CMI7423E_2NB03_S07|10 |41606195 |125 |2026-07-01 01:55:09 UTC|1
970-01-01 00:00:00 UTC|INC000101960687|2026-06-30 19:54:40 UTC|null |U
|CHIANG MAI1 |NORTH |MUEANG CHIANG MAI ||True-L21| |CMI742
3T_2NB03 |2026-07-01|03 |4G |3 |
|07/01/2026 16:20:03|07/01/2026 16:10:20|396272024|ericsson-enm-corba@150.236.2
36.3:6796|SMK8634T_2NB02 |07/01/2026 16:10:20|SMK-A_0040 | SMK0
020 |True_Ericsson_ENM_RAN_6A |ACTIVE
|0 |2026-07-01 16:11:04 UTC|null

|eNode B |0 |null
|10.23.21.8 |396272024 |Contact next level of support and provide log files required by Data Collection Guideline (Mixed Mode with other ME).
|SubNetwork=RadioNode,MeContext=SMK8634T_2NB02,ManagedElement=SMK8634T_2NB02,Equipment=1,FieldReplaceableUnit=RRU-4 |T_SMK8634
|3 |Live |SMK8634|1010400|CORP+RETAINT|Deduplication Automation|SW Error |ericsson-enm-corba@150.236.236.3:6796
|SMK8634T_2NB02 |SW Error
|SMK8634T_2NB02 |0 |689181110 |0 |1970-01-01 00:00:00 UTC|1970-01-01 00:00:00 UTC|null |2026-07-01 16:10:25 UTC|null |U
|SAMUT SAKHON |WEST |MUEANG SAMUT SAKHON |True-LTE FDD|LTE FDD
|2026-07-01|16 |4G |3 |
|07/05/2026 04:35:03|07/04/2026 21:00:02|1500106796|ericsson-enm-corba@150.236.236.58:6794|CMI6272E_1NB05_S04 |07/04/2026 21:00:02|G09,L07,L09,L18,L21,L23,N07,U08,U09,U21 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-05 03:22:46 UTC|CMI6272E_1NB05_S04
|LTE Cell |0 |"MATE info=""Updated by rule: Main_Flow_NOC, module: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1500106796 |No activity, recov fail
|SubNetwork=RadioNode,MeContext=CMI6272T_2NB05,ManagedElement=CMI6272T_2NB05,ENodeBFunction=1,EUtranCellFDD=CMI6272E_1NB05_S04 |null
|4 |Live |CMI6272|1010400|null |Deduplication Automation|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794
|CMI6272E_1NB05_S04 |Suspected Sleeping Cell
|CMI6272E_1NB05_S04|10 |43706659 |125 |2026-07-05 03:01:04 UTC|1970-01-01 00:00:00 UTC|INC000102031988|2026-07-04 21:00:10 UTC|null |U
|CHIANG MAI1 |NORTH |MUEANG CHIANG MAI ||True-L18| |CMI6272T_2NB05 |2026-07-05|04 |4G |3 |
|07/02/2026 12:35:02|07/02/2026 09:21:48|1494631641|ericsson-enm-corba@150.236.236.58:6794|CMI8928E_1NB04_S01 |07/02/2026 09:21:48|CMI-A_0172 |CMI1000 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-02 11:24:12 UTC|CMI8928E_1NB04_S01
|LTE Cell |0 |"MATE info=""Updated by rule: Main_Flow_NOC, module: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1494631641 |No activity, recov succ
|SubNetwork=RadioNode,MeContext=CMI8928T_2NB04,ManagedElement=CMI8928T_2NB04,ENodeBFunction=1,EUtranCellFDD=CMI8928E_1NB04_S01 |GRF-Ban Roichan
|4 |Live |CMI8928|1010400|null
|Deduplication Automation|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794 |CMI8928E_1NB04_S01 |Suspected Sleeping Cell
|CMI8928E_1NB04_S01|10 |42464648 |125 |2026-07-02 10:58:37 UTC|1970-01-01 00:00:00 UTC|INC000101989282|2026-07-02 09:21:55 UTC|null |U
|CHIANG MAI1 |NORTH |HANG DONG ||True-L18| |CMI8928T_2NB04 |2026-07-02|12 |4G |3 |
|07/01/2026 06:25:02|07/01/2026 06:14:50|84577034|nokia-nsn-snmp@150.236.236.48:6791 |07/01/2026 06:14:50|NKR-A_0052 |NKR0043 |Nokia_RAN_Netact5 |ACTIVE
|0 |2026-07-01 06:15:00 UTC|NKR7011-L23
|eNode B |0 |null
|10.34.157.13 |84577034 |High PDU input voltage;6907 unitName=ASIA path=SMOD_R-1(Right) serial_no=6Q182010348 additionalFaultID=6907
|PLMN-PLMN/MRBTS-167357/LNBTS-167357

|T_NKR7011 |3 |Live |NKR701
1|300 |null |Deduplication Automation|BASE STATION NOTIFICATION
|nokia-nsn-snmp@150.236.236.48:6791 |MRBTS-167357/LNBTS-167357|BASE ST
ATION NOTIFICATION;High PDU input voltage;6907 unitName=ASIA path=/SMOD_R-1(Rig
ht) serial_no=6Q182010348 additionalFaultID=6907
|NKR7011-L23 |0 |41824677 |0 |1970-01-01 00:00:00 UTC|1
970-01-01 00:00:00 UTC|null |2026-07-01 06:14:52 UTC|null |U
|NAKHON RATCHASIMA1 |NORTHEAST|MUEANG NAKHON RATCHASIMA||RAN-4G| |MRBT
S-167357/LNBTS-167357|2026-07-01|06 |4G |3 |
|07/02/2026 19:05:02|07/02/2026 11:30:37|1494631551|ericsson-enm-corba@150.23
6.236.58:6794|CMI0170E_1NB02_S02 |07/02/2026 11:30:37|G09,G18,L07,L0
9,L18,L21,N07,U08,U09,U21 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-02 17:52:57 UTC|CMI0170E_1NB02_S02
|LTE Cell |0 |"MATE info="Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1494631551 |No activity, recov succ
|SubNetwork=RadioNode,MeContext=CMI0170T_2NB02,ManagedElement=CMI0170T_2NB02,EN
odeBFunction=1,EUtranCellFDD=CMI0170E_1NB02_S02 |HUIAI_LUK
|4 |Live |CMI0170|1010400|null |Deduplication Automat
ion|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794
|CMI0170E_1NB02_S02 |Suspected Sleeping Cell
|CMI0170E_1NB02_S02|10 |42523706 |125 |2026-07-02 17:31:58 UTC|1
970-01-01 00:00:00 UTC|INC000101995974|2026-07-02 11:30:43 UTC|null |U
|CHIANG MAI2 |NORTH |CHIANG DAO ||True-L18| |CMI017
0T_2NB02 |2026-07-02|19 |4G |3 |
|07/06/2026 21:05:02|07/06/2026 13:23:32|1505154138|ericsson-enm-corba@150.23
6.236.58:6794|CMI8530E_1NB04_S03 |07/06/2026 13:23:32|CMI-A_0461 | CM
I1000 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-06 19:54:36 UTC|CMI8530E_1NB04_S03
|LTE Cell |0 |"MATE info="Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1505154138 |No activity, recov fail
|SubNetwork=RadioNode,MeContext=CMI8530T_2NB04,ManagedElement=CMI8530T_2NB04,EN
odeBFunction=1,EUtranCellFDD=CMI8530E_1NB04_S03 |SHP-Kad Farang
|4 |Live |CMI8530|1010400|null |Deduplication Automat
ion|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794
|CMI8530E_1NB04_S03 |Suspected Sleeping Cell
|CMI8530E_1NB04_S03|10 |44467809 |125 |2026-07-06 19:24:17 UTC|1
970-01-01 00:00:00 UTC|INC000102061183|2026-07-06 13:23:40 UTC|null |U
|CHIANG MAI1 |NORTH |HANG DONG ||True-L18| |CMI853
0T_2NB04 |2026-07-06|21 |4G |3 |
|07/06/2026 06:05:02|07/05/2026 22:25:16|1502651716|ericsson-enm-corba@150.23
6.236.58:6794|CMI2699E_1NB04_S01 |07/05/2026 22:25:16|CMI-A_0386 | CM
I1000 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-06 04:53:23 UTC|CMI2699E_1NB04_S01
|LTE Cell |0 |"MATE info="Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1502651716 |No activity, recov succ
|SubNetwork=RadioNode,MeContext=CMI2699T_2NB04,ManagedElement=CMI2699T_2NB04,EN
odeBFunction=1,EUtranCellFDD=CMI2699E_1NB04_S01 |GRF-Tree site
- Mae Kampong |4 |Live |CMI2699|1010400|null

|Deduplication Automation|Suspected Sleeping Cell|ericsson-enm-corba@150.236.236.58:6794|CMI2699E_1NB04_S01|Suspected Sleeping Cell|CMI2699E_1NB04_S01|10|44147334|125|2026-07-06 04:26:18 UTC|1970-01-01 00:00:00 UTC|INC000102047168|2026-07-05 22:25:22 UTC|null|U|CHIANG MAI|NORTH|MAE ON||True-L18|CMI2699T_2NB04|2026-07-06|06|4G|3|07/07/2026 18:35:03|07/07/2026 10:53:59|1507022877|ericsson-enm-corba@150.236.236.58:6794|CMIC117E_1NB05_R01|07/07/2026 10:53:59|null|True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE|0|2026-07-07 17:22:57 UTC|CMIC117E_1NB05_R01|LTE Cell|0|"MATE info=""Updated by rule: Main_Flow_NOC, module: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTAC-OF-TH_NORTH_ALL.""|10.11.5.8|1507022877|No activity, recov succ|SubNetwork=RadioNode,MeContext=CMI0947T_2NB05,ManagedElement=CMI0947T_2NB05,ENodeBFunction=1,EUtranCellFDD=CMIC117E_1NB05_R01|BAN NONG KEAW PONIT 2|4|Live|CMIC117|1010400|null|Deduplication Automation|Suspected Sleeping Cell|ericsson-enm-corba@150.236.236.58:6794|CMIC117E_1NB05_R01|Suspected Sleeping Cell|CMIC117E_1NB05_R01|10|44969307|125|2026-07-07 16:54:57 UTC|1970-01-01 00:00:00 UTC|INC000102084833|2026-07-07 10:54:04 UTC|null|U|CHIANG MAI|NORTH|CHIANG DAO||True-L18|CMI0947T_2NB05|2026-07-07|18|4G|3|07/02/2026 12:10:02|07/02/2026 09:56:19|1694767035095|SRT0057R_1NB01_S03|zte-ume-corba-3gpp@150.236.236.112:6794|07/02/2026 09:56:19|SRT-A_0042|SRT8361|True-ZTE-UME-5G-UPPER-SOUTH|ACTIVE|1|2026-07-02 10:05:44 UTC|SRT0057R_1NB01_S03|LTE Cell|0|"MATE info=""Updated by rule: Main_Rule_New, module: Multivendor_Unified_2345G_Site_Down_DTAC_TRUE_Rule, instance id: 783, instance name: DTAC-OF-TH_SOUTH_WEST_CENTRAL_ALL.""|10.50.128.45|1694767035095|"AdditionalText=""OFFICEID:12656;OFFICENAME:SRT0057X_2NB01;OBJECTTYPE:CELL;OBJECTID:32;OBJECTNAME:SRT0057R_1NB01_S03;BOARDTYPE:#Remark:G BSCId: 3607; U RNCId: 2673; Cell type: normal cell,Cell ID: 32.LTE eNBId:12656.Location: rack=1,shelf=1,board=1.#AID:1765222293"";RootCauseIndicator=""Aid:1765222293@Restype:LTE-FDD@Position:ENBCUCPFunction=12656,CULTE=1,CUEUtranCellFDDLTE=32@Umeid:umeid"";ManagedObjectClass=""CUEUtranCellFDDLTE""|DC=www.zte.com.cn,SubNetwork=ZTE_UTRAN_SYSTEM,SubNetwork=2672,ManagedElement=ITBBU_12656,ENBCUCPFunction=12656,CULTE=1,CUEUtranCellFDDLTE=32|SHP/Mobys Restaurant|5|Live|SRT0057|1010792|null|Deduplication Automation|LTE cell outage|zte-ume-corba-3gpp@150.236.236.112:6794|SRT0057R_1NB01_S03|LTE cell outage(198094419)|SRT0057R_1NB01_S03|5|689578784|125|2026-07-02 10:05:29 UTC|1970-01-01 00:00:00 UTC|INC000101987474|2026-07-02 09:56:24 UTC|null|U|SURAT THANI|SOUTH|KO SAMUI||True-L18|SRT0057X_2NB01|2026-07-02|12|4G|3|07/06/2026 00:05:02|07/05/2026 16:23:13|1502651392|ericsson-enm-corba@150.236.236.58:6794|CMI2013E_1NB04_S03|07/05/2026 16:23:13|CMI-A_0298|CMI1000|True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE|0|2026-07-05 22:52:16 UTC|CMI2013E_1NB04_S03|LTE Cell|0|"MATE info=""Updated by rule: Main_Flow_NOC, module: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTAC-OF-TH_NORTH_ALL.""|10.11.5.8|1502651392|No activity, recov succ|SubNetwork=RadioNode,MeContext=CMI2013T_2NB04,ManagedElement=CMI2013T_2NB04,EN

odeBFunction=1,EUtranCellFDD=CMI2013E_1NB04_S03 |Doi Suthep
|4 |Live |CMI2013|1010400|null |Deduplication Automat
ion|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794
|CMI2013E_1NB04_S03 |Suspected Sleeping Cell
|CMI2013E_1NB04_S03|10 |44076182 |125 |2026-07-05 22:24:04 UTC|1
970-01-01 00:00:00 UTC|INC000102043638|2026-07-05 16:23:19 UTC|null |U
|CHIANG MAI1 |NORTH |T_Retain_FAC ||True-L18| |CMI201
3T_2NB04 |2026-07-06|00 |4G |3 |
|07/06/2026 00:30:03|07/06/2026 00:20:13|2093576993|ericsson-enm-corba_tdd2@15
0.236.236.15:6791|BKK7479-L23 |07/06/2026 00:20:13|L07,L21,L23,N0
7,U09,U21 |Ericsson_ENM_TDD2 |ACTIVE
|0 |2026-07-06 00:20:26 UTC|null
|eNode B |0 |null
|10.28.76.14 |2093576993 |Contact next level of support and provide log f
iles required by Data Collection Guideline.
|SubNetwork=0NRM_ROOT_M0,SubNetwork=RadioNodeTDD,MeContext=BKK7479-L23,ManagedE
lement=BKK7479-L23,Equipment=1,FieldReplaceableUnit=RRU-1 |THE ICON SIAM
DEPARTMENT AND CARPARK ZONE|3 |Live |BKK7479|1010400|null
|Deduplication Automation|SW Error |ericsson-enm-cor
ba_tdd2@150.236.236.15:6791 |BKK7479-L23 |SW Error
|BKK7479-L23 |0 |44172070 |0 |1970-01-01 00:00:00 UTC|1
970-01-01 00:00:00 UTC|null |2026-07-06 00:20:19 UTC|null |U
|BANGKOK WEST2 |BANGKOK |KHLONG SAN ||RAN-4G| |4G-L23
|2026-07-06|00 |4G |3 |
|07/06/2026 12:55:02|07/06/2026 12:14:40|1694767101718|L23-PKT7006-3C|zte-ume-c
orba-3gpp@150.236.236.112:6794 |07/06/2026 12:14:40|G09,L07,L09,L18,L
21,L23,L26,N07,N26,U08,U09,U21 |True-ZTE-UME-5G-UPPER-SOUTH |ACTIVE
|1 |2026-07-06 12:40:23 UTC|L23-PKT7006-3C
|LTE Cell |0 |"MATE info="Updated by rule: Main_Rule_New, mod
ule: Multivendor_Unified_2345G_Site_Down_DTAC_TRUE_Rule, instance id: 783, inst
ance name: DTAC-OF-TH_SOUTH_WEST_CENTRAL_ALL.""
|10.50.128.45 |1694767101718|"AdditionalText="OFFICEID:97326;OFFICENAME:PKT
7006X_2NB02;OBJECTTYPE:CELL;OBJECTID:196;OBJECTNAME:L23-PKT7006-3C;BOARDTYPE:#R
emark:Cell type: normal cell,70001000,Cell ID: 196.LTE eNBId:97326.Location: ra
ck=1,shelf=1,board=1.#AID:1765393924"";RootCauseIndicator="Aid:1765393924@Rest
ype:LTE-TDD@Position:ENBCUCPFunction=97326,CULTE=1,CUEUtranCellTDDLTE=196@Umei
d:umeid"";ManagedObjectClass="CUEUtranCellTDDLTE""
|DC=www.zte.com.cn,SubNetwork=ZTE_UTRAN_SYSTEM,SubNetwork=2673,ManagedElement=I
TBBU_97326,ENBCUCPFunction=97326,CULTE=1,CUEUtranCellTDDLTE=196 |T_L23-PKT
|5 |Live |L23-PKT|1010792|null |Deduplication Automat
ion|LTE cell outage |zte-ume-corba-3gpp@150.236.236.112:679
4 |L23-PKT7006-3C |LTE cell outage(198094832)
|L23-PKT7006-3C |5 |691617127 |125 |2026-07-06 12:40:51 UTC|1
970-01-01 00:00:00 UTC|INC000102055436|2026-07-06 12:14:47 UTC|null |U
|PHUKET |SOUTH |THALANG ||True-L23| |PKT700
6X_2NB02 |2026-07-06|12 |4G |3 |
|07/02/2026 03:35:02|07/01/2026 19:51:01|1492699748|ericsson-enm-corba@150.23
6.236.58:6794|CMI2013E_1NB04_S01 |07/01/2026 19:51:01|CMI-A_0298 |CM
I1000 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-02 02:22:01 UTC|CMI2013E_1NB04_S01
|LTE Cell |0 |"MATE info="Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1492699748 |Preambles, no activity, recov fail

|SubNetwork=RadioNode,MeContext=CMI2013T_2NB04,ManagedElement=CMI2013T_2NB04,ENodeBFunction=1,EUtranCellFDD=CMI2013E_1NB04_S01|Doi Suthep
|4|Live|CMI2013|1010400|null|Deduplication Automation|Suspected Sleeping Cell|ericsson-enm-corba@150.236.236.58:6794
|CMI2013E_1NB04_S01|Suspected Sleeping Cell
|CMI2013E_1NB04_S01|10|42162079|125|2026-07-02 01:51:59 UTC|1970-01-01 00:00:00 UTC|INC000101978993|2026-07-01 19:51:08 UTC|null|U
|CHIANG MAI1|NORTH|T_Retain_FAC||True-L18|CMI2013T_2NB04|2026-07-02|03|4G|3|
|07/03/2026 08:05:02|07/03/2026 00:22:01|1495405825|ericsson-enm-corba@150.236.236.58:6794|CMI7423E_2NB03_S05|07/03/2026 00:22:01|G09,L09,L18,L21,L23,U08,U21|True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0|2026-07-03 06:55:36 UTC|CMI7423E_2NB03_S05
|LTE Cell|0|"MATE info="Updated by rule: Main_Flow_NOC, module: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA C-OF-TH_NORTH_ALL.""
|10.11.5.8|1495405825|No activity, recov fail
|SubNetwork=RadioNode,MeContext=CMI7423T_2NB03,ManagedElement=CMI7423T_2NB03,ENodeBFunction=1,EUtranCellFDD=CMI7423E_2NB03_S05|KAD DHARA DHEVI
|4|Live|CMI7423|1010400|null|Deduplication Automation|Suspected Sleeping Cell|ericsson-enm-corba@150.236.236.58:6794|CMI7423E_2NB03_S05|Suspected Sleeping Cell
|CMI7423E_2NB03_S05|10|42767384|125|2026-07-03 06:22:47 UTC|1970-01-01 00:00:00 UTC|INC000102005116|2026-07-03 00:22:07 UTC|null|U
|CHIANG MAI1|NORTH|MUEANG CHIANG MAI||True-L21|CMI7423T_2NB03|2026-07-03|08|4G|3|
|07/02/2026 10:45:02|07/02/2026 06:53:09|MATE28834|N1781979628606|786at153.88.78.32|07/02/2026 06:53:09|SSK-A_0022|SSK1332|TRUE-HUAWEI-UTRAN-NEA2|ACTIVE
|0|2026-07-02 07:46:23 UTC|6007/MediationService0805/10.11.7.139/10.140.172.28/NULL|eNode B|0|"Atdelay="900";Current_time="1782975793";LocationFirstOccurence="null";LocationLastOccurence="null";MATE info="Updated by rule: CHILD_CLEAR, module: DTAC_ALL_CHILD_TO_SYN_CLEAR_RULE, instance id: 786, instance name: DTAC-OF-TH_NORTHEAST_ALL.";MATEATTN="TRUE-HUAWEI-UTRAN-NEA2 SSK7270Q_9NB01_CSL Fault_260702 1356_260702 1356 | SSK7270Q_9NB01_S03_GSM Cell out of Service_260702 1356_260702 1356 | SSK7270Q_9NB01_S02_GSM Cell out of Service_260702 1356_260702 1356 | SSK7270Q_9NB01_S01_GSM Cell out of Service_260702 1356_260702 1356 | SSK7270P_9NB01_NE Is Disconnected_260702 1353_260702 1353 | _BN EMS Alarm NE Communication Failure_260702 1355";NMX_Additional="null";TS="CPE-SSK7270 | CPE-SSK7270 | SSK7270P_9NB01 | SSK7270Q_9NB01_S01 | SSK7270Q_9NB01_S02 | SSK7270Q_9NB01_S03 | SSK7270P_9NB01";SS="TRUE";Ci_attach="TRUE";ChildClearTime="20260702 07:46:21"
|10.11.7.6|39160865|eNodeBId=129197, gNodeBId=565657|alarmName: NE Is Disconnected | alarmLocation: neName=SSK7270P_9NB01, neIP=10.140.172.28, neBackupIP=NULL, NE error port=6007, NE error code=N/A, Error message=Network abnormal or protocol error, Mediation partition=MediationService0805, OMC IP=10.11.7.139|appendInfo: Causes=The possible cause is that the network is unreachable or the peer end does not respond to the connection setup request, Advice=Click the cause and suggestion hyperlink of the alarm to open the help docu|L09,G09
|GRF/Ban Nong Koo|4|Live|SSK7270|0|null|MATE|HUAWEI_SITE_DOWN_IPRAN_NODE_DOWN_N_NMX|huawei-u2000-corba-3gpp@150.236.236.113:6791|SSK7270P_9NB01|RE
[2] SSK7270 [A71_ORIGIN]:HUAWEI DOWN IPRAN NODE DOWN

|SSK7270P_9NB01 |10 |42405213 |2127 |2026-07-02 07:21:47 UTC|1
970-01-01 00:00:00 UTC|INC000101983765|2026-07-02 07:03:13 UTC|null |U
|MATE_SI SA KET |NORTHEAST|HUA1 THAP THAN |null |null
|2026-07-02|10 |4G |3 |
|07/02/2026 02:05:02|07/01/2026 19:56:06|1492850502|ericsson-enm-corba@150.23
6.236.58:6794|CMI7430E_1NB03_S03 |07/01/2026 19:56:06|G09,L07,L09,L1
8,L21,N07,U09,U21 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-02 00:54:35 UTC|CMI7430E_1NB03_S03
|LTE Cell |0 |"MATE info=""Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1492850502 |No activity, recov fail
|SubNetwork=RadioNode,MeContext=CMI7430T_2NB03,ManagedElement=CMI7430T_2NB03,EN
odeBFunction=1,EUtranCellFDD=CMI7430E_1NB03_S03 |T_CMI9304
|4 |Live |CMI7430|1010400|null |Deduplication Automat
ion|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794
|CMI7430E_1NB03_S03 |Suspected Sleeping Cell
|CMI7430E_1NB03_S03|10 |42163790 |125 |2026-07-02 00:20:59 UTC|1
970-01-01 00:00:00 UTC|INC000101978219|2026-07-01 19:56:13 UTC|null |U
|CHIANG MAI1 |NORTH |MUEANG CHIANG MAI ||True-L18| |CMI743
0T_2NB03 |2026-07-02|02 |4G |3 |
|07/07/2026 13:35:02|07/07/2026 05:54:54|1507000101|ericsson-enm-corba@150.23
6.236.58:6794|CMI7232E_1NB04_S01 |07/07/2026 05:54:54|CMI-A_0377 |CM
I1000 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-07 12:25:14 UTC|CMI7232E_1NB04_S01
|LTE Cell |0 |"MATE info=""Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1507000101 |No activity, recov fail
|SubNetwork=RadioNode,MeContext=CMI7232T_2NB04,ManagedElement=CMI7232T_2NB04,EN
odeBFunction=1,EUtranCellFDD=CMI7232E_1NB04_S01 |T_CMI7232
|4 |Live |CMI7232|1010400|null |Deduplication Automat
ion|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794
|CMI7232E_1NB04_S01 |Suspected Sleeping Cell
|CMI7232E_1NB04_S01|10 |44810669 |125 |2026-07-07 11:55:49 UTC|1
970-01-01 00:00:00 UTC|INC000102079210|2026-07-07 05:55:01 UTC|null |U
|CHIANG MAI1 |NORTH |MUEANG CHIANG MAI ||True-L18| |CMI723
2T_2NB04 |2026-07-07|13 |4G |3 |
|07/01/2026 18:05:03|07/01/2026 10:22:17|1492431836|ericsson-enm-corba@150.23
6.236.58:6794|CMI6272E_1NB05_S03 |07/01/2026 10:22:17|G09,L07,L09,L1
8,L21,L23,N07,U08,U09,U21 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-01 16:53:17 UTC|CMI6272E_1NB05_S03
|LTE Cell |0 |"MATE info=""Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1492431836 |No activity, recov fail
|SubNetwork=RadioNode,MeContext=CMI6272T_2NB05,ManagedElement=CMI6272T_2NB05,EN
odeBFunction=1,EUtranCellFDD=CMI6272E_1NB05_S03 |null
|4 |Live |CMI6272|1010400|null |Deduplication Automat
ion|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794
|CMI6272E_1NB05_S03 |Suspected Sleeping Cell
|CMI6272E_1NB05_S03|10 |41947583 |125 |2026-07-01 16:23:08 UTC|1
970-01-01 00:00:00 UTC|INC000101973352|2026-07-01 10:22:25 UTC|null |U
|CHIANG MAI1 |NORTH |MUEANG CHIANG MAI ||True-L18| |CMI627

2T_2NB05 |2026-07-01|18 |4G |3 |
|07/02/2026 10:10:03|07/02/2026 08:28:48|84673508|nokia-nsn-snmp@150.236.236.4
8:6791 |07/02/2026 08:28:48|SSK-A_0089 |SSK
1332 |Nokia_RAN_Netact5 |ACTIVE
|0 |2026-07-02 09:57:31 UTC|L23-SSK6702-2D
|eNode B |0 |null
|10.34.157.13 |84673508 |Remote Interference avoidance is active and DwP
TS rollback as SSP5;6127 additionalFaultID=6127 LNCEL-187
|PLMN-PLMN/MRBTS-98847/LNBTS-98847/LNCEL-187
|GRF/Ban Duan Yai |3 |Live |SSK670
2|300 |null |Deduplication Automation|CELL NOTIFICATION
|nokia-nsn-snmp@150.236.236.48:6791 |MRBTS-98847/LNBTS-98847 |CELL NO
TIFICATION;Remote Interference avoidance is active and DwPTS rollback as SSP5;6
127 additionalFaultID=6127 LNCEL-187
|SSK6702-L23 |0 |42441234 |0 |1970-01-01 00:00:00 UTC|1
970-01-01 00:00:00 UTC|null |2026-07-02 08:28:50 UTC|null |U
|SI SA KET |NORTHEAST|WANG HIN ||RAN-4G| |MRBT
S-98847/LNBTS-98847 |2026-07-02|10 |4G |3 |
|07/05/2026 02:35:02|07/04/2026 19:02:49|1500106649|ericsson-enm-corba@150.23
6.236.58:6794|MHS0026E_1NB03_S01 |07/04/2026 19:02:49|G09,L07,L09,L1
8,L21,L23,L26,N07,N26,U09,U21 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-05 01:22:01 UTC|MHS0026E_1NB03_S01
|LTE Cell |0 |"MATE info=""Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1500106649 |No activity, recov succ
|SubNetwork=RadioNode,MeContext=MHS0026T_2NB03,ManagedElement=MHS0026T_2NB03,EN
odeBFunction=1,EUtranCellFDD=MHS0026E_1NB03_S01 |null
|4 |Live |MHS0026|1010400|null |Deduplication Automat
ion|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794
|MHS0026E_1NB03_S01 |Suspected Sleeping Cell
|MHS0026E_1NB03_S01|10 |43684766 |125 |2026-07-05 01:03:51 UTC|1
970-01-01 00:00:00 UTC|INC000102031161|2026-07-04 19:02:58 UTC|null |U
|MAE HONG SON |NORTH |D_Retain_FAC ||True-L18| |MHS002
6T_2NB03 |2026-07-05|02 |4G |3 |
|07/04/2026 02:05:02|07/03/2026 18:25:38|1497530812|ericsson-enm-corba@150.23
6.236.58:6794|LPN6769E_1NB03_S01 |07/03/2026 18:25:38|LPN-B_0460 |LP
N3052 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-04 00:54:30 UTC|LPN6769E_1NB03_S01
|LTE Cell |0 |"MATE info=""Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1497530812 |No activity, recov succ
|SubNetwork=RadioNode,MeContext=LPN6769T_2NB03,ManagedElement=LPN6769T_2NB03,EN
odeBFunction=1,EUtranCellFDD=LPN6769E_1NB03_S01 |3G_CP All Lamp
hon |4 |Live |LPN6769|1010400|null
|Deduplication Automation|Suspected Sleeping Cell |ericsson-enm-cor
ba@150.236.236.58:6794 |LPN6769E_1NB03_S01 |Suspected Sleeping Cell
|LPN6769E_1NB03_S01|10 |43196882 |125 |2026-07-04 00:26:39 UTC|1
970-01-01 00:00:00 UTC|INC000102016824|2026-07-03 18:25:43 UTC|null |U
|LAMPHUN |NORTH |T_Retain_FAC ||True-L18| |LPN676
9T_2NB03 |2026-07-04|02 |4G |3 |
|07/04/2026 10:35:03|07/04/2026 02:52:34|1498359412|ericsson-enm-corba@150.23
6.236.58:6794|CMI6741E_1NB03_S01 |07/04/2026 02:52:34|MHS-B_0170 |MH

S(CMI1000) |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-04 09:23:28 UTC|CMI6741E_1NB03_S01
|LTE Cell |0 |"MATE info=""Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1498359412 |No activity, recov fail
|SubNetwork=RadioNode,MeContext=CMI6741T_2NB03,ManagedElement=CMI6741T_2NB03,EN
odeBFunction=1,EUtranCellFDD=CMI6741E_1NB03_S01 |GRF-Muangket
|4 |Live |CMI6741|1010400|null |Deduplication Automat
ion|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794
|CMI6741E_1NB03_S01 |Suspected Sleeping Cell
|CMI6741E_1NB03_S01|10 |43338419 |125 |2026-07-04 08:53:41 UTC|1
970-01-01 00:00:00 UTC|INC000102022491|2026-07-04 02:52:40 UTC|null |U
|CHIANG MAI |NORTH |MAE TAENG ||True-L18| |CMI674
1T_2NB03 |2026-07-04|10 |4G |3 |
|07/05/2026 20:30:03|07/05/2026 20:12:59|1502808238|ericsson-enm-corba@150.23
6.236.58:6794|CMI0084T_2NB03_S01 |07/05/2026 20:12:59|D18,G09,G18,L0
7,L09,L18,L21,L23,L26,N07,N26,U08,U0|True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-05 20:20:04 UTC|CMI0084T_2NB03_S01
|LTE Cell |0 |null
|10.11.5.8 |1502808238 |Preambles, no activity, recov succ
|SubNetwork=RadioNode,MeContext=CMI0084T_2NB03,ManagedElement=CMI0084T_2NB03,EN
odeBFunction=1,EUtranCellFDD=CMI0084T_2NB03_S01 |DOI_INTANON
|4 |Live |CMI0084|1010400|null |Deduplication Automat
ion|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794
|CMI0084T_2NB03_S01 |Suspected Sleeping Cell
|CMI0084T_2NB03_S01|0 |44122719 |0 |1970-01-01 00:00:00 UTC|1
970-01-01 00:00:00 UTC|null |2026-07-05 20:13:04 UTC|null |U
|CHIANG MAI1 |NORTH |D_Retain_FAC ||True-L21| |CMI008
4T_2NB03 |2026-07-05|20 |4G |3 |
|07/04/2026 08:35:02|07/04/2026 01:00:25|1497900534|ericsson-enm-corba@150.23
6.236.58:6794|CMIC047E_1NB03_S02 |07/04/2026 01:00:25|G09,L07,L09,L1
8,L21,N07,U09,U21 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-04 07:23:02 UTC|CMIC047E_1NB03_S02
|LTE Cell |0 |"MATE info=""Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1497900534 |No activity, recov fail
|SubNetwork=RadioNode,MeContext=CMIC047T_2NB03,ManagedElement=CMIC047T_2NB03,EN
odeBFunction=1,EUtranCellFDD=CMIC047E_1NB03_S02 |null
|4 |Live |CMIC047|1010400|null |Deduplication Automat
ion|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794
|CMIC047E_1NB03_S02 |Suspected Sleeping Cell
|CMIC047E_1NB03_S02|10 |43292872 |125 |2026-07-04 07:01:24 UTC|1
970-01-01 00:00:00 UTC|INC000102021906|2026-07-04 01:00:31 UTC|null |U
|CHIANG MAI1 |NORTH |D_Retain_FAC ||True-L18| |CMIC04
7T_2NB03 |2026-07-04|08 |4G |3 |
|07/06/2026 20:55:02|07/06/2026 13:11:24|1504760466|ericsson-enm-corba@150.23
6.236.58:6794|CMI0083E_1NB02_S03 |07/06/2026 13:11:24|D18,G09,G18,L0
7,L09,L18,L21,L23,N07,U08,U09,U21 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-06 19:42:26 UTC|CMI0083E_1NB02_S03
|LTE Cell |0 |"MATE info=""Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""

|10.11.5.8 |1504760466 |No activity, recov succ
|SubNetwork=RadioNode,MeContext=CMI0083T_2NB02,ManagedElement=CMI0083T_2NB02,ENodeBFunction=1,EUtranCellFDD=CMI0083E_1NB02_S03 |KRITSADAD0I
|4 |Live |CMI0083|1010400|null |Deduplication Automation|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794
|CMI0083E_1NB02_S03 |Suspected Sleeping Cell
|CMI0083E_1NB02_S03|10 |44463630 |125 |2026-07-06 19:12:20 UTC|1970-01-01 00:00:00 UTC|INC000102061261|2026-07-06 13:11:31 UTC|null |U
|CHIANG MAI1 |NORTH |HANG DONG ||True-L18| |CMI0083T_2NB02 |2026-07-06|20 |4G |3 |
|07/01/2026 01:35:02|06/30/2026 17:54:42|1490166792|ericsson-enm-corba@150.236.236.58:6794|CMI6272E_1NB05_S01 |06/30/2026 17:54:42|G09,L07,L09,L18,L21,L23,N07,U08,U09,U21 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-01 00:21:56 UTC|CMI6272E_1NB05_S01
|LTE Cell |0 |"MATE info="Updated by rule: Main_Flow_NOC, module: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1490166792 |No activity, recov fail
|SubNetwork=RadioNode,MeContext=CMI6272T_2NB05,ManagedElement=CMI6272T_2NB05,ENodeBFunction=1,EUtranCellFDD=CMI6272E_1NB05_S01 |null
|4 |Live |CMI6272|1010400|null |Deduplication Automation|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794
|CMI6272E_1NB05_S01 |Suspected Sleeping Cell
|CMI6272E_1NB05_S01|10 |41562840 |125 |2026-06-30 23:55:48 UTC|1970-01-01 00:00:00 UTC|INC000101960026|2026-06-30 17:54:49 UTC|null |U
|CHIANG MAI1 |NORTH |MUEANG CHIANG MAI ||True-L18| |CMI6272T_2NB05 |2026-07-01|01 |4G |3 |
|07/06/2026 20:05:03|07/06/2026 12:27:03|1504721573|ericsson-enm-corba@150.236.236.58:6794|CMI7237E_2NB01_S01 |07/06/2026 12:27:03|CMI-A_0458 |CMI1000 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-06 18:54:53 UTC|CMI7237E_2NB01_S01
|LTE Cell |0 |"MATE info="Updated by rule: Main_Flow_NOC, module: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1504721573 |Preambles, no activity, recov succ
|SubNetwork=RadioNode,MeContext=CMI7237X_2NB01,ManagedElement=CMI7237X_2NB01,ENodeBFunction=1,EUtranCellFDD=CMI7237E_2NB01_S01 |T_CMI7237
|4 |Live |CMI7237|1010400|null |Deduplication Automation|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794
|CMI7237E_2NB01_S01 |Suspected Sleeping Cell
|CMI7237E_2NB01_S01|10 |44448320 |125 |2026-07-06 18:27:46 UTC|1970-01-01 00:00:00 UTC|INC000102061053|2026-07-06 12:27:10 UTC|null |U
|CHIANG MAI1 |NORTH |MUEANG CHIANG MAI ||True-L21| |CMI7237X_2NB01 |2026-07-06|20 |4G |3 |
|07/04/2026 23:25:02|07/04/2026 15:39:50|1499922982|ericsson-enm-corba@150.236.236.58:6794|LPN8637E_1NB04_S02 |07/04/2026 15:39:50|LPN-B_0258 |LPN3052 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-04 22:12:02 UTC|LPN8637E_1NB04_S02
|LTE Cell |0 |"MATE info="Updated by rule: Main_Flow_NOC, module: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1499922982 |No activity, recov fail
|SubNetwork=RadioNode,MeContext=LPN8637T_2NB04,ManagedElement=LPN8637T_2NB04,ENodeBFunction=1,EUtranCellFDD=LPN8637E_1NB04_S02 |Relocate from:

GRF/Ban Nongnuek |4 |Live |LPN8637|1010400|null
|Deduplication Automation|Suspected Sleeping Cell |ericsson-enm-cor
ba@150.236.236.58:6794 |LPN8637E_1NB04_S02 |Suspected Sleeping Cell
|LPN8637E_1NB04_S02|10 |43645824 |125 |2026-07-04 21:40:50 UTC|1
970-01-01 00:00:00 UTC|INC000102029862|2026-07-04 15:39:55 UTC|null |U
|LAMPHUN |NORTH |T_Retain_FAC ||True-L18| |LPN863
7T_2NB04 |2026-07-04|23 |4G |3 |
|07/03/2026 20:35:02|07/03/2026 12:20:48|1497237879|ericsson-enm-corba@150.23
6.236.58:6794|CMI6889P_9NB01_R02 |07/03/2026 12:20:48|G09,U08
|True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE |0 |2026-07-03
19:22:57 UTC|CMI6889P_9NB01_R02 |LTE Cell
|0 |"MATE info="Updated by rule: Main_Flow_NOC, module: Eric_2345G_HW_Al
arm_Correlation_Rule, instance id: 787, instance name: DTAC-OF-TH_NORTH_ALL."
|10.11.5.8 |1497237879 |Preambles, no activity, recov succ
|SubNetwork=RadioNode,MeContext=CMI6838T_2NB01,ManagedElement=CMI6838T_2NB01,EN
odeBFunction=1,EUtranCellFDD=CMI6889P_9NB01_R02 |MeaYaWaterfall
Chomtong |4 |Live |CMI6889|1010400|null
|Deduplication Automation|Suspected Sleeping Cell |ericsson-enm-cor
ba@150.236.236.58:6794 |CMI6889P_9NB01_R02 |Suspected Sleeping Cell
|CMI6889P_9NB01_R02|10 |43090931 |125 |2026-07-03 18:24:26 UTC|1
970-01-01 00:00:00 UTC|INC000102013446|2026-07-03 12:20:55 UTC|null |U
|CHIANG MAI1 |NORTH |CHOM THONG ||True-L09| |CMI683
8T_2NB01 |2026-07-03|20 |4G |3 |
|07/07/2026 05:10:02|07/07/2026 04:19:23|85004847|nokia-nsn-snm@150.236.236.4
8:6791 |07/07/2026 04:19:23|KKN-A_0063 |KKN
0030 |Nokia_RAN_Netact5 |ACTIVE
|0 |2026-07-07 05:00:15 UTC|KKN7900-L23
|eNode B |0 |null
|10.34.157.13 |85004847 |Transport layer connection failure in S1 interf
ace;6202 srcIPAddr=10.223.113.214;dstIPAddr=10.17.252.240;supplDstnIPAddr=10.1
7.253.240;supplAlarmInfo=Transport Network Identifier: 0 additionalFaultID=6202
LNMME-6
|PLMN-PLMN/MRBTS-177827/LNBTS-177827/LNMME-6
|T_KKN7900 |4 |Live |KKN790
0|300 |null |Deduplication Automation|BASE STATION CONNECTIVITY DEGRA
DED |nokia-nsn-snm@150.236.236.48:6791 |MRBTS-177827/LNBTS-177827|BA
SE STATION CONNECTIVITY DEGRADED;Transport layer connection failure in S1 inter
face;6202 srcIPAddr=10.223.113.214;dstIPAddr=10.17.252.240;supplDstnIPAddr=10.1
7.253.240;supplAlarmInfo=Transport Network Identifier: 0 additionalFaultID=6202
LNMME-6|KKN7900-L23 |0 |44780144 |0 |1970-01-01 00:00:0
0 UTC|1970-01-01 00:00:00 UTC|null |2026-07-07 05:00:14 UTC|null
|U |KHON KAEN |NORTHEAST|MUEANG KHON KAEN ||RAN-4G|
|MRBTS-177827/LNBTS-177827|2026-07-07|05 |4G |3 |
|07/01/2026 04:35:03|06/30/2026 20:54:58|1490288645|ericsson-enm-corba@150.23
6.236.58:6794|MHS0043E_1NB02_S03 |06/30/2026 20:54:58|G09,L07,L09,L1
8,L21,N07,U09,U21 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-01 03:22:14 UTC|MHS0043E_1NB02_S03
|LTE Cell |0 |"MATE info="Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL."
|10.11.5.8 |1490288645 |No activity, recov fail
|SubNetwork=RadioNode,MeContext=MHS0043T_2NB02,ManagedElement=MHS0043T_2NB02,EN
odeBFunction=1,EUtranCellFDD=MHS0043E_1NB02_S03 |null
|4 |Live |MHS0043|1010400|null |Deduplication Automat

ion|Suspected Sleeping Cell|ericsson-enm-corba@150.236.236.58:6794
|MHS0043E_1NB02_S03|Suspected Sleeping Cell
|MHS0043E_1NB02_S03|10|41622353|125|2026-07-01 02:55:33 UTC|1
970-01-01 00:00:00 UTC|INC000101962120|2026-06-30 20:55:04 UTC|null|U
|MAE HONG SON|NORTH|D_Retain_FAC||True-L18| |MHS004
3T_2NB02|2026-07-01|04|4G|3|
|07/06/2026 04:35:02|07/05/2026 20:49:59|1502617868|ericsson-enm-corba@150.23
6.236.58:6794|CMI7232E_1NB04_S01|07/05/2026 20:49:59|CMI-A_0377|CM
I1000|True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0|2026-07-06 03:23:09 UTC|CMI7232E_1NB04_S01
|LTE Cell|0|"MATE info="Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8|1502617868|No activity, recov fail
|SubNetwork=RadioNode,MeContext=CMI7232T_2NB04,ManagedElement=CMI7232T_2NB04,EN
odeBFunction=1,EUtranCellFDD=CMI7232E_1NB04_S01|T_CMI7232
|4|Live|CMI7232|1010400|null|Deduplication Automat
ion|Suspected Sleeping Cell|ericsson-enm-corba@150.236.236.58:6794
|CMI7232E_1NB04_S01|Suspected Sleeping Cell
|CMI7232E_1NB04_S01|10|44129373|125|2026-07-06 02:50:36 UTC|1
970-01-01 00:00:00 UTC|INC000102045960|2026-07-05 20:50:07 UTC|null|U
|CHIANG MAI1|NORTH|MUEANG CHIANG MAI||True-L18| |CMI723
2T_2NB04|2026-07-06|04|4G|3|
|07/01/2026 19:10:03|07/01/2026 19:01:35|84633894|nokia-nsn-snm@150.236.236.4
8:6791|07/01/2026 19:01:35|L07,L21,L23,N0
7,U09,U21|Nokia_RAN_Netact5|ACTIVE
|0|2026-07-01 19:01:47 UTC|BRR0045-L23
|eNode B|0|null
|10.34.157.13|84633894|S1 SCTP path failure;6204 srcIPAddr=10.237.19
8.77;dstIPAddr=10.91.55.1;supplDstnIPAddr=;supplAlarmInfo=Transport Network Ide
ntifier: 0 additionalFaultID=6204 LNMME-1
|PLMN-PLMN/MRBTS-400959/LNBTS-400959/LNMME-1
|null|4|Live|BRR004
5|300|null|Deduplication Automation|BASE STATION CONNECTIVITY DEGRA
DED|nokia-nsn-snm@150.236.236.48:6791|MRBTS-400959/LNBTS-400959|BA
SE STATION CONNECTIVITY DEGRADED;S1 SCTP path failure;6204 srcIPAddr=10.237.19
8.77;dstIPAddr=10.91.55.1;supplDstnIPAddr=;supplAlarmInfo=Transport Network Ide
ntifier: 0 additionalFaultID=6204 LNMME-1
|BRR0045-L23|0|42139209|0|1970-01-01 00:00:00 UTC|1
970-01-01 00:00:00 UTC|null|2026-07-01 19:01:38 UTC|null|U
|BURI RAM|NORTHEAST|NON DIN DAENG||RAN-4G| |4G-L23
|2026-07-01|19|4G|3|
|07/01/2026 22:35:03|07/01/2026 14:56:10|1492882314|ericsson-enm-corba@150.23
6.236.58:6794|CMI7301E_2NB05_R02|07/01/2026 14:56:10|G09, L07, L21,
L23, N07, U09, U21|True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0|2026-07-01 21:25:07 UTC|CMI7301E_2NB05_R02
|LTE Cell|0|"MATE info="Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8|1492882314|No activity, recov succ
|SubNetwork=RadioNode,MeContext=CMI0243T_2NB05,ManagedElement=CMI0243T_2NB05,EN
odeBFunction=1,EUtranCellFDD=CMI7301E_2NB05_R02|Remote RRU-Chi
ang Dao Cave|4|Live|CMI7301|1010400|null
|Deduplication Automation|Suspected Sleeping Cell|ericsson-enm-cor

ba@150.236.236.58:6794 |CMI7301E_2NB05_R02 |Suspected Sleeping Cell
|CMI7301E_2NB05_R02|10 |42052554 |125 |2026-07-01 20:56:55 UTC|1
970-01-01 00:00:00 UTC|INC000101976308|2026-07-01 14:56:16 UTC|null |U
|CHIANG MAI2 |NORTH |CHIANG DAO ||True-L21| |CMI024
3T_2NB05 |2026-07-01|22 |4G |3 |
|07/01/2026 19:10:03|07/01/2026 19:01:34|84633945|nokia-nsn-snm@150.236.236.4
8:6791 |07/01/2026 19:01:34|L07,L21,L23,N0
7,U09,U21 |Nokia_RAN_Netact5 |ACTIVE
|0 |2026-07-01 19:01:48 UTC|KKN0380-L23
|eNode B |0 |null
|10.34.157.13 |84633945 |S1 SCTP path failure;6204 srcIPAddr=10.253.11
2.18;dstIPAddr=10.91.55.5;supplDstnIPAddr=;supplAlarmInfo=Transport Network Ide
ntifier: 0 additionalFaultID=6204 LNMME-5
|PLMN-PLMN/MRBTS-403089/LNBTS-403089/LNMME-5
|PTT YEAK BAN THA SALA |4 |Live |KKN038
0|300 |null |Deduplication Automation|BASE STATION CONNECTIVITY DEGRA
DED |nokia-nsn-snm@150.236.236.48:6791 |MRBTS-403089/LNBTS-403089|BA
SE STATION CONNECTIVITY DEGRADED;S1 SCTP path failure;6204 srcIPAddr=10.253.11
2.18;dstIPAddr=10.91.55.5;supplDstnIPAddr=;supplAlarmInfo=Transport Network Ide
ntifier: 0 additionalFaultID=6204 LNMME-5
|KKN0380-L23 |0 |42139343 |0 |1970-01-01 00:00:00 UTC|1
970-01-01 00:00:00 UTC|null |2026-07-01 19:01:39 UTC|null |U
|KHON KAEN |NORTHEAST|NONG RUEA ||RAN-4G| |4G-L23
|2026-07-01|19 |4G |3 |
|07/02/2026 06:45:02|07/02/2026 06:34:57|1104753452|ericsson-enm-corba@150.23
6.236.78:6796|ATG0019T_2NB01 |07/02/2026 06:37:00|G09,G18,L07,L2
1,L23,N07,U08,U09,U21 |True_Ericsson_ENM_UTRAN_CENTRAL_4A |ACTIVE
|0 |2026-07-02 06:37:21 UTC|null
|NR GNodeB - Ericsson|0 |null
|10.23.17.8 |1104753452 |Not functional carrier HW resources [DL/A DL/C
(Band28)] (Mixed Mode with other ME).
|SubNetwork=RadioNode,MeContext=ATG0019T_2NB01,ManagedElement=ATG0019T_2NB01,Eq
uipment=1,FieldReplaceableUnit=RRU-1 |WANGNAMYEN
|3 |Live |ATG0019|1409 |RETAINED |Deduplication Automat
ion|HW Partial Fault |ericsson-enm-corba@150.236.236.78:6796
|ATG0019T_2NB01 |HW Partial Fault
|ATG0019T_2NB01 |0 |689491184 |0 |1970-01-01 00:00:00 UTC|1
970-01-01 00:00:00 UTC|null |2026-07-02 06:35:04 UTC|null |U
|ANG THONG |CENTRAL |SAWAENG HA |True-NR FDD |NR FDD
|2026-07-02|06 |5G |3 |
|07/01/2026 09:55:03|07/01/2026 07:23:32|84580519|nokia-nsn-snm@150.236.236.4
8:6791 |07/01/2026 07:23:32|UDT-A_0012 | UDT
0014 |Nokia_RAN_Netact5 |ACTIVE
|1 |2026-07-01 09:25:42 UTC|L23-UDT8507-2D
|eNode B |0 |"MATE info=""Updated by rule: Main_Rule_New, mod
ule: Multivendor_Unified_2345G_Site_Down_DTAC_TRUE_Rule, instance id: 786, inst
ance name: DTAC-OF-TH_NORTHEAST_ALL.""
|10.34.157.13 |84580519 |Failure in replaceable baseband unit;4341 unitN
ame=ABIA path=/SMOD_R-1(Right)/BBMOD_R-1(Right_Middle) serial_no=K9180929626 ad
ditionalFaultID=3003
|PLMN-PLMN/MRBTS-180017/LNBTS-180017/LNCEL-187
|TBC |5 |Live |UDT850
7|300 |null |Deduplication Automation|CELL FAULTY
|nokia-nsn-snm@150.236.236.48:6791 |MRBTS-180017/LNBTS-180017|CELL FA

ULTY;Failure in replaceable baseband unit;4341 unitName=ABIA path=/SMOD_R-1(Right)
 ht)/BBMOD_R-1(Right_Middle) serial_no=K9180929626 additionalFaultID=3003
 |UDT8507-L23 |5 |41860911 |125 |2026-07-01 07:50:12 UTC|1
 970-01-01 00:00:00 UTC|INC000101966437|2026-07-01 07:23:42 UTC|null |U
 |UDON THANI |NORTHEAST|MUEANG UDON THANI ||RAN-4G| |MRBT
 S-180017/LNBTS-180017|2026-07-01|09 |4G |3 |
 |07/05/2026 12:25:02|07/05/2026 12:15:33|1502080964|ericsson-enm-corba@150.23
 6.236.58:6794|MHS6219P_9NB01 |07/05/2026 12:15:33|G09,L09,U08
 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE |0 |2026-07-05
 12:15:41 UTC|null |eNode B
 |0 |null
 |10.11.5.8 |1502080964 |Restarting LM /c/loadmodules_norepl/CXP901326
 8%15_R96JL (Mixed Mode with other ME).
 |SubNetwork=ERBS,MeContext=MHS6219P_9NB01,ManagedElement=1,Equipment=1,AuxPlugI
 nUnit=RRU-3,DeviceGroup=ru |MAE SUAT SOP M
 OEI |3 |Live |MHS6219|1010400|null
 |Deduplication Automation|General SW Error |ericsson-enm-cor
 ba@150.236.236.58:6794 |MHS6219P_9NB01 |General SW Error
 |MHS6219P_9NB01 |0 |44017194 |0 |1970-01-01 00:00:00 UTC|1
 970-01-01 00:00:00 UTC|null |2026-07-05 12:15:34 UTC|null |U
 |MAE HONG SON |NORTH |SOP MOEI |True-LTE FDD|LTE FD
 D |2026-07-05|12 |4G |3 |
 |07/05/2026 20:30:03|07/05/2026 19:50:59|718285216|ericsson-enm-corba@150.236.2
 36.17:6796|BKFD085-L23 |07/05/2026 19:50:59|L21,L23,U21
 |Ericsson_ENM_TDD3 |ACTIVE |0 |2026-07-05
 20:19:59 UTC|L23-BKFD086-2C |eNode B
 |0 |null
 |10.29.76.14 |718285216 |No activity, recov succ
 |SubNetwork=ONRM_ROOT_M0,SubNetwork=RadioNodeTDD,MeContext=BKFD085-L23,ManagedE
 lement=BKFD085-L23,ENodeBFunction=1,EUtranCellTDD=L23-BKFD086-2C|THAI AIRFORCE
 MUSIUM 2 |4 |Live |BKFD085|1010400|null
 |Deduplication Automation|Suspected Sleeping Cell |ericsson-enm-cor
 ba@150.236.236.17:6796 |BKFD085-L23 |Suspected Sleeping Cell
 |BKFD085-L23 |0 |44118395 |0 |1970-01-01 00:00:00 UTC|1
 970-01-01 00:00:00 UTC|null |2026-07-05 19:51:05 UTC|null |U
 |BANGKOK EAST1 |BANGKOK |SAI MAI |null |4G-L23
 |2026-07-05|20 |4G |3 |
 |07/02/2026 10:10:03|07/02/2026 09:12:07|84676542|nokia-nsn-snm@150.236.236.4
 8:6791 |07/02/2026 09:12:07|L23
 |Nokia_RAN_Netact5 |ACTIVE |0 |2026-07-02
 09:57:11 UTC|L23-MHRC008-3C |eNode B
 |0 |null
 |10.34.157.13 |84676542 |Remote Interference avoidance is active and DwP
 TS rollback as SSP5;6127 additionalFaultID=6127 LNCEL-196
 |PLMN-PLMN/MRBTS-404817/LNBTS-404817/LNCEL-196
 |null |3 |Live |MHRC00
 8|300 |null |Deduplication Automation|CELL NOTIFICATION
 |nokia-nsn-snm@150.236.236.48:6791 |MRBTS-404817/LNBTS-404817|CELL NO
 TIFICATION;Remote Interference avoidance is active and DwPTS rollback as SSP5;6
 127 additionalFaultID=6127 LNCEL-196
 |MHRC008-L23 |0 |42460230 |0 |1970-01-01 00:00:00 UTC|1
 970-01-01 00:00:00 UTC|null |2026-07-02 09:12:15 UTC|null |U
 |MAHA SARAKHAM |NORTHEAST|D_Retain_FAC ||RAN-4G| |4G-L23
 |2026-07-02|10 |4G |3 |

|07/05/2026 07:15:02|07/05/2026 07:04:39|1114219423|ericsson-enm-corba@150.23
6.236.78:6796|LBR0092T_2NB01 |07/05/2026 07:04:39|G09,L07,L21,N0
7,U09,U21 |True_Ericsson_ENM_UTRAN_CENTRAL_4A |ACTIVE
|0 |2026-07-05 07:04:47 UTC|null
|NR GNodeB - Ericsson|0 |null
|10.23.17.8 |1114219423 |Contact next level of support and provide log f
iles required by Data Collection Guideline.
|SubNetwork=RadioNode,MeContext=LBR0092T_2NB01,ManagedElement=LBR0092T_2NB01,EN
odeBFunction=1,SectorCarrier=L07G-S2C1 |null
|3 |Live |LBR0092|1409 |null |Deduplication Automat
ion|SW Error |ericsson-enm-corba@150.236.236.78:6796
|LBR0092T_2NB01 |SW Error
|LBR0092T_2NB01 |0 |690993053 |0 |1970-01-01 00:00:00 UTC|1
970-01-01 00:00:00 UTC|null |2026-07-05 07:04:45 UTC|null |U
|LOP BURI |CENTRAL |PHATTHANA NIKHOM |True-NR FDD |NR FDD
|2026-07-05|07 |5G |3 |
|07/07/2026 18:55:02|07/07/2026 02:43:26|MATE42467|N1781979628606|786at153.88.7
8.32 |07/07/2026 02:43:26|G09,L07,L09,L21,L
23,N07,U09,U21 |TRUE-HUAWEI-UTRAN-NEA2 |ACTIVE
|0 |2026-07-07 16:02:13 UTC|70/793/5198/NKR0329C_9NB01_S01/520-0
5-5df7-d284 |LTE Cell |0 |"Current_time2=""178339291
3"";EMS_NAME=""TRUE-HUAWEI-UTRAN-NEA2"";LocationFirstOccurence=""null"";Locatio
nLastOccurence=""null"";MATE info=""Updated by rule: ChildClear_Rule, module: U
nified_Multivendor_2345G_Route_Cell_Down_Rule, instance id: 786, instance name:
DTAC-OF-TH_NORTHEAST_ALL."";MATEATTR=""|| Incident [Route Cell Sites Down: BANH
UAISOKNOI = 6 Sites (Total 15 Base Stations: 2G-G18=0,2G-D18=0,2G-G09=4, 3G-U0
8=0,3G-U09=4,3G-U21=5, 4G-E18=0,4G-L21=0,4G-L23=2,4G-L26=0,4G-L18=0,4G-L09=0,4
G-L07=0,5G-N07=0,5G-N23=0,5G-N26=0)] at 09:43_Service Impact : 2G/3G/4G servic
es in some areas of_PAK CHONG(MATE_NAKHON RATCHASIMA1),_Status : Escalated to F
SO.||NMA0329-U21,, ,NKR0329-U09,NKR0329-U21,NKR0329-G09,,NKR0414-U09,NKR0414-U2
1,NKR0414-L23,NKR0414-G09,,NKR0235-U09,NKR0235-U21,NKR0235-G09,,NKR0319-U09,NKR
0319-U21,NKR0319-L23,NKR0319-G09,, "";NEW_SER=""SAME"";NMX_Additional=""null"";T
S=""NKR0319-L23 | NKR0319-L23 | NKR0319C_9NB01_S01 | NKR0319D_2NB01_S01 | NKR03
19D_2NB01_S02 | NKR0319D_2NB01_S03 | NKR0319D_9NB01_S01 | NKR0319D_9NB01_S02 |
NKR0319D_9NB01_S03 | NKR0319Q_9NB01 | NKR0319X_2NB01 | NKR0319C_9NB01_S01 | NKR
0319-L23 | NKR0319-L23 | NKR0319C_9NB01_S01 | NKR0319D_2NB01_S01 | NKR0319D_2NB
01_S02 | NKR0319D_2NB01_S03 | NKR0319D_9NB01_S01 | NKR0319D_9NB01_S02 | NKR0319
D_9NB01_S03 | NKR0319Q_9NB01 | NKR0319X_2NB01 | NKR0319C_9NB01_S01 | NKR0235C_9
NB01_S01 | NKR0235C_9NB01_S01 | NKR0235D_2NB01_S01 | NKR0235D_2NB01_S02 | NKR02
35D_9NB01_S01 | NKR0235D_9NB01_S02 | NKR0235Q_9NB01_S01 | NKR0235Q_9NB01_S02 |
NKR0235X_2NB01 | NKR0235Q_9NB01_S01 | NKR0414-PK-01 | NKR0414-PK-01 | NKR0414-A
C-01 | NKR0414-AG-01 | NKR0414-L23 | NKR0414C_9NB01_S01 | NKR0414D_2NB01_S01 |
NKR0414D_2NB01_S02 | NKR0414D_2NB01_S03 | NKR0414D_9NB01_S01 | NKR0414D_9NB01_S
02 | NKR0414D_9NB01_S03 | NKR0414Q_9NB01_S01 | NKR0414Q_9NB01_S02 | NKR0414Q_9N
B01_S03 | NKR0414X_2NB01 | NKR0414Q_9NB01_S02 | NKR0329C_9NB01_S01 | NKR0329C_9
NB01_S01 | NKR0329D_2NB01_S01 | NKR0329D_2NB01_S02 | NKR0329D_2NB01_S03 | NKR03
29D_9NB01_S01 | NKR0329D_9NB01_S02 | NKR0329D_9NB01_S03 | NKR0329Q_9NB01_S01 |
NKR0329Q_9NB01_S02 | NKR0329Q_9NB01_S03 | NKR0329X_2NB01 | NKR0329C_9NB01_S01 |
NMA0235-AC-01 | NMA0235-AC-01 | MRBTS-406798 | MRBTS-406798 | NMA0329-AC-01 | N
MA0329-AC-01 | NMA0329-U21 | NMA0329-UL | NMA0329-U21"";TS1=""NMA0329 | NMA0235
| NKR0329 | NKR0414 | NKR0235 | NKR0319 | NKR0319"";Ci_attach=""TRUE""|10.1
1.7.6 |40486603 |NKR0319-L23 | NKR0319-L23 | NKR0319C_9NB01_S01 | NKR
0319D_2NB01_S01 | NKR0319D_2NB01_S02 | NKR0319D_2NB01_S03 | NKR0319D_9NB01_S01
| NKR0319D_9NB01_S02 | NKR0319D_9NB01_S03 | NKR0319Q_9NB01 | NKR0319X_2NB01 | N

NKR0319C_9NB01_S01 | NKR0319-L23 | NKR0319-L23 | NKR0319C_9NB01_S01 | NKR0319D_2NB01_S01 | NKR0319D_2NB01_S02 | NKR0319D_2NB01_S03 | NKR0319D_9NB01_S01 | NKR0319D_9NB01_S02 | NKR0319D_9NB01_S03 | NKR0319Q_9NB01 | NKR0319X_2NB01 | NKR0319C_9NB01_S01 | NKR0235C_9NB01_S01 | NKR0235C_9NB01_S01 | NKR0235D_2|G09,U21,U09 |BANHUAISOKNOI |4 |null |NKR0329|0 |null |MATE |Route_Cell_Down_Major_NMX |huawei-u2000-corba-3gpp@150.236.236.113:6791|NKR0329C_9NB01_S01 |[NW]:NKR0329(RETAINED):ROUTE CELL DOWN (15) 6 PCNs |NKR0329C_9NB01_S01|10 |44701512 |127 |2026-07-07 02:55:47 UTC|1970-01-01 00:00:00 UTC|INC000102067966|2026-07-07 02:55:13 UTC|null |U |MATE_NAKHON RATCHASIMA1|NORTHEAST|PAK CHONG |null |null |2026-07-07|18 |4G |3 | |07/06/2026 00:45:02|07/06/2026 00:16:09|BKC0154B-L23|BKC0154B-L23|911003066|huawei-u2000-corba-3gpp@150.236.236.15:6793|07/06/2026 00:16:09|PNC03_0682 | AN-P NC-03 |Huawei_RAN_U2020_RST1 |ACTIVE |0 |2026-07-06 00:33:16 UTC|eX2 |eNode B |0 |null |10.27.193.5 |911003066 |NeType: BTS3900| NeLocation: | vendor: | neName: BKC0154B-L23| alarmName: User Plane Fault| alarmLocation: Service Type=eX2| appendInfo: RAT_INFO=L, AFFECTED_RAT=L, DID=NULL, Cumulative Duration(s)=90, eNodeBId=87630 |Service Type=eX2 |T_BKC0154 |4 |Live |BKC0154|8891 |null |Deduplication Automation|User Plane Fault |huawei-u2000-corba-3gpp@150.236.236.15:6793 |BKC0154B-L23 |User Plane Fault |BKC0154B-L23 |0 |44175285 |0 |1970-01-01 00:00:00 UTC|1970-01-01 00:00:00 UTC|null |2026-07-06 00:35:13 UTC|null |U |BANGKOK EAST2 |BANGKOK |PATHUM WAN |RAN-BSC |BKC0154B-L23 |2026-07-06|00 |4G |3 | |07/01/2026 01:35:02|06/30/2026 17:54:43|1490166793|ericsson-enm-corba@150.236.236.58:6794|CMI6272E_1NB05_S02 |06/30/2026 17:54:43|G09,L07,L09,L18,L21,L23,N07,U08,U09,U21 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE |0 |2026-07-01 00:21:56 UTC|CMI6272E_1NB05_S02 |LTE Cell |0 |"MATE info=""Updated by rule: Main_Flow_NOC, module: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA C-OF-TH_NORTH_ALL."" |10.11.5.8 |1490166793 |No activity, recov fail |SubNetwork=RadioNode,MeContext=CMI6272T_2NB05,ManagedElement=CMI6272T_2NB05,ENodeBFunction=1,EUtranCellFDD=CMI6272E_1NB05_S02 |null |4 |Live |CMI6272|1010400|null |Deduplication Automation|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794 |CMI6272E_1NB05_S02 |Suspected Sleeping Cell |CMI6272E_1NB05_S02|10 |41562841 |125 |2026-06-30 23:55:48 UTC|1970-01-01 00:00:00 UTC|INC000101960026|2026-06-30 17:54:49 UTC|null |U |CHIANG MAI1 |NORTH |MUEANG CHIANG MAI ||True-L18| |CMI6272T_2NB05 |2026-07-01|01 |4G |3 | |07/02/2026 10:10:03|07/02/2026 08:50:13|84675034|nokia-nsn-snmp@150.236.236.48:6791 |07/02/2026 08:50:13|L07,L21,L23,N07,U09,U21 |Nokia_RAN_Netact5 |Out of Service|0 |2026-07-02 09:57:36 UTC|L23-SSK0221-3B |eNode B |0 |null |10.34.157.13 |84675034 |Remote Interference avoidance is active and DWP TS rollback as SSP5;6127 additionalFaultID=6127 LNCEL-195

|PLMN-PLMN/MRBTS-401108/LNBTS-401108/LNCEL-195
|BAN RAHAN |3 |Live |SSK022
1|300 |null |Deduplication Automation|CELL NOTIFICATION
|nokia-nsn-snmpp@150.236.236.48:6791 |MRBTS-401108/LNBTS-401108|CELL NO
TIFICATION;Remote Interference avoidance is active and DwPTS rollback as SSP5;6
127 additionalFaultID=6127 LNCEL-195
|SSK0221-L23 |0 |42451058 |0 |1970-01-01 00:00:00 UTC|1
970-01-01 00:00:00 UTC|null |2026-07-02 08:50:16 UTC|null |U
|SI SA KET |NORTHEAST|KHUN HAN ||RAN-4G| |4G-L23
|2026-07-02|10 |4G |3 |
|07/07/2026 05:10:02|07/07/2026 04:19:26|85004849|nokia-nsn-snmpp@150.236.236.4
8:6791 |07/07/2026 04:19:26|KKN-A_0063 |KKN
0030 |Nokia_RAN_Netact5 |ACTIVE
|0 |2026-07-07 05:00:15 UTC|KKN7900-L23
|eNode B |0 |null
|10.34.157.13 |85004849 |Transport layer connection failure in S1 interf
ace;6202 srcIPAddr=10.223.113.214;dstIPAddr=10.80.4.5;supplDstnIPAddr=;supplAla
rmInfo=Transport Network Identifier: 0 additionalFaultID=6202 LNMME-10
|PLMN-PLMN/MRBTS-177827/LNBTS-177827/LNMME-10
|T_KKN7900 |4 |Live |KKN790
0|300 |null |Deduplication Automation|BASE STATION CONNECTIVITY DEGRA
DED |nokia-nsn-snmpp@150.236.236.48:6791 |MRBTS-177827/LNBTS-177827|BA
SE STATION CONNECTIVITY DEGRADED;Transport layer connection failure in S1 inter
face;6202 srcIPAddr=10.223.113.214;dstIPAddr=10.80.4.5;supplDstnIPAddr=;supplAl
armInfo=Transport Network Identifier: 0 additionalFaultID=6202 LNMME-10
|KKN7900-L23 |0 |44780146 |0 |1970-01-01 00:00:00 UTC|1
970-01-01 00:00:00 UTC|null |2026-07-07 05:00:14 UTC|null |U
|KHON KAEN |NORTHEAST|MUEANG KHON KAEN ||RAN-4G| |MRBT
S-177827/LNBTS-177827|2026-07-07|05 |4G |3 |
|07/03/2026 18:05:02|07/03/2026 10:24:43|1497248716|ericsson-enm-corba@150.23
6.236.58:6794|CMI6741E_1NB03_S02 |07/03/2026 10:24:43|MHS-B_0170 |MH
S(CMI1000) |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-03 16:52:13 UTC|CMI6741E_1NB03_S02
|LTE Cell |0 |"MATE info=""Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1497248716 |No activity, recov fail
|SubNetwork=RadioNode,MeContext=CMI6741T_2NB03,ManagedElement=CMI6741T_2NB03,EN
odeBFunction=1,EUtranCellFDD=CMI6741E_1NB03_S02 |GRF-Muangket
|4 |Live |CMI6741|1010400|null |Deduplication Automat
ion|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794
|CMI6741E_1NB03_S02 |Suspected Sleeping Cell
|CMI6741E_1NB03_S02|10 |43039033 |125 |2026-07-03 16:25:56 UTC|1
970-01-01 00:00:00 UTC|INC000102012968|2026-07-03 10:24:49 UTC|null |U
|CHIANG MAI |NORTH |MAE TAENG ||True-L18| |CMI674
1T_2NB03 |2026-07-03|18 |4G |3 |
|07/01/2026 06:25:02|07/01/2026 06:13:48|84576646|nokia-nsn-snmpp@150.236.236.4
8:6791 |07/01/2026 06:13:48|PCB-A_0106 |PCB
8030 |Nokia_RAN_Netact5 |ACTIVE
|0 |2026-07-01 06:15:07 UTC|PCB8567-L23
|eNode B |0 |null
|10.34.157.13 |84576646 |Transport layer connection failure in S1 interf
ace;6202 srcIPAddr=10.223.92.146;dstIPAddr=10.91.55.5;supplDstnIPAddr=10.91.5
5.6;supplAlarmInfo=Transport Network Identifier: 0 additionalFaultID=6202 LNMME

E-5

|PLMN-PLMN/MRBTS-403792/LNBTS-403792/LNMME-5
|GRF-Watnonsawang |4 |Live |PCB856
7|300 |null |Deduplication Automation|BASE STATION CONNECTIVITY DEGRA
DED |nokia-nsn-snm@150.236.236.48:6791 |MRBTS-403792/LNBTS-403792|BA
SE STATION CONNECTIVITY DEGRADED;Transport layer connection failure in S1 inter
face;6202 srcIPAddr=10.223.92.146;dstIPAddr=10.91.55.5;supplDstnIPAddr=10.91.5
5.6;supplAlarmInfo=Transport Network Identifier: 0 additionalFaultID=6202 LNMME-5
E-5 |PCB8567-L23 |0 |41822500 |0 |1970-01-01 00:0
0:00 UTC|1970-01-01 00:00:00 UTC|null |2026-07-01 06:13:54 UTC|null
|U |PHETCHABUN |NORTH |MUEANG PHETCHABUN ||RAN-4G|
|4G-L23 |2026-07-01|06 |4G |3 |
|07/06/2026 13:35:03|07/06/2026 05:54:49|1504497042|ericsson-enm-corba@150.23
6.236.58:6794|CMI6272E_1NB05_S03 |07/06/2026 05:54:49|G09,L07,L09,L1
8,L21,L23,N07,U08,U09,U21 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-06 12:22:49 UTC|CMI6272E_1NB05_S03
|LTE Cell |0 |"MATE info=""Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1504497042 |No activity, recov fail
|SubNetwork=RadioNode,MeContext=CMI6272T_2NB05,ManagedElement=CMI6272T_2NB05,EN
odeBFunction=1,EUtranCellFDD=CMI6272E_1NB05_S03 |null
|4 |Live |CMI6272|1010400|null |Deduplication Automat
ion|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794
|CMI6272E_1NB05_S03 |Suspected Sleeping Cell
|CMI6272E_1NB05_S03|10 |44281263 |125 |2026-07-06 11:55:49 UTC|1
970-01-01 00:00:00 UTC|INC000102054504|2026-07-06 05:54:55 UTC|null |U
|CHIANG MAI1 |NORTH |MUEANG CHIANG MAI ||True-L18| |CMI627
2T_2NB05 |2026-07-06|13 |4G |3 |
|07/02/2026 01:05:02|07/01/2026 17:24:41|1492699612|ericsson-enm-corba@150.23
6.236.58:6794|CMI0934E_1NB04_S01 |07/01/2026 17:24:41|G09,L07,L09,L1
8,L21,L23,L26,N07,N26,U09,U21 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-01 23:54:36 UTC|CMI0934E_1NB04_S01
|LTE Cell |0 |"MATE info=""Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1492699612 |No activity, recov succ
|SubNetwork=RadioNode,MeContext=CMI0934T_2NB04,ManagedElement=CMI0934T_2NB04,EN
odeBFunction=1,EUtranCellFDD=CMI0934E_1NB04_S01 |BAN NONG KON K
HARU |4 |Live |CMI0934|1010400|null
|Deduplication Automation|Suspected Sleeping Cell |ericsson-enm-cor
ba@150.236.236.58:6794 |CMI0934E_1NB04_S01 |Suspected Sleeping Cell
|CMI0934E_1NB04_S01|10 |42097323 |125 |2026-07-01 23:25:17 UTC|1
970-01-01 00:00:00 UTC|INC000101977315|2026-07-01 17:24:46 UTC|null |U
|CHIANG MAI2 |NORTH |SAN SAI ||True-L18| |CMI093
4T_2NB04 |2026-07-02|01 |4G |3 |
|07/06/2026 17:15:02|07/06/2026 16:57:09|1760717219|ericsson-enm-corba@150.23
6.236.3:6795|PSN6744P_9NB01 |07/06/2026 17:06:37|PLK-B_0218 |PS
N0013 |True_Ericsson_ENM_UTRAN_SOUTH_3A |ACTIVE
|0 |2026-07-06 17:06:42 UTC|null
|NR GNodeB - Ericsson|0 |null
|10.11.6.8 |1760717219 |Contact next level of support and provide log f
iles required by Data Collection Guideline.
|SubNetwork=RadioNode,MeContext=PSN6744P_9NB01,ManagedElement=PSN6744P_9NB01,EN

odeBFunction=1,SectorCarrier=L07G-S1C1 |Nakhonchum
|3 |Live |PSN6744|1409 |RETAINT |Deduplication Automat
ion|SW Error |ericsson-enm-corba@150.236.236.3:6795
|PSN6744P_9NB01 |SW Error
|PSN6744P_9NB01 |0 |691734144 |0 |1970-01-01 00:00:00 UTC|1
970-01-01 00:00:00 UTC|null |2026-07-06 16:57:15 UTC|null |U
|PHITSANULOK |NORTH |NAKHON THAI |True-NR FDD |NR FDD
|2026-07-06|17 |5G |3 |
|07/03/2026 20:05:03|07/03/2026 11:55:22|1497746131|ericsson-enm-corba@150.23
6.236.58:6794|CMI7692E_1NB01_S04 |07/03/2026 11:55:22|L18,L21,L23,U0
8,U21 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-03 18:52:55 UTC|CMI7692E_1NB01_S04
|LTE Cell |0 |"MATE info="Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1497746131 |No activity, recov fail
|SubNetwork=RadioNode,MeContext=CMI7692T_2NB01,ManagedElement=CMI7692T_2NB01,EN
odeBFunction=1,EUtranCellFDD=CMI7692E_1NB01_S04 |T_CMI7692
|4 |Live |CMI7692|1010400|null |Deduplication Automat
ion|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794
|CMI7692E_1NB01_S04 |Suspected Sleeping Cell
|CMI7692E_1NB01_S04|10 |43078628 |125 |2026-07-03 18:15:32 UTC|1
970-01-01 00:00:00 UTC|INC000102013542|2026-07-03 11:55:28 UTC|null |U
|CHIANG MAI1 |NORTH |T_Retain_FAC ||True-L18| |CMI769
2T_2NB01 |2026-07-03|20 |4G |3 |
|07/04/2026 00:25:02|07/03/2026 16:35:04|1497829349|ericsson-enm-corba@150.23
6.236.58:6794|CMIA161E_1NB03_S04 |07/03/2026 16:35:04|CMI-A_0138 |CM
I1000 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-03 23:12:18 UTC|CMIA161E_1NB03_S04
|LTE Cell |0 |"MATE info="Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1497829349 |No activity, recov succ
|SubNetwork=RadioNode,MeContext=CMIA161T_2NB03,ManagedElement=CMIA161T_2NB03,EN
odeBFunction=1,EUtranCellFDD=CMIA161E_1NB03_S04 |CHANGHUEAK MUE
ANG CHIANG MAI |4 |Live |CMIA161|1010400|null
|Deduplication Automation|Suspected Sleeping Cell |ericsson-enm-cor
ba@150.236.236.58:6794 |CMIA161E_1NB03_S04 |Suspected Sleeping Cell
|CMIA161E_1NB03_S04|10 |43170940 |125 |2026-07-03 22:35:46 UTC|1
970-01-01 00:00:00 UTC|INC000102015480|2026-07-03 16:35:10 UTC|null |U
|CHIANG MAI1 |NORTH |MUEANG CHIANG MAI ||True-L18| |CMIA16
1T_2NB03 |2026-07-04|00 |4G |3 |
|07/01/2026 19:10:03|07/01/2026 19:01:34|84633770|nokia-nsn-snmpp@150.236.236.4
8:6791 |07/01/2026 19:01:34|KPP-A_0096 |KPP
1144 |Nokia_RAN_Netact5 |ACTIVE
|0 |2026-07-01 19:01:45 UTC|KPP6720-L23
|eNode B |0 |null
|10.34.157.13 |84633770 |S1 SCTP path failure;6204 srcIPAddr=10.223.87.2
10;dstIPAddr=10.91.55.3;supplDstnIPAddr=;supplAlarmInfo=Transport Network Ident
ifier: 0 additionalFaultID=6204 LNMME-3
|PLMN-PLMN/MRBTS-344212/LNBTS-344212/LNMME-3
|T_KPP6720 |4 |Live |KPP672
0|300 |null |Deduplication Automation|BASE STATION CONNECTIVITY DEGRA
DED |nokia-nsn-snmpp@150.236.236.48:6791 |MRBTS-344212/LNBTS-344212|BA

SE STATION CONNECTIVITY DEGRADED;S1 Sctp path failure;6204 srcIPAddr=10.223.8
7.210;dstIPAddr=10.91.55.3;supplDstnIPAddr=;supplAlarmInfo=Transport Network Id
entifier: 0 additionalFaultID=6204 LNMME-3
|KPP6720-L23 |0 |42139108 |0 |1970-01-01 00:00:00 UTC|1
970-01-01 00:00:00 UTC|null |2026-07-01 19:01:37 UTC|null |U
|KAMPHAENG PHET |NORTH |MUEANG KAMPHAENG PHET ||RAN-4G| |MRBT
S-344212/LNBTS-344212|2026-07-01|19 |4G |3 |
|07/02/2026 10:10:03|07/02/2026 09:53:19|84679318|nokia-nsn-snm@150.236.236.4
8:6791 |07/02/2026 09:53:19|UBR-A_0132 |UBR
1642 |Nokia_RAN_Netact5 |ACTIVE
|0 |2026-07-02 09:57:21 UTC|UBR7695-L23
|eNode B |0 |null
|10.34.157.13 |84679318 |Transport layer connection failure in S1 interf
ace;6202 srcIPAddr=10.223.7.118;dstIPAddr=10.81.98.65;supplDstnIPAddr=;supplAla
rmInfo=Transport Network Identifier: 0 additionalFaultID=6202 LNMME-16
|PLMN-PLMN/MRBTS-171887/LNBTS-171887/LNMME-16
|T_UBR7695 |4 |Live |UBR769
5|300 |null |Deduplication Automation|BASE STATION CONNECTIVITY DEGRA
DED |nokia-nsn-snm@150.236.236.48:6791 |MRBTS-171887/LNBTS-171887|BA
SE STATION CONNECTIVITY DEGRADED;Transport layer connection failure in S1 interf
ace;6202 srcIPAddr=10.223.7.118;dstIPAddr=10.81.98.65;supplDstnIPAddr=;supplAl
armInfo=Transport Network Identifier: 0 additionalFaultID=6202 LNMME-16
|UBR7695-L23 |0 |42480376 |0 |1970-01-01 00:00:00 UTC|1
970-01-01 00:00:00 UTC|null |2026-07-02 09:56:56 UTC|null |U
|UBON RATCHATHANI |NORTHEAST|MUEANG UBON RATCHATHANI ||RAN-4G| |MRBT
S-171887/LNBTS-171887|2026-07-02|10 |4G |3 |
|07/02/2026 22:05:03|07/02/2026 18:39:40|1495443555|ericsson-enm-corba@150.23
6.236.58:6794|CMI6989R_1NB01_S03 |07/02/2026 18:39:40|CMI-A_0153 |CM
I1000 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-02 20:53:24 UTC|CMI6989R_1NB01_S03
|LTE Cell |0 |"MATE info="Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1495443555 |No activity, recov succ
|SubNetwork=RadioNode,MeContext=CMI6989P_9NB01,ManagedElement=CMI6989P_9NB01,EN
odeBFunction=1,EUtranCellFDD=CMI6989R_1NB01_S03 |700th annivers
ary stadium chiangmai |4 |Live |CMI6989|1010400|null
|Deduplication Automation|Suspected Sleeping Cell |ericsson-enm-cor
ba@150.236.236.58:6794 |CMI6989R_1NB01_S03 |Suspected Sleeping Cell
|CMI6989R_1NB01_S03|10 |42677209 |125 |2026-07-02 20:19:42 UTC|1
970-01-01 00:00:00 UTC|INC000101997856|2026-07-02 18:39:46 UTC|null |U
|CHIANG MAI2 |NORTH |MAE RIM ||True-L18| |CMI698
9P_9NB01 |2026-07-02|22 |4G |3 |
|07/02/2026 10:35:02|07/02/2026 02:53:27|1494257275|ericsson-enm-corba@150.23
6.236.58:6794|CMI7692E_1NB01_S02 |07/02/2026 02:53:27|L18,L21,L23,U0
8,U21 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-02 09:23:53 UTC|CMI7692E_1NB01_S02
|LTE Cell |0 |"MATE info="Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1494257275 |No activity, recov fail
|SubNetwork=RadioNode,MeContext=CMI7692T_2NB01,ManagedElement=CMI7692T_2NB01,EN
odeBFunction=1,EUtranCellFDD=CMI7692E_1NB01_S02 |T_CMI7692
|4 |Live |CMI7692|1010400|null |Deduplication Automat

ion|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794
|CMI7692E_1NB01_S02 |Suspected Sleeping Cell
|CMI7692E_1NB01_S02|10 |42288913 |125 |2026-07-02 08:54:07 UTC|1
970-01-01 00:00:00 UTC|INC000101985601|2026-07-02 02:53:34 UTC|null |U
|CHIANG MAI1 |NORTH |T_Retain_FAC ||True-L18| |CMI769
2T_2NB01 |2026-07-02|10 |4G |3 |
|07/02/2026 10:10:03|07/02/2026 09:54:34|4060042487|ericsson-enm-corba_fdd1@15
0.236.236.15:6790|BKB1045F-L23 |07/02/2026 09:54:34|HAM04_0795 |AN-
HAM-04 |Ericsson_ENM_FDD1 |ACTIVE
|0 |2026-07-02 09:57:09 UTC|null
|eNode B |0 |null
|10.27.72.14 |4060042487 |null
|SubNetwork=ONRM_ROOT_M0,SubNetwork=RadioNodeTDD,MeContext=BKB1045F-L23,Managed
Element=BKB1045F-L23,Equipment=1,FieldReplaceableUnit=1 |SUAN LUANG SUA
N LUANG |2 |Live |BKB1045|1010400|null
|Deduplication Automation|Temperature High |ericsson-enm-cor
ba_fdd1@150.236.236.15:6790 |BKB1045F-L23 |Temperature High
|BKB1045F-L23 |0 |42479255 |0 |1970-01-01 00:00:00 UTC|1
970-01-01 00:00:00 UTC|null |2026-07-02 09:54:40 UTC|null |U
|BANGKOK EAST1 |BANGKOK |SUAN LUANG ||RAN-4G| |BKB104
5F-L23 |2026-07-02|10 |4G |3 |
|07/05/2026 00:50:02|07/05/2026 00:40:43|2094035611|ericsson-enm-corba_tdd2@15
0.236.236.15:6791|PTT0093B-L23 |07/05/2026 00:40:43|L07,L09,L18,L2
1,L23,L26,N07,N26,U09,U21 |Ericsson_ENM_TDD2 |ACTIVE
|0 |2026-07-05 00:40:49 UTC|null
|eNode B |0 |null
|10.28.76.14 |2094035611 |Contact next level of support and provide log f
iles required by Data Collection Guideline.
|SubNetwork=ONRM_ROOT_M0,SubNetwork=RadioNodeTDD,MeContext=PTT0093B-L23,Managed
Element=PTT0093B-L23,Equipment=1,FieldReplaceableUnit=RRU-2 |LAMLOOKKA
|3 |Live |PTT0093|1010400|null |Deduplication Automat
ion|SW Error |ericsson-enm-corba_tdd2@150.236.236.1
5:6791 |PTT0093B-L23 |SW Error
|PTT0093B-L23 |0 |43758181 |0 |1970-01-01 00:00:00 UTC|1
970-01-01 00:00:00 UTC|null |2026-07-05 00:40:49 UTC|null |U
|PATHUM THANI |BANGKOK |LAM LUK KA |null |4G-L23
|2026-07-05|00 |4G |3 |
|07/02/2026 21:05:02|07/02/2026 13:22:00|1494615528|ericsson-enm-corba@150.23
6.236.58:6794|CMI2600E_1NB04_S02 |07/02/2026 13:22:00|CMI-B_0298 |CM
I1000 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-02 19:53:45 UTC|CMI2600E_1NB04_S02
|LTE Cell |0 |"MATE info=""Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1494615528 |Preambles, no activity, recov succ
|SubNetwork=RadioNode,MeContext=CMI2600T_2NB04,ManagedElement=CMI2600T_2NB04,EN
odeBFunction=1,EUtranCellFDD=CMI2600E_1NB04_S02 |T_CMI2600
|4 |Live |CMI2600|1010400|null |Deduplication Automat
ion|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794
|CMI2600E_1NB04_S02 |Suspected Sleeping Cell
|CMI2600E_1NB04_S02|10 |42572613 |125 |2026-07-02 19:22:57 UTC|1
970-01-01 00:00:00 UTC|INC000101996397|2026-07-02 13:22:07 UTC|null |U
|CHIANG MAI1 |NORTH |MUEANG CHIANG MAI ||True-L18| |CMI260
0T_2NB04 |2026-07-02|21 |4G |3 |

|07/07/2026 21:10:02|07/07/2026 20:59:05|405367454|ericsson-enm-corba@150.236.236.3:6796|SMK0499H_7NB01|07/07/2026 20:59:05|L07,L21,L23,N07,U09,U21|True_Ericsson_ENM_RAN_6A|ACTIVE
|0|2026-07-07 21:00:49 UTC|null
|NR GNodeB - Ericsson|0|null
|10.23.21.8|405367454|Contact next level of support and provide log files required by Data Collection Guideline.
|SubNetwork=RadioNode,MeContext=SMK0499H_7NB01,ManagedElement=SMK0499H_7NB01,Equipment=1,AntennaUnitGroup=3,AntennaNearUnit=2|null
|3|Live|SMK0499|1409|RETAIND|Deduplication Automation|SW Error|ericsson-enm-corba@150.236.236.3:6796
|SMK0499H_7NB01|SW Error
|SMK0499H_7NB01|0|692340211|0|1970-01-01 00:00:00 UTC|1970-01-01 00:00:00 UTC|null|2026-07-07 20:59:10 UTC|null|U
|SAMUT SAKHON|WEST|KRATHUM BAEN|True-NR FDD|NR FDD
|2026-07-07|21|5G|3|
|07/02/2026 04:05:02|07/02/2026 03:29:07|MATE35160|N1781980311286|783at153.88.78.42|07/02/2026 03:29:07|G09,L07,L09,L21,N07,U09,U21|True_ZTE_ElasticNet_UTRAN|ACTIVE
|0|2026-07-02 03:42:33 UTC|NRTC019X_2NB01
|eNode B|0|"Current_time2=""1782963643"";EMS_NAME=""True_ZTE_ElasticNet_UTRAN"";LocationFirstOccurence=""null"";LocationLastOccurence=""null"";MATE info=""Updated by rule: ChildClear_Rule, module: Unified_Multivendor_2345G_Route_Cell_Down_Rule, instance id: 783, instance name: DTAC-OF-TH_SOUTH_WEST_CENTRAL_ALL."";MATEATTR=""|| Incident [Route Cell Sites Down: T_NRTC019 = 3 Sites (Total 5 Base Stations: 2G-G18=0,2G-D18=0,2G-G09=2, 3G-U08=0,3G-U09=0,3G-U21=3, 4G-E18=0,4G-L21=0,4G-L23=0,4G-L26=0,4G-L18=0,4G-L09=0,4G-L07=0,5G-N07=0,5G-N23=0,5G-N26=0)] at 10:29_Service Impact : 2G/3G services in some areas of_KAPHO(MATE_NARATHIWAT),D_Retain_FAC(MATE_NARATHIWAT),BACHO(MATE_NARATHIWAT),_Status : Escalated to FS0.||PTN0112-U21,,NRTC019-U21,NRTC019-G09,,NRTC121-U21,NRTC121-G09,, "";NEW_SER=""SAME"";NMX_Additional=""null"";TS=""NRTC121-AC-01 | NRTC121-AC-01 | NRTC121Q_9NB01 | NRTC121X_2NB01 | NRTC121C_9NB01_S01 | NRTC121Q_9NB01_S01 | NRTC121X_2NB01 | NRTC121-AC-01 | NRTC121-AC-01 | NRTC121Q_9NB01 | NRTC121X_2NB01 | NRTC121C_9NB01_S01 | NRTC121Q_9NB01_S01 | NRTC121X_2NB01 | NRTC019-AC-01 | NRTC019-AC-01 | NRTC019Q_9NB01 | NRTC019X_2NB01 | NRTC019C_9NB01_S01 | NRTC019Q_9NB01_S01 | NRTC019X_2NB01 | PTN0112-AC-01 | PTN0112-AC-01 | PTN0112X_2NB01 | PTN0112X_2NB01"";TS1=""PTN0112 | NRTC019 | NRTC121 | NRTC121"";Ci_attach=""TRUE""
|10.50.234.142|-2|NRTC121-AC-01 | NRTC121-AC-01 | NRTC121Q_9NB01 | NRTC121X_2NB01 | NRTC121C_9NB01_S01 | NRTC121Q_9NB01_S01 | NRTC121X_2NB01 | NRTC121-AC-01 | NRTC121-AC-01 | NRTC121Q_9NB01 | NRTC121X_2NB01 | NRTC121C_9NB01_S01 | NRTC121Q_9NB01_S01 | NRTC121X_2NB01 | NRTC019-AC-01 | NRTC019-AC-01 | NRTC019Q_9NB01 | NRTC019X_2NB01 | NRTC019C_9NB01_S01 | NRTC019Q_9NB01_S01 | NRTC019X_2NB01 | PTN0112-AC-01 | PTN0112-AC-01 | PTN0112X_2NB01 | PTN0112X_2NB01
|G09,U21
|T_NRTC019|4|null|NRTC019|0|null|MATE|Route_Cell_Down_NMX
|zte-ume-corba-3gpp@150.236.236.5:6793|NRTC019X_2NB01|[NW]:NRTC019(RETAIND):ROUTE CELL DOWN (5) 3 PCNs
|NRTC019X_2NB01|5|689418746|127|2026-07-02 03:41:05 UTC|1970-01-01 00:00:00 UTC|INC000101980399|2026-07-02 03:40:43 UTC|null|U
|MATE_NARATHIWAT|SOUTH|D_Retain_FAC|null|null
|2026-07-02|04|4G|3|
|07/05/2026 17:35:02|07/05/2026 11:10:35|1502660013|ericsson-enm-corba@150.23

6.236.58:6794|CMI0909E_1NB02_S03 |07/05/2026 11:10:35|G09,L07,L18,L2
1,L23,N07,U09,U21 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-05 16:23:19 UTC|CMI0909E_1NB02_S03
|LTE Cell |0 |"MATE info=""Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1502660013 |No activity, recov fail
|SubNetwork=RadioNode,MeContext=CMI0909T_2NB02,ManagedElement=CMI0909T_2NB02,EN
odeBFunction=1,EUtranCellFDD=CMI0909E_1NB02_S03 |CHAIPATTANA FO
UNDATION |4 |Live |CMI0909|1010400|null
|Deduplication Automation|Suspected Sleeping Cell |ericsson-enm-cor
ba@150.236.236.58:6794 |CMI0909E_1NB02_S03 |Suspected Sleeping Cell
|CMI0909E_1NB02_S03|10 |43997068 |125 |2026-07-05 15:51:30 UTC|1
970-01-01 00:00:00 UTC|INC000102040286|2026-07-05 11:10:42 UTC|null |U
|CHIANG MAI2 |NORTH |D_Retain_FAC ||True-L18| |CMI090
9T_2NB02 |2026-07-05|17 |4G |3 |
|07/03/2026 12:15:02|07/01/2026 04:52:17|1492103015|ericsson-enm-corba@150.23
6.236.58:6794|CMIA002G_7NB02_S02 |07/01/2026 04:52:17|CMI-A_1017 | CM
I1000 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|1 |2026-07-03 11:02:02 UTC|CMIA002G_7NB02_S02
|LTE Cell |0 |"MATE info=""Updated by rule: Main_Rule_New, mod
ule: Multivendor_Unified_2345G_Site_Down_DTAC_TRUE_Rule, instance id: 787, inst
ance name: DTAC-OF-TH_NORTH_ALL.""
|10.11.5.8 |1492103015 |This is a secondary alarm, use alarm correlatio
n information in Additional Information to find primary alarm.
|SubNetwork=RadioNode,MeContext=CMIA002H_7NB02,ManagedElement=CMIA002H_7NB02,EN
odeBFunction=1,EUtranCellFDD=CMIA002G_7NB02_S02 |CMRU Saluang D
ormitory |5 |Live |CMIA002|1010400|null
|Deduplication Automation|Service Unavailable |ericsson-enm-cor
ba@150.236.236.58:6794 |CMIA002G_7NB02_S02 |Service Unavailable
|CMIA002G_7NB02_S02|10 |41783076 |125 |2026-07-01 05:03:37 UTC|1
970-01-01 00:00:00 UTC|INC000101963557|2026-07-01 04:52:24 UTC|null |U
|CHIANG MAI2 |NORTH |MAE RIM ||True-L07| |CMIA00
2H_7NB02 |2026-07-03|12 |4G |3 |
|07/06/2026 21:05:02|07/06/2026 13:23:31|1505154135|ericsson-enm-corba@150.23
6.236.58:6794|CMI7232E_1NB04_S01 |07/06/2026 13:23:31|CMI-A_0377 | CM
I1000 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-06 19:54:40 UTC|CMI7232E_1NB04_S01
|LTE Cell |0 |"MATE info=""Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1505154135 |No activity, recov fail
|SubNetwork=RadioNode,MeContext=CMI7232T_2NB04,ManagedElement=CMI7232T_2NB04,EN
odeBFunction=1,EUtranCellFDD=CMI7232E_1NB04_S01 |T_CMI7232
|4 |Live |CMI7232|1010400|null |Deduplication Automat
ion|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794
|CMI7232E_1NB04_S01 |Suspected Sleeping Cell
|CMI7232E_1NB04_S01|10 |44467766 |125 |2026-07-06 19:24:17 UTC|1
970-01-01 00:00:00 UTC|INC000102061276|2026-07-06 13:23:37 UTC|null |U
|CHIANG MAI1 |NORTH |MUEANG CHIANG MAI ||True-L18| |CMI723
2T_2NB04 |2026-07-06|21 |4G |3 |
|07/03/2026 16:35:02|07/03/2026 09:03:14|1497207419|ericsson-enm-corba@150.23
6.236.58:6794|MHS8521E_1NB03_S01 |07/03/2026 09:03:14|MHS-A_1014 | MH
S(CMI1000) |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE

|0 |2026-07-03 15:21:43 UTC|MHS8521E_1NB03_S01
|LTE Cell |0 |"MATE info=""Updated by rule: Main_Flow_NOC, module: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1497207419 |No activity, recov succ
|SubNetwork=RadioNode,MeContext=MHS8521T_2NB03,ManagedElement=MHS8521T_2NB03,ENodeBFunction=1,EUtranCellFDD=MHS8521E_1NB03_S01 |GRF/Pha Bong
|4 |Live |MHS8521|1010400|null |Deduplication Automation|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794
|MHS8521E_1NB03_S01 |Suspected Sleeping Cell
|MHS8521E_1NB03_S01|10 |43007081 |125 |2026-07-03 15:03:56 UTC|1970-01-01 00:00:00 UTC|INC000102011588|2026-07-03 09:03:19 UTC|null |U
|MAE HONG SON |NORTH |MUEANG MAE HONG SON ||True-L18| |MHS8521T_2NB03 |2026-07-03|16 |4G |3 |
|07/05/2026 12:00:03|07/05/2026 11:50:41|400832503|ericsson-enm-corba@150.236.236.3:6796|SMK0499H_7NB01 |07/05/2026 11:50:41|L07,L21,L23,N07,U09,U21 |True_Ericsson_ENM_RAN_6A |ACTIVE
|0 |2026-07-05 11:51:51 UTC|null
|NR GNodeB - Ericsson|0 |null
|10.23.21.8 |400832503 |Contact next level of support and provide log files required by Data Collection Guideline.
|SubNetwork=RadioNode,MeContext=SMK0499H_7NB01,ManagedElement=SMK0499H_7NB01,Equipment=1,AntennaUnitGroup=3,AntennaNearUnit=2 |null
|3 |Live |SMK0499|1409 |RETAIND |Deduplication Automation|SW Error |ericsson-enm-corba@150.236.236.3:6796
|SMK0499H_7NB01 |SW Error
|SMK0499H_7NB01 |0 |691103219 |0 |1970-01-01 00:00:00 UTC|1970-01-01 00:00:00 UTC|null |2026-07-05 11:50:46 UTC|null |U
|SAMUT SAKHON |WEST |KRATHUM BAEN |True-NR FDD |NR FDD
|2026-07-05|12 |5G |3 |
|07/04/2026 08:35:02|07/04/2026 01:00:25|1497900536|ericsson-enm-corba@150.236.236.58:6794|CMIC047E_1NB03_S04 |07/04/2026 01:00:25|G09,L07,L09,L18,L21,N07,U09,U21 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-04 07:23:02 UTC|CMIC047E_1NB03_S04
|LTE Cell |0 |"MATE info=""Updated by rule: Main_Flow_NOC, module: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1497900536 |No activity, recov fail
|SubNetwork=RadioNode,MeContext=CMIC047T_2NB03,ManagedElement=CMIC047T_2NB03,ENodeBFunction=1,EUtranCellFDD=CMIC047E_1NB03_S04 |null
|4 |Live |CMIC047|1010400|null |Deduplication Automation|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794
|CMIC047E_1NB03_S04 |Suspected Sleeping Cell
|CMIC047E_1NB03_S04|10 |43292874 |125 |2026-07-04 07:01:24 UTC|1970-01-01 00:00:00 UTC|INC000102021906|2026-07-04 01:00:31 UTC|null |U
|CHIANG MAI1 |NORTH |D_Retain_FAC ||True-L18| |CMIC047T_2NB03 |2026-07-04|08 |4G |3 |
|07/01/2026 09:55:03|07/01/2026 07:23:32|84580523|nokia-nsn-snmp@150.236.236.48:6791 |07/01/2026 07:23:32|UDT-A_0012 |UDT0014 |Nokia_RAN_Netact5 |ACTIVE
|1 |2026-07-01 09:25:42 UTC|L23-UDT8507-2C
|eNode B |0 |"MATE info=""Updated by rule: Main_Rule_New, module: Multivendor_Unified_2345G_Site_Down_DTAC_TRUE_Rule, instance id: 786, instance name: DTAC-OF-TH_NORTHEAST_ALL.""

|10.34.157.13 |84580523 |Failure in replaceable baseband unit;4341 unitName=ABIA path=/SMOD_R-1(Right)/BBMOD_R-1(Right_Middle) serial_no=K9180929626 additionalFaultID=3003
|PLMN-PLMN/MRBTS-180017/LNBTS-180017/LNCEL-186
|TBC |5 |Live |UDT850
7|300 |null |Deduplication Automation|CELL FAULTY
|nokia-nsn-snmp@150.236.236.48:6791 |MRBTS-180017/LNBTS-180017|CELL FAULTY;Failure in replaceable baseband unit;4341 unitName=ABIA path=/SMOD_R-1(Right)/BBMOD_R-1(Right_Middle) serial_no=K9180929626 additionalFaultID=3003
|UDT8507-L23 |5 |41860915 |125 |2026-07-01 07:50:12 UTC|1970-01-01 00:00:00 UTC|INC000101966437|2026-07-01 07:23:43 UTC|null |U
|UDON THANI |NORTHEAST|MUEANG UDON THANI ||RAN-4G| |MRBTS-180017/LNBTS-180017|2026-07-01|09 |4G |3 |
|07/05/2026 15:55:02|07/05/2026 15:44:41|401010524|ericsson-enm-corba@150.236.236.3:6796|SMK0499H_7NB01 |07/05/2026 15:44:41|L07,L21,L23,N07,U09,U21 |True_Ericsson_ENM_RAN_6A |ACTIVE
|0 |2026-07-05 15:45:10 UTC|null
|NR GNodeB - Ericsson|0 |null
|10.23.21.8 |401010524 |Contact next level of support and provide log files required by Data Collection Guideline.
|SubNetwork=RadioNode,MeContext=SMK0499H_7NB01,ManagedElement=SMK0499H_7NB01,Equipment=1,AntennaUnitGroup=3,AntennaNearUnit=1 |null
|3 |Live |SMK0499|1409 |RETAINED |Deduplication Automation|SW Error |ericsson-enm-corba@150.236.236.3:6796
|SMK0499H_7NB01 |SW Error
|SMK0499H_7NB01 |0 |691176566 |0 |1970-01-01 00:00:00 UTC|1970-01-01 00:00:00 UTC|null |2026-07-05 15:44:49 UTC|null |U
|SAMUT SAKHON |WEST |KRATHUM BAEN |True-NR FDD |NR FDD
|2026-07-05|15 |5G |3 |
|07/07/2026 13:35:02|07/07/2026 05:54:53|1507000100|ericsson-enm-corba@150.236.236.58:6794|CMI7232E_1NB04_S04 |07/07/2026 05:54:53|CMI-A_0377 |CMI1000 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-07 12:25:14 UTC|CMI7232E_1NB04_S04
|LTE Cell |0 |"MATE info=""Updated by rule: Main_Flow_NOC, module: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1507000100 |No activity, recov fail
|SubNetwork=RadioNode,MeContext=CMI7232T_2NB04,ManagedElement=CMI7232T_2NB04,ENodeBFunction=1,EUtranCellFDD=CMI7232E_1NB04_S04 |T_CMI7232
|4 |Live |CMI7232|1010400|null |Deduplication Automation|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794
|CMI7232E_1NB04_S04 |Suspected Sleeping Cell
|CMI7232E_1NB04_S04|10 |44810631 |125 |2026-07-07 11:55:49 UTC|1970-01-01 00:00:00 UTC|INC000102079210|2026-07-07 05:54:58 UTC|null |U
|CHIANG MAI1 |NORTH |MUEANG CHIANG MAI ||True-L18| |CMI7232T_2NB04 |2026-07-07|13 |4G |3 |
|07/05/2026 01:05:02|07/04/2026 17:28:40|1500207411|ericsson-enm-corba@150.236.236.58:6794|CMIC047E_1NB03_S02 |07/04/2026 17:28:40|G09,L07,L09,L18,L21,N07,U09,U21 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-04 23:54:28 UTC|CMIC047E_1NB03_S02
|LTE Cell |0 |"MATE info=""Updated by rule: Main_Flow_NOC, module: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1500207411 |No activity, recov fail

|SubNetwork=RadioNode,MeContext=CMIC047T_2NB03,ManagedElement=CMIC047T_2NB03,ENodeBFunction=1,EUtranCellFDD=CMIC047E_1NB03_S02|null
|4|Live|CMIC047|1010400|null|Deduplication Automation|Suspected Sleeping Cell|ericsson-enm-corba@150.236.236.58:6794
|CMIC047E_1NB03_S02|Suspected Sleeping Cell
|CMIC047E_1NB03_S02|10|43667396|125|2026-07-04 23:29:15 UTC|1970-01-01 00:00:00 UTC|INC000102030734|2026-07-04 17:28:46 UTC|null|U
|CHIANG MAI1|NORTH|D_Retain_FAC||True-L18|CMIC047T_2NB03|2026-07-05|01|4G|3|
|07/03/2026 11:05:03|07/03/2026 03:28:03|1496233040|ericsson-enm-corba@150.236.236.58:6794|CMI7151E_1NB04_S03|07/03/2026 03:28:03|CMI-A_0298|CMI1000|True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0|2026-07-03 09:53:34 UTC|CMI7151E_1NB04_S03
|LTE Cell|0|"MATE info="Updated by rule: Main_Flow_NOC, module: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA C-OF-TH_NORTH_ALL.""
|10.11.5.8|1496233040|Preambles, no activity, recov succ
|SubNetwork=RadioNode,MeContext=CMI7151T_2NB04,ManagedElement=CMI7151T_2NB04,ENodeBFunction=1,EUtranCellFDD=CMI7151E_1NB04_S03|Phuping
|4|Live|CMI7151|1010400|null|Deduplication Automation|Suspected Sleeping Cell|ericsson-enm-corba@150.236.236.58:6794
|CMI7151E_1NB04_S03|Suspected Sleeping Cell
|CMI7151E_1NB04_S03|10|42834880|125|2026-07-03 09:28:44 UTC|1970-01-01 00:00:00 UTC|INC000102007385|2026-07-03 03:28:10 UTC|null|U
|CHIANG MAI1|NORTH|T_Retain_FAC||True-L18|CMI7151T_2NB04|2026-07-03|11|4G|3|
|07/06/2026 10:20:02|07/06/2026 10:08:37|400729871|ericsson-enm-corba@150.236.236.3:6796|SMK0499H_7NB01|07/06/2026 10:08:37|L07,L21,L23,N07,U09,U21|True_Ericsson_ENM_RAN_6A|ACTIVE
|0|2026-07-06 10:09:31 UTC|null
|NR GNodeB - Ericsson|0|null
|10.23.21.8|400729871|Contact next level of support and provide log files required by Data Collection Guideline.
|SubNetwork=RadioNode,MeContext=SMK0499H_7NB01,ManagedElement=SMK0499H_7NB01,Equipment=1,AntennaUnitGroup=3,AntennaNearUnit=1|null
|3|Live|SMK0499|1409|RETAIND|Deduplication Automation|SW Error|ericsson-enm-corba@150.236.236.3:6796
|SMK0499H_7NB01|SW Error
|SMK0499H_7NB01|0|691552362|0|1970-01-01 00:00:00 UTC|1970-01-01 00:00:00 UTC|null|2026-07-06 10:08:44 UTC|null|U
|SAMUT SAKHON|WEST|KRATHUM BAEN|True-NR FDD|NR FDD
|2026-07-06|10|5G|3|
|07/07/2026 00:35:02|07/06/2026 17:00:34|1504926638|ericsson-enm-corba@150.236.236.58:6794|MHS0064E_1NB02_S03|07/06/2026 17:00:34|G09,L07,L09,L18,L21,L23,N07,U09,U21|True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0|2026-07-06 23:22:09 UTC|MHS0064E_1NB02_S03
|LTE Cell|0|"MATE info="Updated by rule: Main_Flow_NOC, module: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA C-OF-TH_NORTH_ALL.""
|10.11.5.8|1504926638|No activity, recov succ
|SubNetwork=RadioNode,MeContext=MHS0064T_2NB02,ManagedElement=MHS0064T_2NB02,ENodeBFunction=1,EUtranCellFDD=MHS0064E_1NB02_S03|null
|4|Live|MHS0064|1010400|null|Deduplication Automation|Suspected Sleeping Cell|ericsson-enm-corba@150.236.236.58:6794

|MHS0064E_1NB02_S03 |Suspected Sleeping Cell
|MHS0064E_1NB02_S03|10 |44537265 |125 |2026-07-06 23:01:08 UTC|1
970-01-01 00:00:00 UTC|INC000102065264|2026-07-06 17:00:40 UTC|null |U
|MAE HONG SON |NORTH |MUEANG MAE HONG SON ||True-L18| |MHS006
4T_2NB02 |2026-07-07|00 |4G |3 |
|07/07/2026 16:05:03|07/07/2026 13:41:44|1507629186|ericsson-enm-corba@150.23
6.236.58:6794|CMI7423E_2NB03_S05 |07/07/2026 13:41:44|G09,L09,L18,L2
1,L23,U08,U21 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-07 14:52:26 UTC|CMI7423E_2NB03_S05
|LTE Cell |0 |"MATE info="Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1507629186 |No activity, recov succ
|SubNetwork=RadioNode,MeContext=CMI7423T_2NB03,ManagedElement=CMI7423T_2NB03,EN
odeBFunction=1,EUtranCellFDD=CMI7423E_2NB03_S05 |KAD DHARA DHEV
I |4 |Live |CMI7423|1010400|null
|Deduplication Automation|Suspected Sleeping Cell |ericsson-enm-cor
ba@150.236.236.58:6794 |CMI7423E_2NB03_S05 |Suspected Sleeping Cell
|CMI7423E_2NB03_S05|10 |45036635 |125 |2026-07-07 14:33:29 UTC|1
970-01-01 00:00:00 UTC|INC000102081929|2026-07-07 13:41:49 UTC|null |U
|CHIANG MAI1 |NORTH |MUEANG CHIANG MAI ||True-L21| |CMI742
3T_2NB03 |2026-07-07|16 |4G |3 |
|07/07/2026 18:35:03|07/07/2026 11:02:36|1507000934|ericsson-enm-corba@150.23
6.236.58:6794|MHS0024E_1NB02_S02 |07/07/2026 11:02:36|G09,L07,L09,L1
8,L21,N07,U09,U21 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-07 17:23:13 UTC|MHS0024E_1NB02_S02
|LTE Cell |0 |"MATE info="Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1507000934 |No activity, recov succ
|SubNetwork=RadioNode,MeContext=MHS0024T_2NB02,ManagedElement=MHS0024T_2NB02,EN
odeBFunction=1,EUtranCellFDD=MHS0024E_1NB02_S02 |null
|4 |Live |MHS0024|1010400|null |Deduplication Automat
ion|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794
|MHS0024E_1NB02_S02 |Suspected Sleeping Cell
|MHS0024E_1NB02_S02|10 |44974197 |125 |2026-07-07 17:03:09 UTC|1
970-01-01 00:00:00 UTC|INC000102085223|2026-07-07 11:02:43 UTC|null |U
|MAE HONG SON |NORTH |D_Retain_FAC ||True-L18| |MHS002
4T_2NB02 |2026-07-07|18 |4G |3 |
|07/03/2026 08:15:02|07/03/2026 08:04:58|84758943|nokia-nsn-snp@150.236.236.4
8:6791 |07/03/2026 08:04:58|NKR-A_0052 |NKR
0043 |Nokia_RAN_Netact5 |ACTIVE
|0 |2026-07-03 08:05:10 UTC|NKR7011-L23
|eNode B |0 |null
|10.34.157.13 |84758943 |High PDU input voltage;6907 unitName=ASIA pat
h=/SMOD_R-1(Right) serial_no=6Q182010348 additionalFaultID=6907
|PLMN-PLMN/MRBTS-167357/LNBTS-167357
|T_NKR7011 |3 |Live |NKR701
1|300 |null |Deduplication Automation|BASE STATION NOTIFICATION
|nokia-nsn-snp@150.236.236.48:6791 |MRBTS-167357/LNBTS-167357|BASE ST
ATION NOTIFICATION;High PDU input voltage;6907 unitName=ASIA path=/SMOD_R-1(Rig
ht) serial_no=6Q182010348 additionalFaultID=6907
|NKR7011-L23 |0 |42976456 |0 |1970-01-01 00:00:00 UTC|1
970-01-01 00:00:00 UTC|null |2026-07-03 08:05:01 UTC|null |U

|NAKHON RATCHASIMA1 |NORTHEAST|MUEANG NAKHON RATCHASIMA| |RAN-4G| |MRBT
S-167357/LNBTS-167357|2026-07-03|08 |4G |3 |
|07/07/2026 05:10:02|07/07/2026 04:58:44|85004836|nokia-nsn-snm@150.236.236.4
8:6791 |07/07/2026 04:58:44|D18,G09,G18,L0
7,L09,L18,L21,L23,N07,U08,U09,U21 |Nokia_RAN_Netact5 |ACTIVE
|0 |2026-07-07 05:00:15 UTC|PCT0029-L23
|eNode B |0 |null
|10.34.157.13 |85004836 |Transport layer connection failure in S1 interf
ace;6202 srcIPAddr=10.254.136.36;dstIPAddr=10.78.51.1;supplDstnIPAddr=;supplAla
rmInfo=Transport Network Identifier: 0 additionalFaultID=6202 LNMME-10
|PLMN-PLMN/MRBTS-361022/LNBTS-361022/LNMME-10
|BANG_MUL_NAK |4 |Live |PCT002
9|300 |null |Deduplication Automation|BASE STATION CONNECTIVITY DEGRA
DED |nokia-nsn-snm@150.236.236.48:6791 |MRBTS-361022/LNBTS-361022|BA
SE STATION CONNECTIVITY DEGRADED;Transport layer connection failure in S1 inter
face;6202 srcIPAddr=10.254.136.36;dstIPAddr=10.78.51.1;supplDstnIPAddr=;supplAl
armInfo=Transport Network Identifier: 0 additionalFaultID=6202 LNMME-10
|PCT0029-L23 |0 |44780136 |0 |1970-01-01 00:00:00 UTC|1
970-01-01 00:00:00 UTC|null |2026-07-07 05:00:14 UTC|null |U
|PHICHIT |NORTH |BANG MUN NAK ||RAN-4G| |MRBT
S-361022/LNBTS-361022|2026-07-07|05 |4G |3 |
|07/07/2026 23:05:02|07/07/2026 15:28:34|1507548443|ericsson-enm-corba@150.23
6.236.58:6794|CMI6335E_1NB04_S03 |07/07/2026 15:28:34|CMI-A_0501 |CM
I1000 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-07 21:54:48 UTC|CMI6335E_1NB04_S03
|LTE Cell |0 |"MATE info=""Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1507548443 |Preambles, no activity, recov succ
|SubNetwork=RadioNode,MeContext=CMI6335T_2NB04,ManagedElement=CMI6335T_2NB04,EN
odeBFunction=1,EUtranCellFDD=CMI6335E_1NB04_S03 |T_CMI6335
|4 |Live |CMI6335|1010400|null |Deduplication Automat
ion|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794
|CMI6335E_1NB04_S03 |Suspected Sleeping Cell
|CMI6335E_1NB04_S03|10 |45068033 |125 |2026-07-07 21:29:19 UTC|1
970-01-01 00:00:00 UTC|INC000102089439|2026-07-07 15:28:40 UTC|null |U
|CHIANG MAI1 |NORTH |MUEANG CHIANG MAI ||True-L18| |CMI633
5T_2NB04 |2026-07-07|23 |4G |3 |
|07/07/2026 19:35:03|07/07/2026 11:52:01|1507115774|ericsson-enm-corba@150.23
6.236.58:6794|CMI0141G_7NB01_S01 |07/07/2026 11:52:01|G09,G18,L07,L2
1,L23,N07,U08,U09,U21 |True_Ericsson_ENM5_WEST_UPPER_NORTH|ACTIVE
|0 |2026-07-07 18:23:18 UTC|CMI0141G_7NB01_S01
|LTE Cell |0 |"MATE info=""Updated by rule: Main_Flow_NOC, mod
ule: Eric_2345G_HW_Alarm_Correlation_Rule, instance id: 787, instance name: DTA
C-OF-TH_NORTH_ALL.""
|10.11.5.8 |1507115774 |Preambles, no activity, recov fail
|SubNetwork=RadioNode,MeContext=CMI0141P_9NB01,ManagedElement=CMI0141P_9NB01,EN
odeBFunction=1,EUtranCellFDD=CMI0141G_7NB01_S01 |PHRAMAHATHAT
|4 |Live |CMI0141|1010400|null |Deduplication Automat
ion|Suspected Sleeping Cell |ericsson-enm-corba@150.236.236.58:6794
|CMI0141G_7NB01_S01 |Suspected Sleeping Cell
|CMI0141G_7NB01_S01|10 |44999176 |125 |2026-07-07 17:52:49 UTC|1
970-01-01 00:00:00 UTC|INC000102085447|2026-07-07 11:52:07 UTC|null |U
|CHIANG MAI1 |NORTH |D_Retain_FAC ||True-L07| |CMI014

[illegible]


```
26/07/16 05:33:39 WARN DAGScheduler: Broadcasting large task binary with size 167.0 KiB
26/07/16 05:34:33 WARN DAGScheduler: Broadcasting large task binary with size 5.1 MiB
26/07/16 05:36:40 WARN DAGScheduler: Broadcasting large task binary with size 2.9 MiB
[Stage 709:=====>(66 + 1) / 67]
Duplicates removed : 1,144,619
```

In [52]: *##keeping only the required site ids in the data which are present in the tick*

```
# List of cell IDs to keep
site_ids = ['CMI7658', 'CNT0039', 'CCS0331', 'CMIA002', 'LBR8549', 'PCR0036',

# Filter the DataFrame
alarm_df_filtered = alarm_df_dedup.filter(
    F.col("SITEID").isin(site_ids)
)

print(f"Rows after filtering: {alarm_df_filtered.count():,}")

alarm_df_filtered.show(10, truncate=False)
```

```
26/07/16 05:36:55 WARN DAGScheduler: Broadcasting large task binary with size 6.6 MiB
26/07/16 05:38:37 WARN DAGScheduler: Broadcasting large task binary with size 4.4 MiB
```

Rows after filtering: 149,143

```
26/07/16 05:38:43 WARN DAGScheduler: Broadcasting large task binary with size 6.6 MiB
26/07/16 05:40:33 WARN DAGScheduler: Broadcasting large task binary with size 4.4 MiB
```


[illegible]

DE|null|Input Power Failure|ericsson-enm-corb
a@150.236.236.78:6796|ATG8505T_2NB01|Input Power Failure|ATG85
05T_2NB01|0|688818615|0|1970-01-01 00:00:00 UTC|1970-0
1-01 00:00:00 UTC|null|2026-07-01 02:18:22 UTC|null|I|2026-07-01
02:18:37.645003 UTC|ANG THONG|CENTRAL|CHAIYO|True-LTE FDD|LTE FDD
|2026-07-01|02|4G|
|07/01/2026 02:25:02|07/01/2026 02:18:16|1101718629|ericsson-enm-corba@150.23
6.236.78:6796|ATG8505T_2NB01|07/01/2026 02:18:16|ATG-A_0019|ATG0011|Tru
e_Ericsson_ENM_UTRAN_CENTRAL_4A|ACTIVE|0|1970-01-01 00:00:00
UTC|null|eNode B|3|null|10.23.17.8|11017186
29|Input voltage outside unit operating range.
|SubNetwork=RadioNode,MeContext=ATG8505T_2NB01,ManagedElement=ATG8505T_2NB01,Eq
uipment=1,FieldReplaceableUnit=RRU-5|Liab Kh
long /Chaiphum|3|Live|ATG8505|1010400|HSP3+CORP+RETAINT+IN
DE|null|Input Power Failure|ericsson-enm-corb
a@150.236.236.78:6796|ATG8505T_2NB01|Input Power Failure|ATG85
05T_2NB01|0|688818616|0|1970-01-01 00:00:00 UTC|1970-0
1-01 00:00:00 UTC|null|2026-07-01 02:18:22 UTC|null|I|2026-07-01
02:18:37.637008 UTC|ANG THONG|CENTRAL|CHAIYO|True-LTE FDD|LTE FDD
|2026-07-01|02|4G|
|07/01/2026 02:25:02|07/01/2026 02:18:16|1101718623|ericsson-enm-corba@150.23
6.236.78:6796|ATG8505R_1NB01_S04|07/01/2026 02:18:16|ATG-A_0019|ATG0011|Tru
e_Ericsson_ENM_UTRAN_CENTRAL_4A|ACTIVE|1|1970-01-01 00:00:00
UTC|ATG8505R_1NB01_S04|LTE Cell|5|null|10.23.17.8|11017186
23|This is a secondary alarm, use alarm correlation information in Additional I
nformation to find primary alarm.|SubNetwork=RadioNode,MeContext=ATG8505T_2NB0
1,ManagedElement=ATG8505T_2NB01,ENodeBFunction=1,EUtranCellFDD=ATG8505R_1NB01_S
04|Liab Khlong /Chaiphum|5|Live|AT
G8505|1010400|null|null|Service Unavailab
le|ericsson-enm-corba@150.236.236.78:6796|ATG8505R_1NB01_S04|Servi
ce Unavailable|ATG8505R_1NB01_S04|0|688818610|0
|1970-01-01 00:00:00 UTC|1970-01-01 00:00:00 UTC|null|2026-07-01 02:18:21 UT
C|null|I|2026-07-01 02:18:37.646002 UTC|ANG THONG|CENTRAL|CHAIYO
||True-L18||ATG8505T_2NB01|2026-07-01|02|4G|
|07/01/2026 03:20:02|07/01/2026 02:18:16|1101718626|ericsson-enm-corba@150.23
6.236.78:6796|ATG8505T_2NB01|07/01/2026 02:18:16|ATG-A_0019|ATG0011|Tru
e_Ericsson_ENM_UTRAN_CENTRAL_4A|ACTIVE|0|2026-07-01 03:09:26
UTC|null|eNode B|0|null|10.23.17.8|11017186
26|Input voltage outside unit operating range.
|SubNetwork=RadioNode,MeContext=ATG8505T_2NB01,ManagedElement=ATG8505T_2NB01,Eq
uipment=1,FieldReplaceableUnit=RRU-2|Liab Kh
long /Chaiphum|3|Live|ATG8505|1010400|HSP3+CORP+RETAINT+IN
DE|Deduplication Automation|Input Power Failure|ericsson-enm-corb
a@150.236.236.78:6796|ATG8505T_2NB01|Input Power Failure|ATG85
05T_2NB01|0|688818613|0|1970-01-01 00:00:00 UTC|1970-0
1-01 00:00:00 UTC|null|2026-07-01 02:18:21 UTC|null|U|2026-07-01
03:10:06.707001 UTC|ANG THONG|CENTRAL|CHAIYO|True-LTE FDD|LTE FDD
|2026-07-01|03|4G|
|07/01/2026 03:20:02|07/01/2026 02:18:19|1101718640|ericsson-enm-corba@150.23
6.236.78:6796|ATG8505T_2NB01|07/01/2026 02:18:19|ATG-A_0019|ATG0011|Tru
e_Ericsson_ENM_UTRAN_CENTRAL_4A|ACTIVE|0|2026-07-01 03:09:26
UTC|null|eNode B|0|null|10.23.17.8|11017186
40|Input voltage outside unit operating range.
|SubNetwork=RadioNode,MeContext=ATG8505T_2NB01,ManagedElement=ATG8505T_2NB01,Eq

uipment=1,FieldReplaceableUnit=RRU-25 |Liab Kh
long /Chaiphum|3 |Live |ATG8505|1010400|HSP3+CORP+RETAINT+IN
DE|Deduplication Automation|Input Power Failure |ericsson-enm-corb
a@150.236.236.78:6796|ATG8505T_2NB01 |Input Power Failure |ATG85
05T_2NB01 |0 |688818645 |0 |1970-01-01 00:00:00 UTC|1970-0
1-01 00:00:00 UTC|null |2026-07-01 02:18:25 UTC|null |U |2026-07-01
03:11:08.327002 UTC|ANG THONG|CENTRAL|CHAIYO |True-LTE FDD|LTE FDD
|2026-07-01|03 |4G |
|07/02/2026 03:50:02|07/02/2026 03:26:32|1103545946|ericsson-enm-corba@150.23
6.236.78:6796|ATG6722T_2NB02 |07/02/2026 03:32:37|ATG-A_0016 | ATG0011|Tru
e_Ericsson_ENM_UTRAN_CENTRAL_4A|ACTIVE |0 |2026-07-02 03:40:59
UTC|null |eNode B |0 |null |10.23.17.8 |11035459
46|X2 link problem to one or several neighbouring eNodeBs.
|SubNetwork=RadioNode,MeContext=ATG6722T_2NB02,ManagedElement=ATG6722T_2NB02,EN
odeBFunction=1 |T_ATG67
22 |3 |Live |ATG6722|1010400|CONSOT
|Deduplication Automation|External Link Failure |ericsson-enm-corba@1
50.236.236.78:6796|ATG6722T_2NB02 |External Link Failure |ATG6722
T_2NB02 |0 |689410076 |0 |1970-01-01 00:00:00 UTC|1970-01-0
1 00:00:00 UTC|null |2026-07-02 03:26:39 UTC|null |U |2026-07-02 0
3:41:28.027002 UTC|ANG THONG|CENTRAL|WISSET CHAI CHAN|True-LTE FDD|LTE FDD
|2026-07-02|03 |4G |
|07/07/2026 03:30:02|07/06/2026 23:12:06|1118294553|ericsson-enm-corba@150.23
6.236.78:6796|ATG8505T_2NB01 |07/06/2026 23:12:06|ATG-A_0019 | ATG0011|Tru
e_Ericsson_ENM_UTRAN_CENTRAL_4A|ACTIVE |0 |2026-07-07 03:22:35
UTC|null |eNode B |0 |null |10.23.17.8 |11182945
53|X2 link problem to one or several neighbouring gNodeBs.
|SubNetwork=RadioNode,MeContext=ATG8505T_2NB01,ManagedElement=ATG8505T_2NB01,EN
odeBFunction=1 |Liab Kh
long /Chaiphum|3 |Live |ATG8505|1010400|CORP+HSP3+INDE+RETAI
NT|Deduplication Automation|External Link to GNodeB Failure |ericsson-enm-corb
a@150.236.236.78:6796|ATG8505T_2NB01 |External Link to GNodeB Failure |ATG85
05T_2NB01 |0 |691845154 |0 |1970-01-01 00:00:00 UTC|1970-0
1-01 00:00:00 UTC|null |2026-07-06 23:12:12 UTC|null |U |2026-07-07
03:23:36.124003 UTC|ANG THONG|CENTRAL|CHAIYO |True-LTE FDD|LTE FDD
|2026-07-07|03 |4G |
|07/07/2026 03:35:03|07/06/2026 23:12:06|1118294553|ericsson-enm-corba@150.23
6.236.78:6796|ATG8505T_2NB01 |07/06/2026 23:12:06|ATG-A_0019 | ATG0011|Tru
e_Ericsson_ENM_UTRAN_CENTRAL_4A|ACTIVE |0 |2026-07-07 03:22:35
UTC|null |eNode B |0 |null |10.23.17.8 |11182945
53|X2 link problem to one or several neighbouring gNodeBs.
|SubNetwork=RadioNode,MeContext=ATG8505T_2NB01,ManagedElement=ATG8505T_2NB01,EN
odeBFunction=1 |Liab Kh
long /Chaiphum|3 |Live |ATG8505|1010400|CORP+HSP3+INDE+RETAI
NT|Deduplication Automation|External Link to GNodeB Failure |ericsson-enm-corb
a@150.236.236.78:6796|ATG8505T_2NB01 |External Link to GNodeB Failure |ATG85
05T_2NB01 |0 |691845154 |0 |1970-01-01 00:00:00 UTC|1970-0
1-01 00:00:00 UTC|null |2026-07-06 23:12:12 UTC|null |U |2026-07-07
03:25:35.591006 UTC|ANG THONG|CENTRAL|CHAIYO |True-LTE FDD|LTE FDD
|2026-07-07|03 |4G |
|07/02/2026 03:40:03|07/02/2026 03:30:10|1105839164|ericsson-enm-corba@150.23
6.236.78:6796|ATG8505T_2NB01 |07/02/2026 03:30:10|ATG-A_0019 | ATG0011|Tru
e_Ericsson_ENM_UTRAN_CENTRAL_4A|ACTIVE |0 |1970-01-01 00:00:00
UTC|null |eNode B |2 |null |10.23.17.8 |11058391


```

        " | ",
        F.col("X733SPECIFICPROB"),
        F.col("SUMMARY")
    )
)

# Verify
alarm_df_filtered.select(
    "X733SPECIFICPROB",
    "SUMMARY",
    "ALARM_DESCRIPTION"
).show(10, truncate=False)

```

26/07/16 05:40:39 WARN DAGScheduler: Broadcasting large task binary with size 6.6 MiB

26/07/16 05:42:22 WARN DAGScheduler: Broadcasting large task binary with size 4.4 MiB

```

+-----+-----+
+-----+-----+
|X733SPECIFICPROB          |SUMMARY                      |ALARM_DESCRIPTION
|
+-----+-----+
+-----+-----+
|Service Unavailable       |Service Unavailable         |Service Unavailable
| Service Unavailable      | |                           |
|Service Degraded          |Service Degraded           |Service Degraded | S
ervice Degraded           | |                           |
|Service Unavailable       |Service Unavailable         |Service Unavailable
| Service Unavailable      | |                           |
|Service Unavailable       |Service Unavailable         |Service Unavailable
| Service Unavailable      | |                           |
|Service Unavailable       |Service Unavailable         |Service Unavailable
| Service Unavailable      | |                           |
|PLMN Service Redundancy Lost|PLMN Service Redundancy Lost|PLMN Service Redunda
ncy Lost | PLMN Service Redundancy Lost|
|Service Degraded          |Service Degraded           |Service Degraded | S
ervice Degraded           | |                           |
|Service Unavailable       |Service Unavailable         |Service Unavailable
| Service Unavailable      | |                           |
|Service Unavailable       |Service Unavailable         |Service Unavailable
| Service Unavailable      | |                           |
|Service Degraded          |Service Degraded           |Service Degraded | S
ervice Degraded           | |                           |
+-----+-----+
+-----+-----+
only showing top 10 rows

```

In [54]: `alarm_pd = alarm_df_filtered.toPandas()`

```
26/07/16 05:42:27 WARN DAGScheduler: Broadcasting large task binary with size
6.6 MiB
26/07/16 05:44:16 WARN DAGScheduler: Broadcasting large task binary with size
4.4 MiB
```

```
In [55]: filtered_ticket_df.shape
```

```
Out[55]: (329, 36)
```

```
In [56]: num_rows = alarm_df_filtered.count()
num_cols = len(alarm_df_filtered.columns)

print(f"Shape: ({num_rows}, {num_cols})")
```

```
26/07/16 05:44:26 WARN DAGScheduler: Broadcasting large task binary with size
6.6 MiB
26/07/16 05:46:06 WARN DAGScheduler: Broadcasting large task binary with size
4.4 MiB
Shape: (149143, 48)
```

```
In [57]: alarm_df_filtered.printSchema()
```

```

root
|-- RECTIMESTAMP: string (nullable = true)
|-- FIRSTOCCURRENCE: string (nullable = true)
|-- IDENTIFIER: string (nullable = true)
|-- LASTOCCURRENCE: string (nullable = true)
|-- SERVICE: string (nullable = true)
|-- EMS_NAME: string (nullable = true)
|-- NODESTATUS: string (nullable = true)
|-- SERVICEAFFECTING: string (nullable = true)
|-- CLEARTIME: string (nullable = true)
|-- OBJECT: string (nullable = true)
|-- NODETYPE: string (nullable = true)
|-- SEVERITY: string (nullable = true)
|-- EXTENDEDATTR: string (nullable = true)
|-- ELEMENTMANAGERIP: string (nullable = true)
|-- EVENTID: string (nullable = true)
|-- ADDITIONALSTRING: string (nullable = true)
|-- ALERTKEY: string (nullable = true)
|-- SITENAME: string (nullable = true)
|-- ORIGINALSEVERITY: string (nullable = true)
|-- SITESTATUS: string (nullable = true)
|-- SITEID: string (nullable = true)
|-- CLASS: string (nullable = true)
|-- OBJECTSTATUS: string (nullable = true)
|-- CLEAREDDBY: string (nullable = true)
|-- X733SPECIFICPROB: string (nullable = true)
|-- MANAGER: string (nullable = true)
|-- NODEALIAS: string (nullable = true)
|-- SUMMARY: string (nullable = true)
|-- NODE: string (nullable = true)
|-- TT_FLAG: string (nullable = true)
|-- SERVERSERIAL: string (nullable = true)
|-- TMTRANSSTATE: string (nullable = true)
|-- TTCREATIONTIME: string (nullable = true)
|-- TMTRANSBEGINTIME: string (nullable = true)
|-- TT_ID: string (nullable = true)
|-- RECEIVEDATPROBE: string (nullable = true)
|-- physicalCard: string (nullable = true)
|-- OP_TYPE: string (nullable = true)
|-- OP_TIME_DWH: string (nullable = true)
|-- CIRCLE: string (nullable = true)
|-- REGION: string (nullable = true)
|-- LOCATION: string (nullable = true)
|-- BACKHAULTYPE: string (nullable = true)
|-- CONTROLNE: string (nullable = true)
|-- DATE_ID: string (nullable = false)
|-- HOUR_ID: string (nullable = false)
|-- Technology: string (nullable = true)
|-- ALARM_DESCRIPTION: string (nullable = false)

```

```

In [58]: filtered_df['SITE_KEY'] = filtered_df['SITE'].str[:7] # or use regex: str.ext

# Step 2: Join filtered_df with filtered_ticket_df (329 rows - tiny, pure pandas)

```

```
merged_ticket = filtered_df.merge(
    filtered_ticket_df,
    left_on='SITE_KEY',
    right_on='HPD_CI',
    how='left'
)
```

In [59]: merged_ticket.shape

Out[59]: (6298302, 119)

In [60]: merged_ticket.head()

Out[60]:

	COLLECTTIME	CELLID	SITE	NR_SA_Paging_Discard_
0	2026-07-01 19:00:00+00:00	ATG0011H_7NB04_S02	ATG0011H_7NB04	
1	2026-07-01 19:45:00+00:00	ATG0011H_7NB04_S02	ATG0011H_7NB04	
2	2026-07-01 19:30:00+00:00	ATG0011H_7NB04_S02	ATG0011H_7NB04	
3	2026-07-02 12:45:00+00:00	ATG0011H_7NB04_S02	ATG0011H_7NB04	
4	2026-07-02 12:15:00+00:00	ATG0011H_7NB04_S02	ATG0011H_7NB04	

5 rows × 119 columns

In [61]: kpi_complete_outage_df['SITE_KEY'] = kpi_complete_outage_df['SITE'].str[:7]

In [62]: *# Number of duplicate SITEIDs*
duplicate_count = alarm_pd.duplicated(subset=["SITEID"]).sum()
print(f"Duplicate SITEIDs: {duplicate_count}")

Duplicate SITEIDs: 148822

In [63]: alarm_pd_unique = alarm_pd.drop_duplicates(
subset=["SITEID", "NODE"],
keep="first"
)

print("Original shape:", alarm_pd.shape)
print("After removing duplicates:", alarm_pd_unique.shape)

Original shape: (149143, 48)

After removing duplicates: (3563, 48)

In [64]: *# Merge KPI data with Ticket data*
consolidated_df1 = kpi_complete_outage_df.merge(
filtered_ticket_df,
left_on="SITE_KEY",


```
        right_on="HPD_CI",
        how="left",
        suffixes=("", "_ticket")
    )
```

In [65]: consolidated_df1.shape

Out[65]: (21230, 119)

```
In [67]: # Merge with Alarm data
consolidated_df = consolidated_df1.merge(
    alarm_pd_unique,
    left_on="SITE_KEY",
    right_on="SITEID",
    how="left",
    suffixes=("", "_alarm")
)
```

In [122... consolidated_df.shape

Out[122... (83322, 167)

```
In [111... merged_df = kpi_complete_outage_df.merge(
    filtered_ticket_df,
    left_on="SITE_KEY",
    right_on="HPD_CI",
    how="left",
    suffixes=("", "_ticket")
)
```

```
In [112... merged_df = merged_df.merge(
    alarm_pd,
    left_on="SITE_KEY",
    right_on="SITEID",
    how="left",
    suffixes=("", "_ticket")
)
```

In [123... merged_df.shape

Out[123... (3645646, 167)

In [124... merged_df1 = merged_df

In [125... merged_df1.shape

Out[125... (3645646, 167)

```
In [126... #prints a summary of the complete site outage KPI dataset
#by displaying the number of unique sites, unique cells, and the total number
print("=*50)
```

```

print("COMPLETE SITE OUTAGE SUMMARY")
print("="*50)

print("Distinct Sites :",
      kpi_complete_outage_df["SITE_KEY"].nunique())

print("Distinct Cells :",
      kpi_complete_outage_df["CELLID"].nunique())

print("Rows :",
      len(kpi_complete_outage_df))

```

```

=====
COMPLETE SITE OUTAGE SUMMARY
=====
Distinct Sites : 610
Distinct Cells : 1901
Rows : 21230

```

```

In [127... #identifies all cells with 0% NR_Cell_Availability (complete cell outages),
#groups them by CELLID, counts how many outage records each cell has, and returns
cell_outages = (
    merged_df[
        filtered_df["NR_Cell_Availability"] == 0
    ]
    .groupby("CELLID")
    .agg(
        outage_count=("NR_Cell_Availability", "count"),
        site=("SITE_KEY", "first")
    )
    .reset_index()
)

```

```

/tmp/ipykernel_1154/167009037.py:4: UserWarning: Boolean Series key will be reindexed to match DataFrame index.
merged_df[

```

```

In [128... """sorts the cell_outages DataFrame in descending order of outage_count,
so that the cells with the highest number of outage occurrences appear at the top

cell_outages = cell_outages.sort_values(
    "outage_count",
    ascending=False
)

```

```

In [129... top20_cells = cell_outages.head(20)

top20_cells

```

Out[129...

	CELLID	outage_count	site
58	MHS0059H_7NB02_S01	3167	MHS0059
61	MHS0059H_7NB02_S04	2372	MHS0059
59	MHS0059H_7NB02_S02	1216	MHS0059
115	SRB0067H_7NB02_S04	1001	SRB0067
60	MHS0059H_7NB02_S03	972	MHS0059
112	SRB0067H_7NB02_S01	912	SRB0067
114	SRB0067H_7NB02_S03	742	SRB0067
95	SMK0084H_7NB02_S03	727	SMK0084
132	TAK7313L_6MM06_R11	615	TAK6708
96	SMK0084H_7NB02_S04	587	SMK0084
113	SRB0067H_7NB02_S02	567	SRB0067
94	SMK0084H_7NB02_S02	538	SMK0084
127	TAK6708H_7NB09_S02	467	TAK6708
93	SMK0084H_7NB02_S01	275	SMK0084
109	SRB0018L_68T01_S13	220	SRB0018
84	PSN6716H_7NB01_S03	124	PSN6716
64	N07-BKK3100-1C	104	BKK3100
62	N07-BKK3100-1A	104	BKK3100
63	N07-BKK3100-1B	103	BKK3100
57	MHS0018H_7NB01_S03	74	MHS0018

```
In [130... print(cell_outages["outage_count"].describe())
```

```
count      137.000000
mean       116.941606
std        391.749184
min         1.000000
25%         3.000000
50%         7.000000
75%        24.000000
max        3167.000000
Name: outage_count, dtype: float64
```

```
In [131... import matplotlib.pyplot as plt
        """creates a bar chart showing the Top 20 cells with the highest number of out
        height corresponds to the number of outage occurrences (outage_count), making
        affected cells."""
        top20_cells.plot()
```

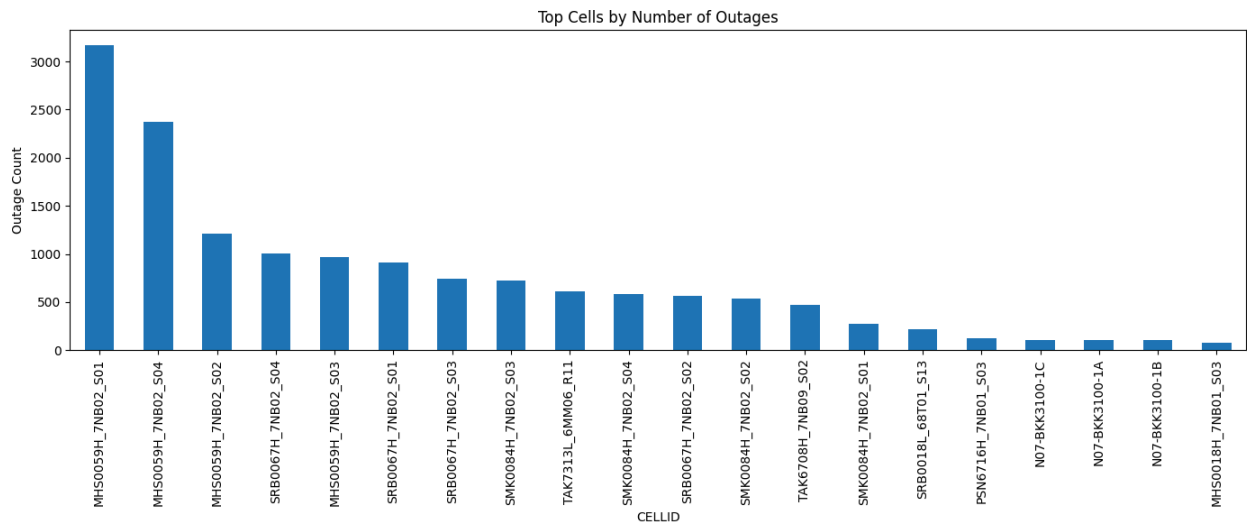
```

x="CELLID",
y="outage_count",
kind="bar",
figsize=(14,6),
legend=False
)

plt.title("Top Cells by Number of Outages")
plt.ylabel("Outage Count")
plt.tight_layout()

plt.show()

```



```

In [132... filtered_ticket_df.head(20).to_csv("ticket_consolidated.csv",index=False)

```

```

In [136... merged_df1["EVENT_START_TIME"] = pd.to_datetime(
    merged_df1["EVENT_START_TIME"],
    errors="coerce"
)

merged_df1["EVENT_END_TIME"] = pd.to_datetime(
    merged_df1["EVENT_END_TIME"],
    errors="coerce"
)

merged_df1["LAST_RESOLVED_DATE.UTC"] = pd.to_datetime(
    merged_df1["LAST_RESOLVED_DATE.UTC"],
    errors="coerce"
)

```

```

In [137... merged_df1["OUTAGE_DURATION"] = (
    merged_df1["LAST_RESOLVED_DATE.UTC"]
    .fillna(merged_df1["EVENT_END_TIME"])
    - merged_df1["EVENT_START_TIME"]
)

```

```
In [134... alarm_pd.columns
```

```
Out[134... Index(['RECTIMESTAMP', 'FIRSTOCCURRENCE', 'IDENTIFIER', 'LASTOCCURRENCE',  
        'SERVICE', 'EMS_NAME', 'NODESTATUS', 'SERVICEAFFECTING', 'CLEARTIME',  
        'OBJECT', 'NODETYPE', 'SEVERITY', 'EXTENDEDATTR', 'ELEMENTMANAGERIP',  
        'EVENTID', 'ADDITIONALSTRING', 'ALERTKEY', 'SITENAME',  
        'ORIGINALSEVERITY', 'SITESTATUS', 'SITEID', 'CLASS', 'OBJECTSTATUS',  
        'CLEAREDBY', 'X733SPECIFICPROB', 'MANAGER', 'NODEALIAS', 'SUMMARY',  
        'NODE', 'TT_FLAG', 'SERVERSERIAL', 'TMTRANSSTATE', 'TTCREATIONTIME',  
        'TMTRANSBEGINTIME', 'TT_ID', 'RECEIVEDATPROBE', 'physicalCard',  
        'OP_TYPE', 'OP_TIME_DWH', 'CIRCLE', 'REGION', 'LOCATION',  
        'BACKHAULTYPE', 'CONTROLNE', 'DATE_ID', 'HOUR_ID', 'Technology',  
        'ALARM_DESCRIPTION'],  
        dtype='object')
```

```
In [138... merged_df1["OUTAGE_DURATION_MIN"] = (  
    merged_df1["OUTAGE_DURATION"]  
    .dt.total_seconds() / 60  
)
```

```
In [139... """function categorizes an outage duration (in minutes) into predefined time b  
If the duration is missing (NaN), it labels the outage as "Ongoing"; otherwise  
it assigns it to ranges such as ≤1 Hour, 1-2 Hours, 2-3 Hours, 3-4 Hours, 4-24  
def bucket(x):  
    if pd.isna(x):  
        return "Ongoing"  
  
    elif x <= 60:  
        return "<=1 Hour"  
  
    elif x <= 120:  
        return "1-2 Hours"  
  
    elif x <= 180:  
        return "2-3 Hours"  
  
    elif x <= 240:  
        return "3-4 Hours"  
  
    elif x <= 1440:  
        return "4-24 Hours"  
  
    elif x <= 2880:  
        return "1-2 Days"  
  
    elif x <= 10080:  
        return "2-7 Days"  
  
    else:  
        return ">7 Days"
```

```
In [140... """creates a new column OUTAGE_BUCKET in merged_df1 by applying the bucket() f  
in the OUTAGE_DURATION_MIN column.
```

As a result, every outage is classified into a meaningful duration category (e

```
merged_df1["OUTAGE_BUCKET"] = (  
    merged_df1["OUTAGE_DURATION_MIN"]  
    .apply(bucket)  
)
```

```
In [171]... #counts the number of records in each OUTAGE_BUCKET category and displays the  
merged_df1["OUTAGE_BUCKET"].value_counts(dropna=False)
```

```
Out[171]... Ongoing          1794321  
2-3 Hours          1498787  
<=1 Hour           289233  
4-24 Hours         32349  
1-2 Hours          24534  
3-4 Hours           4775  
1-2 Days            1647  
Name: OUTAGE_BUCKET, dtype: int64
```

```
In [142]... # For each outage duration bucket (for example, 1-2 hrs, 2-3 hrs, 4-8 hrs), yo
```

"""code creates a summary of outage duration buckets by first removing duplicate combinations to avoid double counting, and then grouping the data by OUTAGE_BUCKET

For each duration bucket, it calculates the number of unique affected cells (Cells) and the number of unique incidents (Incidents)"""

```
bucket_summary = (  
    merged_df1[  
        ["INCIDENT_NUMBER",  
         "CELLID",  
         "OUTAGE_BUCKET"]  
    ]  
    .drop_duplicates()  
)  
  
bucket_summary = (  
    bucket_summary  
    .groupby("OUTAGE_BUCKET")  
    .agg(  
        Cells=("CELLID", "nunique"),  
        Incidents=("INCIDENT_NUMBER", "nunique")  
    )  
    .reset_index()  
)  
  
bucket_summary
```

Out[142...

	OUTAGE_BUCKET	Cells	Incidents
0	1-2 Days	3	1
1	1-2 Hours	14	5
2	2-3 Hours	21	6
3	3-4 Hours	7	3
4	4-24 Hours	5	2
5	<=1 Hour	41	12
6	Ongoing	1810	45

In [167...

```
cell_bucket_counts = (  
    merged_df1  
        .groupby(["CELLID", "OUTAGE_BUCKET"])  
        .size()  
        .reset_index(name="COUNT")  
    )  
  
cell_bucket_counts.to_csv("cell_bucket_wise_outages.csv", index = False)
```

In [172...

```
cell_bucket_summary = (  
    merged_df1  
        .groupby(["SITE", "CELLID", "OUTAGE_BUCKET"])  
        .size()  
        .unstack(fill_value=0)  
    )  
  
cell_bucket_counts.to_csv("site_cell_bucket_wise_outages.csv", index = False)
```

In [143...

```
merged_df.columns.tolist()
```

```
Out[143]: ['COLLECTTIME',
           'CELLID',
           'SITE',
           'NR_SA_Paging_Discard_Rate',
           'NR_DL_Active_UEs',
           'NR_UL_Active_UEs',
           'NR_DL_MAC_Volume_MB',
           'NR_UL_MAC_Volume_MB',
           'NR_DL_Active_UEs_True',
           'NR_UL_Active_UEs_True',
           'NR_DL_RBSym_Util',
           'NR_UL_RBSym_Util',
           'NR_Msg2_Attempt_SR',
           'NR_Msg2_Attempts',
           'NR_RACH_SR',
           'NR_RACH_Att',
           'NR_UL_MAC_Cell_Thp_total_time_Mbps',
           'NR_DL_MAC_Cell_Thp_total_time_Mbps',
           'NR_DL_HARQ_BLER_256QAM',
           'NR_DL_RLC_BLER',
           'NR_Avg_DL_MAC_Thp_Mbps',
           'NR_Avg_UL_MAC_UE_Thp_Mbps',
           'NR_DL_MAC_Cell_Thp_Mbps',
           'NR_UL_MAC_Cell_Thp_Mbps',
           'NR_UL_HARQ_BLER_64_QAM',
           'NR_UL_RLC_BLER',
           'NR_UL_HARQ_BLER',
           'NR_DL_HARQ_BLER',
           'NR_UL_DTX_Rate',
           'NR_DL_DTX_Rate',
           'NR_Cell_Availability',
           'NR_Avg_UL_PUCCH_SINR_dB',
           'NR_Avg_UL_PUSCH_SINR_dB',
           'NR_RANK_Below3_Rate',
           'NR_Avg_CQI',
           'NR_Avg_CQI_64QAM',
           'NR_Avg_CQI_256QAM',
           'NR_Avg_UL_Pathloss_dB',
           'NR_Avg_UL_Power_Headroom_dB',
           'NR_Avg_UL_Pathloss_95percentile_dB',
           'NR_UL_RSSI',
           'NR_Avg_MCS_Table1',
           'NR_Avg_MCS_Table2',
           'NR_TA',
           'DL_Thp_Bad_Sessions',
           'DL_Thp_Fair_Sessions',
           'DL_Thp_Good_Sessions',
           'DL_Thp_Excellent_Sessions',
           'UL_Thp_Bad_Sessions',
           'UL_Thp_Fair_Sessions',
           'UL_Thp_Good_Sessions',
           'UL_Thp_Excellent_Sessions',
           'NR_Avg_UL_MAC_Thp_EBS_Mbps',
           'NR_Avg_DL_MAC_Thp_EBS_Mbps',
```


'DL_Total_Sessions',
'UL_Total_Sessions',
'NR_Avg_DL_Latency_ms',
'NR_Cell_Availability_Num',
'NR_ENDC_Setup_Att',
'NR_ENDC_Setup_SR',
'NR_SA_RRC_ATT',
'NR_SA_RRC_SSSR',
'NR_SA_NG_Sig_SR',
'NR_ENDC_Inter_sgNodeB_PSCell_Change_SR',
'NR_ENDC_Intra_sgNodeB_PSCell_Change_SR',
'NR_SA_NR_to_LTE_RWR_Ratio',
'NR_SA_INTER_GNB_HO_SR',
'NR_SA_INTRA_GNB_HO_SR',
'NR_SA_HO_Prep_Succ_Rate',
'NR_SA_HO_Exec_Succ_Rate',
'NR_SA_HO_SR',
'NR_ENDC_SARR',
'NR_ENDC_RRC_CU_Max',
'NR_SA_RRC_CU_Max',
'NR_ENDC_RRC_CU_Avg',
'NR_SA_DRB_EST_SR',
'NR_SA_Drop_Rate',
'NR_SgNB_Abnormal_RR',
'NR_RRC_SSR',
'ESS_TOTAL_PRB_AVG_UTIL_DL',
'ESS_NR_PRB_AVG_UTIL_DL',
'essEnabled',
'SITE_KEY',
'INCIDENT_NUMBER',
'VENDOR_TICKET_NUMBER',
'ASSIGNED_GROUP',
'IMPACT',
'PRIORITY',
'URGENCY',
'COMPANY',
'DESCRIPTION',
'DETAILED_DESCRIPTION',
'HPD_CI',
'NODETYPE',
'OP_TIME',
'OP_TYPE',
'SUBMIT_DATE',
'submit_date_UTC',
'submit_date_BKK',
'Categorization_Tier_1',
'Categorization_Tier_2',
'Categorization_Tier_3',
'RESOLUTION_CATEGORY',
'RESOLUTION_CATEGORY_TIER_2',
'RESOLUTION_CATEGORY_TIER_3',
'INC_STATUS',
'STATUS',
'REPORTED_SOURCE',

'EVENT_END_TIME',
'EVENT_START_TIME',
'PENDING_TIME',
'PENDING_DUE_TIME',
'rn',
'SEVERITY',
'LAST_RESOLVED_DATE_UTC',
'LAST_RESOLVED_DATE_BKK',
'CLOSED_DATE_UTC',
'CLOSED_DATE_BKK',
'Technology',
'RECTIMESTAMP',
'FIRSTOCCURRENCE',
'IDENTIFIER',
'LASTOCCURRENCE',
'SERVICE',
'EMS_NAME',
'NODESTATUS',
'SERVICEAFFECTING',
'CLEARTIME',
'OBJECT',
'NODETYPE_ticket',
'SEVERITY_ticket',
'EXTENDEDATTR',
'ELEMENTMANAGERIP',
'EVENTID',
'ADDITIONALSTRING',
'ALERTKEY',
'SITENAME',
'ORIGINALSEVERITY',
'SITESTATUS',
'SITEID',
'CLASS',
'OBJECTSTATUS',
'CLEAREDBY',
'X733SPECIFICPROB',
'MANAGER',
'NODEALIAS',
'SUMMARY',
'NODE',
'TT_FLAG',
'SERVERSERIAL',
'TMTRANSSTATE',
'TTCREATIONTIME',
'TMTRANSBEGINTIME',
'TT_ID',
'RECEIVEDATPROBE',
'physicalCard',
'OP_TYPE_ticket',
'OP_TIME_DWH',
'CIRCLE',
'REGION',
'LOCATION',
'BACKHAULTYPE',

```
'CONTROLNE',
'DATE_ID',
'HOURL_ID',
'Technology_ticket',
'ALARM_DESCRIPTION',
'OUTAGE_DURATION',
'OUTAGE_DURATION_MIN',
'OUTAGE_BUCKET']
```

```
In [144... # KPI availability column
avail_col = 'NR_Cell_Availability'

# Step 1: Find cells that have 0 availability for ALL timestamps
cell_zero_availability = (
    merged_df.groupby('CELLID')[avail_col]
    .apply(lambda x: (x == 0).all())
)

zero_avail_cells = cell_zero_availability[
    cell_zero_availability
].index.tolist()

print(f"Cells with 0% availability throughout: {len(zero_avail_cells)}")
```

Cells with 0% availability throughout: 1901

```
In [148... """creates a mapping between each CELLID and its corresponding SITE by selecti
removing duplicate combinations.

The resulting cell_site_map provides a unique Cell-to-Site reference,"""
cell_site_map = (
    merged_df[['CELLID', 'SITE']]
    .drop_duplicates()
)

print(cell_site_map.head())
```

	CELLID	SITE
0	ATG0019H_7NB01_S01	ATG0019T_2NB01
1	ATG0019H_7NB01_S03	ATG0019T_2NB01
2	ATG0019H_7NB01_S02	ATG0019T_2NB01
84	ATG0022H_7NB03_S02	ATG0022P_9NB03
85	ATG0022H_7NB03_S04	ATG0022P_9NB03

```
In [185... """identifies fully down sites by first calculating the total number of cells
associated with each site and the number of cells with 0% availability for each

A site is classified as fully down if all of its associated cells have 0% avail
(i.e., the total cell count equals the zero-availability cell count)"""

# Total cells in every site
total_cells_per_site = (
    cell_site_map.groupby('SITE')['CELLID']
    .nunique()
```

```

)

# Zero availability cells in every site
zero_cells_per_site = (
    cell_site_map[cell_site_map['CELLID'].isin(zero_avail_cells)]
    .groupby('SITE')['CELLID']
    .nunique()
    .reindex(total_cells_per_site.index, fill_value=0)
)

# Sites where all associated cells are zero
fully_down_sites = total_cells_per_site.index[
    total_cells_per_site == zero_cells_per_site
].tolist()

fully_down_sites_names = [site[:7] for site in fully_down_sites]

print(f"Fully Down Sites: {len(fully_down_sites)}")
print(f"Fully Down Sites: {fully_down_sites}")

```

Fully Down Sites: 648

Fully Down Sites: ['ATG0013P_9NB02', 'ATG0019T_2NB01', 'ATG0022L_6NB01', 'ATG0022P_9NB03', 'ATG0024T_2NB01', 'ATG0105T_2NB01', 'ATG1101P_9NB02', 'ATG1532P_9NB01', 'ATG6004P_9NB01', 'ATG6722P_9NB01', 'ATG7106H_7NB03', 'ATG7107P_9NB01', 'ATG7130P_9NB01', 'ATG8505L_6NB02', 'ATG8512P_9NB02', 'ATG8527P_9NB01', 'ATG8539P_9NB01', 'AYT0026P_9NB01', 'AYT0066P_9NB01', 'AYT0093P_9NB02', 'AYT0096P_9NB01', 'AYT0112T_2NB01', 'AYT0143T_2NB01', 'AYT0175L_6NB04', 'AYT0226T_2NB01', 'AYT0228P_9NB01', 'AYT0318P_9NB02', 'AYT0369T_2NB01', 'AYT0381P_9NB01', 'AYT1105L_6NB03', 'AYT1304L_6NB04', 'AYT1307H_7NB04', 'AYT1307L_6NB02', 'AYT1411P_9NB01', 'AYT6761P_9NB01', 'AYT6769L_6NB03', 'AYT7158T_2NB01', 'AYT7160H_7NB02', 'AYT7191L_6NB03', 'AYT7204P_9NB01', 'AYT7211P_9NB01', 'AYT7674L_6NB01', 'AYT8524L_6NB02', 'AYT8547L_6NB05', 'AYT8579P_9NB01', 'AYT8605L_6NB02', 'AYT8657H_7NB04', 'AYTEV44L_6NB02', 'BKFTS02L_6NB04', 'BKFTS02P_9NB04', 'BKK3100B', 'CMI0157P_9NB02', 'CMI0233P_9NB01', 'CMI0254L_6NB06', 'CMI0254P_9NB01', 'CMI0384P_9NB01', 'CMI0425P_9NB02', 'CMI0606P_9NB02', 'CMI0608P_9NB02', 'CMI0723P_9NB02', 'CMI0772P_9NB02', 'CMI0805P_9NB02', 'CMI0836P_9NB02', 'CMI0876P_9NB01', 'CMI0900P_9NB02', 'CMI0929P_9NB02', 'CMI1537H_7NB02', 'CMI1537L_6NB03', 'CMI1819H_7NB04', 'CMI1912P_9NB03', 'CMI1936L_6NB02', 'CMI1936P_9NB04', 'CMI1991P_9NB01', 'CMI1995P_9NB04', 'CMI2699L_6NB03', 'CMI6333L_6NB04', 'CMI6470L_6NB02', 'CMI6583P_9NB02', 'CMI6706R_1NB02', 'CMI6737H_7NB02', 'CMI6745L_6NB02', 'CMI6755P_9NB03', 'CMI6782L_6NB02', 'CMI6827P_9NB02', 'CMI6843R_1NB03', 'CMI6931L_6NB03', 'CMI6999L_6NB05', 'CMI7300L_6NB03', 'CMI7504P_9NB01', 'CMI7510H_7NB02', 'CMI7562H_7NB04', 'CMI7619R_1NB01', 'CMI7624L_6NB02', 'CMI7628P_9NB02', 'CMI7644R_1NB01', 'CMI7759L_6NB02', 'CMI8519P_9NB02', 'CMI8527P_9NB01', 'CMI8610L_6NB02', 'CMI8799L_6NB03', 'CMI8799P_9NB04', 'CMI8800P_9NB03', 'CMI8863L_6NB02', 'CMIA002H_7NB02', 'CMIA060L_6NB02', 'CMIA069P_9NB03', 'CMIA113R_1NB04', 'CMIA284R_1NB01', 'CMIA290R_1NB01', 'CMIA304R_1NB01', 'CMIA323P_9NB02', 'CMIC067L_6NB01', 'CNT6704L_6NB06', 'CNT6704P_9NB01', 'CNT7154R_1NB01', 'CNT8522H_7NB01', 'CNT8541P_9NB02', 'CRI0066P_9NB02', 'CRI0126P_9NB02', 'CRI0144P_9NB01', 'CRI0377P_9NB02', 'CRI0465P_9NB02', 'CRI1717P_9NB01', 'CRI1975L_6NB08', 'CRI1975P_9NB01', 'CRI6301L_6NB02', 'CRI6330H_7NB02', 'CRI6373H_7NB02', 'CRI6403L_6NB02', 'CRI6403L_6NB03', 'CRI6428L_6NB02', 'CRI6462L_6NB01', 'CRI6462P_9NB03', 'CRI6527L_6NB02', 'CRI6701H_7NB02', 'CRI6715P_9NB01', 'CRI6716P_9NB01', 'CRI6717H_7NB02', 'CRI6720L_6NB02', 'CRI6721P_9NB01', 'CRI6738P_9NB01', 'CRI6756L_6NB05', 'CRI6756L_6NB06', 'CRI6756P_9NB01', 'CRI6762P_9NB01', 'CRI6763R_1NB02', 'CRI6775R_1NB02', 'CRI6788H_7NB02', 'CRI6894T_2NB01', 'CRI7197P_9NB03', 'CRI7649P_9NB03', 'CRI7904H_7NB03', 'CRI8571L_6NB02', 'CRI8655P_9NB02', 'CRIC008P_9NB01', 'KCN0004H_7NB01', 'KCN0024P_9NB02', 'KCN0026P_9NB03', 'KCN0036P_9NB02', 'KCN0038T_2NB01', 'KCN0039P_9NB01', 'KCN0040T_2NB01', 'KCN0051P_9NB02', 'KCN0063H_7NB01', 'KCN0076T_2NB01', 'KCN0084T_2NB01', 'KCN0086P_9NB02', 'KCN0087T_2NB01', 'KCN0104P_9NB02', 'KCN0132P_9NB02', 'KCN0136P_9NB02', 'KCN0143T_2NB01', 'KCN0180H_7NB01', 'KCN0204T_2NB01', 'KCN0273P_9NB02', 'KCN0279T_2NB01', 'KCN0293T_2NB01', 'KCN0301T_2NB01', 'KCN0354T_2NB01', 'KCN1034P_9NB01', 'KCN1301P_9NB01', 'KCN1502H_7NB04', 'KCN1502L_6NB02', 'KCN6037P_9NB01', 'KCN6046R_1NB02', 'KCN6705T_2NB01', 'KCN6717T_2NB01', 'KCN6751R_1NB02', 'KCN6767P_9NB01', 'KCN6776H_7NB02', 'KCN6802P_9NB01', 'KCN6827H_7NB02', 'KCN7185P_9NB02', 'KCN7401L_6NB03', 'KCN7905H_7NB02', 'KCN7906L_6NB02', 'KCN7918H_7NB02', 'KCN8501T_2NB01', 'KCN8551T_2NB01', 'KCNC001H_7NB01', 'KCNC002P_9NB02', 'KCNC003T_2NB01', 'KCNC005T_2NB01', 'KPP0048P_9NB01', 'KPP0198P_9NB01', 'KPP0224P_9NB02', 'KPP6702H_7NB03', 'KPP6716P_9NB03', 'KPP6743P_9NB01', 'LBR0003T_2NB01', 'LBR0015T_2NB01', 'LBR0028P_9NB03', 'LBR0037P_9NB03', 'LBR0056L_6NB01', 'LBR0072T_2NB01', 'LBR0077T_2NB01', 'LBR0078T_2NB01', 'LBR0079T_2NB01', 'LBR0082T_2NB01', 'LBR0085P_9NB02', 'LBR0091P_9NB01', 'LBR0099T_2NB01', 'LBR0227T_2NB01', 'LBR0264T_2NB01', 'LBR1021P_9NB01', 'LBR1401P_9NB01', 'LBR1402L_6NB02', 'LBR1534R_1NB02', 'LBR6021P_9NB01', 'LB

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```
In [186... fully_down_df = merged_df[
    merged_df['SITE'].isin(fully_down_sites)
].copy()

print("Rows:", fully_down_df.shape[0])
print("Sites:", fully_down_df['SITE'].nunique())
print("Cells:", fully_down_df['CELLID'].nunique())
```

Rows: 3645646
Sites: 648
Cells: 1901

```
In [152... ticket_df = filtered_ticket_df.copy()

ticket_df['EVENT_START_TIME'] = pd.to_datetime(
    ticket_df['EVENT_START_TIME'],
    errors='coerce'
)

ticket_df['EVENT_END_TIME'] = pd.to_datetime(
    ticket_df['EVENT_END_TIME'],
    errors='coerce'
)

ticket_df['LAST_RESOLVED_DATE_UTC'] = pd.to_datetime(
    ticket_df['LAST_RESOLVED_DATE_UTC'],
    errors='coerce'
)
```

```
In [155... ticket_df["is_unresolved"] = (
    ticket_df["EVENT_END_TIME"].isna() &
    ticket_df["LAST_RESOLVED_DATE_UTC"].isna()
)
print(ticket_df['is_unresolved'].value_counts())
```

```
True      190
False     139
Name: is_unresolved, dtype: int64
```

```
In [156... ticket_df['OUTAGE_DURATION_HOURS'] = (
    ticket_df['EVENT_END_TIME']
    .fillna(ticket_df['EVENT_END_TIME'])
    .sub(ticket_df['EVENT_START_TIME'])
    .dt.total_seconds() / 3600
)
```

```
In [157... ticket_df['OUTAGE_DURATION_HOURS'].describe()
```

```
Out[157... count      131.000000
mean         2.466785
std          3.483675
min          0.260833
25%          0.713889
50%          1.387778
75%          2.596389
max          23.224444
Name: OUTAGE_DURATION_HOURS, dtype: float64
```

```
In [158... ticket_df['OUTAGE_DURATION_HOURS'].quantile(
    [0.25, 0.5, 0.75, 0.9, 0.95, 0.99]
)
```



```
Out[158... 0.25      0.713889
           0.50      1.387778
           0.75      2.596389
           0.90      4.862778
           0.95      8.563750
           0.99     16.829389
           Name: OUTAGE_DURATION_HOURS, dtype: float64
```

```
In [161... def outage_bucket(hours):
            if pd.isna(hours):
                return "Ongoing"
            elif hours < 1:
                return "<1 Hour"
            elif hours < 3:
                return "1-3 Hours"
            elif hours < 6:
                return "3-6 Hours"
            elif hours < 12:
                return "6-12 Hours"
            elif hours < 24:
                return "12-24 Hours"
            elif hours < 72:
                return "1-3 Days"
            elif hours < 168:
                return "3-7 Days"
            else:
                return ">7 Days"

ticket_df["OUTAGE_BUCKET"] = ticket_df["OUTAGE_DURATION_HOURS"].apply(outage_b
```

```
In [164... bucket_summary = (
    ticket_df.groupby("OUTAGE_BUCKET")
    .agg(
        Tickets=("INCIDENT_NUMBER", "count"),
        Resolved=("is_unresolved", lambda x: (~x).sum()),
        Unresolved=("is_unresolved", "sum"),
        Avg_Duration_Hours=("OUTAGE_DURATION_HOURS", "mean")
    )
    .sort_values("Avg_Duration_Hours")
)

bucket_summary
```

Out[164...

	Tickets	Resolved	Unresolved	Avg_Duration_Hours
OUTAGE_BUCKET				
<1 Hour	51	51	0	0.631269
1-3 Hours	54	54	0	1.807870
3-6 Hours	15	15	0	3.905278
6-12 Hours	6	6	0	8.247546
12-24 Hours	5	5	0	17.052944
Ongoing	198	8	190	NaN

In [165...

```
bucket_percent = (  
    ticket_df["OUTAGE_BUCKET"]  
    .value_counts(normalize=True)  
    .mul(100)  
    .round(2)  
    .rename("Percentage")  
)  
  
bucket_percent
```

Out[165...

```
Ongoing      60.18  
1-3 Hours    16.41  
<1 Hour      15.50  
3-6 Hours     4.56  
6-12 Hours    1.82  
12-24 Hours   1.52  
Name: Percentage, dtype: float64
```

Alarm Data EDA

In [173...

```
print("Alarm rows:", alarm_pd.shape[0])  
  
print("\nUnique Sites:", alarm_pd['SITEID'].nunique())  
  
print("\nUnique Alarm Types:", alarm_pd['X733SPECIFICPROB'].nunique())  
  
alarm_pd[['SITEID',  
          'OP_TIME_DWH',  
          'X733SPECIFICPROB',  
          'SEVERITY']].head(20)
```

Alarm rows: 149143

Unique Sites: 321

Unique Alarm Types: 150

Out[173...

	SITEID	OP_TIME_DWH	X733SPECIFICPROB	SEVERITY
0	ATG8505	2026-07-01 02:18:37.645003+00:00	Input Power Failure	3
1	ATG8505	2026-07-01 02:18:37.637008+00:00	Input Power Failure	3
2	ATG8505	2026-07-01 02:18:37.646002+00:00	Service Unavailable	5
3	ATG8505	2026-07-01 03:10:06.707001+00:00	Input Power Failure	0
4	ATG8505	2026-07-01 03:11:08.327002+00:00	Input Power Failure	0
5	ATG6722	2026-07-02 03:41:28.027002+00:00	External Link Failure	0
6	ATG8505	2026-07-07 03:23:36.124003+00:00	External Link to GNodeB Failure	0
7	ATG8505	2026-07-07 03:25:35.591006+00:00	External Link to GNodeB Failure	0
8	ATG8505	2026-07-02 03:31:02.654000+00:00	Node Group Sync Loss of All SoCC	2
9	ATG8505	2026-07-02 03:28:04.097007+00:00	Input Power Failure	3
10	ATG8505	2026-07-02 03:28:04.088000+00:00	Service Unavailable	5
11	ATG8505	2026-07-02 04:47:03.395008+00:00	Link Failure	0
12	ATG8505	2026-07-02 04:46:33.991003+00:00	Input Power Failure	0
13	ATG8505	2026-07-02 04:48:32.696007+00:00	Input Power Failure	0
14	ATG8505	2026-07-02 04:47:31.784004+00:00	Link Failure	0
15	ATG8505	2026-07-02 04:46:33.993010+00:00	Input Power Failure	0
16	ATG8505	2026-07-02 04:46:36.220001+00:00	Link Failure	3
17	ATG8505	2026-07-02 04:46:36.227002+00:00	Link Failure	3
18	ATG8505	2026-07-02 04:47:03.589004+00:00	Link Failure	3
19	ATG8505	2026-07-02 04:52:32.938007+00:00	External Alarm Door alarm	0

```
In [174... top_alarms = (
    alarm_pd['X733SPECIFICPROB']
    .value_counts()
    .head(20)
    .reset_index()
)

top_alarms.columns = ['Alarm', 'Count']

top_alarms
```

```
Out[174... 
```

	Alarm	Count
0	Service Unavailable	40859
1	Service Degraded	17867
2	Input Power Failure	9883
3	No Connection	7480
4	External Link Failure	7299
5	Resource Activation Timeout	6156
6	NE and OSS alarms are not in sync	5860
7	Suspected Sleeping Cell	5562
8	Ethernet Link Failure	3775
9	Cell Unavailable	3617
10	BASE STATION CONNECTIVITY DEGRADED	3370
11	XnC Link to GNodeB Failure	3321
12	Link Failure	3034
13	CELL FAULTY	3033
14	PLMN Service Redundancy Lost	2422
15	External Alarm Rectifier Non Urgent Alarm	2332
16	Temperature High	1809
17	External Link to GNodeB Failure	1702
18	Heartbeat Failure	1637
19	HW Fault	1606

```
In [175... alarm_pd['SEVERITY'].value_counts()
```

```
Out[175... 0    94533
           3    25535
           5    21833
           4     4605
           2     2603
           1       34
           Name: SEVERITY, dtype: int64
```

```
In [176... alarms_per_site = (
            alarm_pd.groupby('SITEID')
            .size()
            .sort_values(ascending=False)
        )

alarms_per_site
```

```
Out[176... SITEID
KCN0361    19797
CMI8895    18814
NKT0147     9005
TAK6708     5648
CBR1120     5201
...
CMI7534         4
BKE8607         2
LPN7604         2
BKF1042         2
CRI6244         1
Length: 321, dtype: int64
```

Link Alarms with Tickets

```
In [177... alarm_pd['OP_TIME_DWH'] = pd.to_datetime(
    alarm_pd['OP_TIME_DWH']
)

ticket_df['EVENT_START_TIME'] = pd.to_datetime(
    ticket_df['EVENT_START_TIME']
)
```

```
In [187... # Keep only tickets belonging to fully down sites
ticket_down = filtered_ticket_df[
    filtered_ticket_df['HPD_CI'].isin(fully_down_sites_names)
].copy()
```

```
In [189... print(f"Tickets after filtering: {ticket_down.shape[0]}")
ticket_down
```

Tickets after filtering: 74

Out[189...

	INCIDENT_NUMBER	VENDOR_TICKET_NUMBER	ASSIGNED_GROUP
16961	INC000101963557	SCCD_TT202607103555_A23450645	TRUE-TH-BBT-NOR1-CMI2-NOP
18455	INC000101974653	SCCD_TT202607108266_A23455380	TRUE-TH-BBT-NOR1-LPG-NOP
18535	INC000101974947	SCCD_TT202607108399_A23455514	TRUE-TH-WW-CEN-AYT-NOP
19272	INC000101977681	SCCD_TT202607109337_A23456419	TRUE-TH-BBT-NOR1-CMI2-NOP
19389	INC000101978819	SCCD_TT202607109475_A23456551	TRUE-TH-WW-CEN-SRB-NOP
...	...	...	...
50380	INC000102094051	SCCD_TT202607154677_A23501489	TRUE-TH-WW-WES-SPB-NOP
51411	INC000102097009	SCCD_TT202607155646_A23502475	TRUE-TH-WW-WES-SMK-NOP
53659	INC000102106670	SCCD_TT202607158897_A23505780	TRUE-TH-BBT-NOR2-PCB-NOP
54066	INC000102108451	SCCD_TT202607159428_A23506312	TRUE-TH-BBT-NOR2-TAK-NOP
56007	INC000102119260	SCCD_TT202607161395_A23508267	TRUE-TH-WW-CEN-ATG-NOP

74 rows × 36 columns

```
In [190... # Keep only alarms belonging to fully down sites
alarm_down = alarm_pd[
    alarm_pd['SITEID'].isin(fully_down_sites)
].copy()
```

```
In [191... print(f"Tickets after filtering: {ticket_down.shape[0]}")
```

Tickets after filtering: 74

```
In [193... ticket_df.columns
```

```

Out[193...] Index(['INCIDENT_NUMBER', 'VENDOR_TICKET_NUMBER', 'ASSIGNED_GROUP', 'IMPACT',
                  'PRIORITY', 'URGENCY', 'COMPANY', 'DESCRIPTION', 'DETAILED_DESCRIPTION',
                  'HPD_CI', 'NODETYPE', 'OP_TIME', 'OP_TYPE', 'SUBMIT_DATE',
                  'submit_date_UTC', 'submit_date_BKK', 'Categorization_Tier_1',
                  'Categorization_Tier_2', 'Categorization_Tier_3', 'RESOLUTION_CATEGORY_TIER_1',
                  'RESOLUTION_CATEGORY_TIER_2', 'RESOLUTION_CATEGORY_TIER_3',
                  'INC_STATUS', 'STATUS', 'REPORTED_SOURCE', 'EVENT_END_TIME',
                  'EVENT_START_TIME', 'PENDING_TIME', 'PENDING_DUE_TIME', '\n',
                  'SEVERITY', 'LAST_RESOLVED_DATE_UTC', 'LAST_RESOLVED_DATE_BKK',
                  'CLOSED_DATE_UTC', 'CLOSED_DATE_BKK', 'Technology', 'is_unresolved',
                  'OUTAGE_DURATION_HOURS', 'OUTAGE_BUCKET'],
                  dtype='object')

```

```

In [194...] alarm_ticket = alarm_pd.merge(
    ticket_df[
        ['INCIDENT_NUMBER',
         'HPD_CI',
         'EVENT_START_TIME',
         'EVENT_END_TIME',
         'OUTAGE_DURATION_HOURS',
         'OUTAGE_BUCKET',
         'is_unresolved']
    ],
    left_on='SITEID',
    right_on='HPD_CI',
    how='inner'
)
print(alarm_ticket.shape)

```

(149143, 55)

```

In [195...] alarm_ticket = alarm_ticket[
    alarm_ticket['OP_TIME_DWH']
    <=
    alarm_ticket['EVENT_START_TIME']
]

```

```

In [196...] alarm_count = (
    alarm_ticket
    .groupby('INCIDENT_NUMBER')
    .size()
    .reset_index(name='Alarm_Count')
)

alarm_count.head()

```

Out[196...

	INCIDENT_NUMBER	Alarm_Count
0	INC000101962626	2
1	INC000101965073	12
2	INC000101965452	59
3	INC000101967857	17
4	INC000101968618	17

	INCIDENT_NUMBER	Alarm_Count
0	INC000101962626	2
1	INC000101965073	12
2	INC000101965452	59
3	INC000101967857	17
4	INC000101968618	17

```
In [207... alarm_count = (  
    alarm_ticket  
    .groupby(["INCIDENT_NUMBER", "SITEID"])  
    .size()  
    .reset_index(name="Alarm_Count")  
)  
  
alarm_count.head()
```

Out[207...

	INCIDENT_NUMBER	SITEID	Alarm_Count
0	INC000101962626	CCS0331	2
1	INC000101965073	LBR8549	12
2	INC000101965452	PCR0036	59
3	INC000101967857	CMI8935	17
4	INC000101968618	TAK7125	17

	INCIDENT_NUMBER	SITEID	Alarm_Count
0	INC000101962626	CCS0331	2
1	INC000101965073	LBR8549	12
2	INC000101965452	PCR0036	59
3	INC000101967857	CMI8935	17
4	INC000101968618	TAK7125	17

```
In [197... filtered_ticket_df = filtered_ticket_df.merge(  
    alarm_count,  
    on='INCIDENT_NUMBER',  
    how='left'  
)  
  
filtered_ticket_df['Alarm_Count'] = (  
    filtered_ticket_df['Alarm_Count']  
    .fillna(0)  
    .astype(int)  
)
```

```
In [198... filtered_ticket_df.columns
```



```
Out[198... Index(['INCIDENT_NUMBER', 'VENDOR_TICKET_NUMBER', 'ASSIGNED_GROUP', 'IMPACT',
        'PRIORITY', 'URGENCY', 'COMPANY', 'DESCRIPTION', 'DETAILED_DESCRIPTION',
        'HPD_CI', 'NODETYPE', 'OP_TIME', 'OP_TYPE', 'SUBMIT_DATE',
        'submit_date_UTC', 'submit_date_BKK', 'Categorization_Tier_1',
        'Categorization_Tier_2', 'Categorization_Tier_3', 'RESOLUTION_CATEGORY_TIER_1',
        'RESOLUTION_CATEGORY_TIER_2', 'RESOLUTION_CATEGORY_TIER_3',
        'INC_STATUS', 'STATUS', 'REPORTED_SOURCE', 'EVENT_END_TIME',
        'EVENT_START_TIME', 'PENDING_TIME', 'PENDING_DUE_TIME', '\n',
        'SEVERITY', 'LAST_RESOLVED_DATE_UTC', 'LAST_RESOLVED_DATE_BKK',
        'CLOSED_DATE_UTC', 'CLOSED_DATE_BKK', 'Technology', 'Alarm_Count'],
        dtype='object')
```

```
In [208... alarm_pd.columns
```

```
Out[208... Index(['RECTIMESTAMP', 'FIRSTOCCURRENCE', 'IDENTIFIER', 'LASTOCCURRENCE',
        'SERVICE', 'EMS_NAME', 'NODESTATUS', 'SERVICEAFFECTING', 'CLEARTIME',
        'OBJECT', 'NODETYPE', 'SEVERITY', 'EXTENDEDATTR', 'ELEMENTMANAGERIP',
        'EVENTID', 'ADDITIONALSTRING', 'ALERTKEY', 'SITENAME',
        'ORIGINALSEVERITY', 'SITESTATUS', 'SITEID', 'CLASS', 'OBJECTSTATUS',
        'CLEAREDBY', 'X733SPECIFICPROB', 'MANAGER', 'NODEALIAS', 'SUMMARY',
        'NODE', 'TT_FLAG', 'SERVERSERIAL', 'TMTRANSSTATE', 'TTCREATIONTIME',
        'TMTRANSBEGINTIME', 'TT_ID', 'RECEIVEDATPROBE', 'physicalCard',
        'OP_TYPE', 'OP_TIME_DWH', 'CIRCLE', 'REGION', 'LOCATION',
        'BACKHAULTYPE', 'CONTROLNE', 'DATE_ID', 'HOUR_ID', 'Technology',
        'ALARM_DESCRIPTION'],
        dtype='object')
```

```
In [209... site_cell_map = (
    filtered_df[['SITE', 'CELLID']]
    .drop_duplicates()
)

print(site_cell_map.shape)
site_cell_map.head()
```

```
(10490, 2)
```

```
Out[209...      SITE      CELLID
0  ATG0011H_7NB04  ATG0011H_7NB04_S02
473  ATG0011H_7NB04  ATG0011H_7NB04_S03
946  ATG0011L_6NB02  ATG0011L_6MM02_S13
1475  ATG0019T_2NB01  ATG0019H_7NB01_S01
2078  ATG0019T_2NB01  ATG0019H_7NB01_S03
```

```
In [212... filtered_ticket_df.columns
```

```
Out[212... Index(['INCIDENT_NUMBER', 'VENDOR_TICKET_NUMBER', 'ASSIGNED_GROUP', 'IMPACT',
        'PRIORITY', 'URGENCY', 'COMPANY', 'DESCRIPTION', 'DETAILED_DESCRIPTION',
        'HPD_CI', 'NODETYPE', 'OP_TIME', 'OP_TYPE', 'SUBMIT_DATE',
        'submit_date.UTC', 'submit_date.BKK', 'Categorization_Tier_1',
        'Categorization_Tier_2', 'Categorization_Tier_3', 'RESOLUTION_CATEGORY_TIER_1',
        'RESOLUTION_CATEGORY_TIER_2', 'RESOLUTION_CATEGORY_TIER_3',
        'INC_STATUS', 'STATUS', 'REPORTED_SOURCE', 'EVENT_END_TIME',
        'EVENT_START_TIME', 'PENDING_TIME', 'PENDING_DUE_TIME', '\n',
        'SEVERITY', 'LAST_RESOLVED_DATE.UTC', 'LAST_RESOLVED_DATE.BKK',
        'CLOSED_DATE.UTC', 'CLOSED_DATE.BKK', 'Technology', 'Alarm_Count'],
        dtype='object')
```

```
In [214... site_ticket_summary = (
    ticket_df
    .groupby('HPD_CI')
    .agg(
        Ticket_Count=('INCIDENT_NUMBER', 'nunique'),
        Total_Outage_Hours=('OUTAGE_DURATION_HOURS', 'sum'),
        Avg_Outage_Hours=('OUTAGE_DURATION_HOURS', 'mean')
    )
    .reset_index()
)

site_ticket_summary.head(100)
```

Out[214...

	HPD_CI	Ticket_Count	Total_Outage_Hours	Avg_Outage_Hours
0	ATG6722	1	0.000000	NaN
1	ATG8505	1	0.000000	NaN
2	AYT0347	1	0.325833	0.325833
3	AYT0369	1	0.000000	NaN
4	AYT0405	1	0.343333	0.343333
...	...	...	...	...
95	CMIC059	1	0.000000	NaN
96	CMIC081	1	1.377222	1.377222
97	CNT0039	1	0.000000	NaN
98	CRI0103	1	1.444444	1.444444
99	CRI0469	1	8.631389	8.631389

100 rows × 4 columns

```
In [215... site_alarm_summary = (
    alarm_pd
    .groupby('SITEID')
```

```

    .agg(
        Alarm_Count=('EVENTID', 'count'),
        Unique_Alarm_Types=('X733SPECIFICPROB', 'nunique')
    )
    .reset_index()
)

site_alarm_summary.head()

```

Out[215...

	SITEID	Alarm_Count	Unique_Alarm_Types
0	ATG6722	247	7
1	ATG8505	469	22
2	AYT0347	13	2
3	AYT0369	233	6
4	AYT0405	34	3

In [216...

```

site_availability = (
    filtered_df
    .groupby('SITE')
    .agg(
        Avg_Availability=('NR_Cell_Availability', 'mean'),
        Min_Availability=('NR_Cell_Availability', 'min')
    )
    .reset_index()
)

site_availability.head()

```

Out[216...

	SITE	Avg_Availability	Min_Availability
0	ATG0003T_2NB01	99.841479	41.667
1	ATG0011H_7NB04	99.769085	76.556
2	ATG0011L_6NB02	95.090253	29.889
3	ATG0013P_9NB02	99.437674	0.000
4	ATG0019T_2NB01	86.098951	0.000

In [217...

```

site_score = (
    site_availability
    .merge(
        site_ticket_summary,
        left_on='SITE',
        right_on='HPD_CI',
        how='left'
    )
    .merge(
        site_alarm_summary,

```

```

        left_on='SITE',
        right_on='SITEID',
        how='left'
    )
)

```

```

In [ ]: site_score = site_score.fillna({
    'Ticket_Count': 0,
    'Total_Outage_Hours': 0,
    'Avg_Outage_Hours': 0,
    'Alarm_Count': 0,
    'Unique_Alarm_Types': 0
})

```

```

In [ ]: cell_score = (
    site_cell_map
    .merge(
        site_score,
        on='SITE',
        how='left'
    )
)

cell_score.tail(100)

```

```

In [ ]: from sklearn.preprocessing import MinMaxScaler

cols = [
    'Ticket_Count',
    'Total_Outage_Hours',
    'Alarm_Count',
    'Avg_Availability'
]

scaler = MinMaxScaler()

cell_score[cols] = scaler.fit_transform(cell_score[cols])

```

```

In [ ]: cell_score['Importance_Score'] = (
    0.40 * cell_score['Ticket_Count']
    + 0.30 * cell_score['Total_Outage_Hours']
    + 0.20 * cell_score['Alarm_Count']
    + 0.10 * (1 - cell_score['Avg_Availability'])
)

```

```

In [ ]: cell_score['Cumulative_Score'] = (
    cell_score['Importance_Score'].cumsum()
)

total_score = cell_score['Importance_Score'].sum()

cell_score['Cumulative_%'] = (

```

```
cell_score['Cumulative_Score']  
/ total_score  
* 100  
)  
  
cell_score.head()
```

```
In [ ]: for coverage in [80, 85, 90, 95]:  
        k = (  
            cell_score[  
                cell_score['Cumulative_%'] >= coverage  
            ]  
            .index[0] + 1  
        )  
  
        print(f"Top {k} cells explain {coverage}% of importance")
```

```
In [ ]:
```